



Frameworkx Standard

Information Framework (SID)

Product Business Entities

Information Framework Suite

GB922 Product

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Notice

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1. Business Entities

1.1. Product Domain

1.1.1. Introduction

In looking at 'Product' for the SID model, the modelers looked at Product concepts in the eTOM and member company offerings and implementations. The key concepts that were looked at were the detailed specification of products (ProductSpecifications); the way in which they are offered into the market (ProductOfferings); and the way in which they are maintained and perform while in use (Products). The SID model reflects Product entities from a business-oriented point-of-view and uses the best of standard modeling patterns for this area.

1.1.2. Overview of 'Product'

An overview of how the main Product domain business entities are related is shown in the figure below. This model is developed as we work through this document.

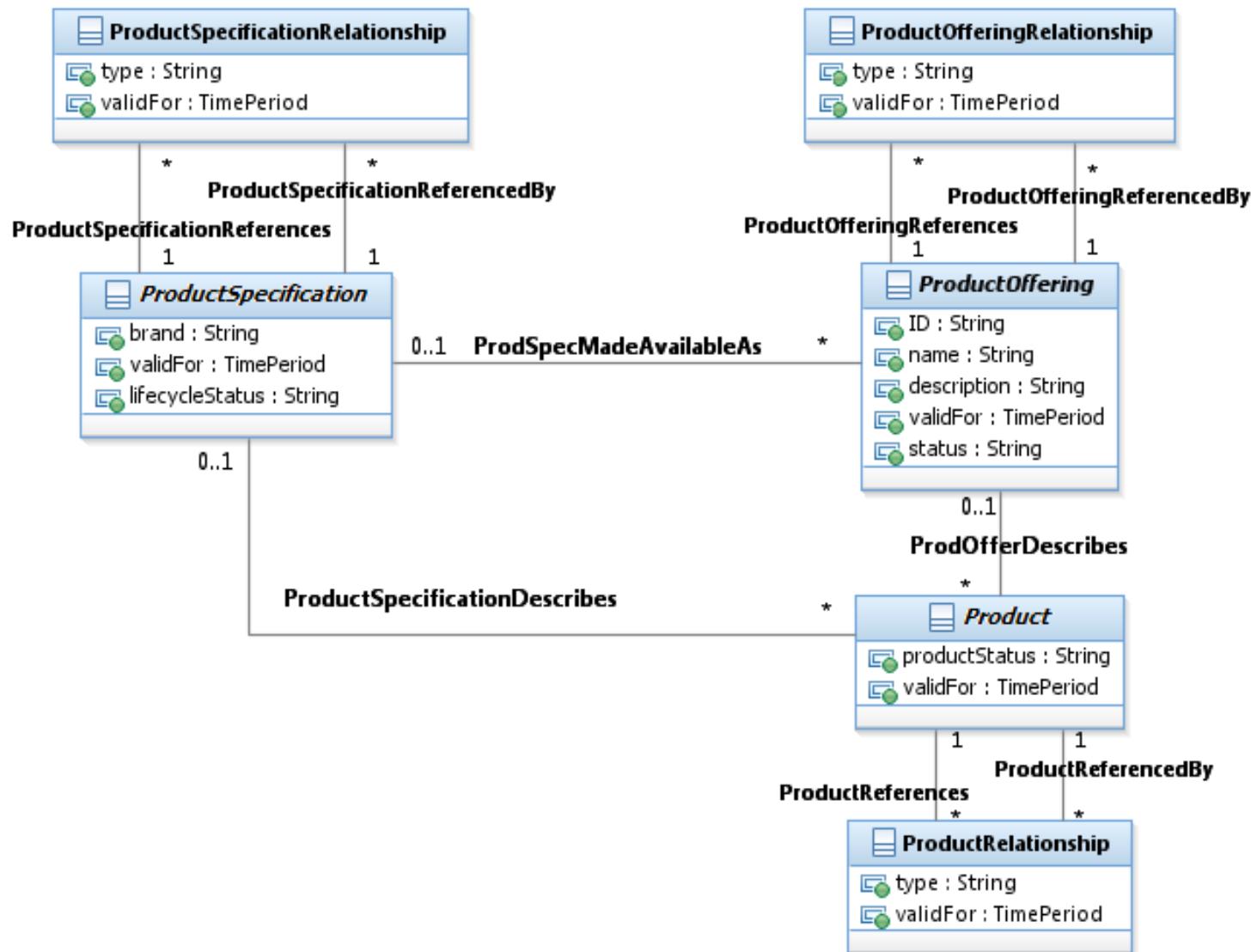


Figure Pr.01 - Primary Product Domain Business Entities

The Product domain is essential to the overall SID model. Customers need to be able to express their needs in plain English in order to determine which ProductOfferings can support their requirements and Service Providers need to be able to match these requirements to technical specifications to realize the ProductOffering.

A ProductOffering represents what is externally presented to the market for the market's use.

A ProductOffering can be assembled from a reusable ProductSpecification (sometimes referred to as a product spec).

For example, an email specification may be used in a number of different ProductOfferings. For example, an offering may be bundled with a Broadband Internet offering, while another may represent stand-alone email.

A single ProductOffering (called a BundledProductOffering), such as home-based internet service, may represent the aggregation of internet connection, email service, and web hosting ProductOfferings. In the case of the home-based internet service, the ProductOffering may not be related to a ProductSpecification, since it merely represents a grouping of other offerings.

A Product represents the subscription of a ProductOffering by a Party playing a PartyRole, such as a Customer. For example, Helen has subscribed to company ABC's internet ProductOffering.

The association between ProductSpecification and Product allows ProductSpecifications, particularly specifications that are part of a CompositeProductSpecification, which are not marketed as a ProductOffering to be instantiated as Products and related to customers or other involved parties. If this association did not exist, then these specifications would have to be instantiated as offerings. This enables a customer or other party to report problems about a product which was not marketed as an offering or for a product to appear on a bill, and so forth. For example, a composite high speed internet specification may include email and personal web pages atomic specifications which are not instantiated as separate offerings, but must be instantiated as Products. This is also necessary if a ProductSpecification not instantiated as a ProductOffering has configurable characteristics or prices associated with one or more characteristics.

The following figure gives an overview of the relationships between Product, Service and Resource Specifications

A **ProductOffering** represents how the ProductSpecification is sold (i.e. packaging rules, prices, alterations, commitments...).

The **ProductSpecification** represents a product specification as **perceived by the business user** and specifies what the marketing operator wants to sell at a functional level (i.e. what are the capacities, which usages are available...).

It can represent:

- material goods (Terminal, phone, mobile, fax, modem)
- or non tangible goods (like an Anti-Spam service for email).

When the ProductSpecification represents a tangible goods, it is realized as ResourceSpecification. Corresponding Resources are stored in warehouses or bought to a supplier on demand.

When the ProductSpecification is a non-tangible goods, it is either realized as a CustomerFacingService (when the Service Provider is able to realize it) or bought as a ProductSpecification proposed by a Supplier (when the Service Provider is not able to realize it). For example, a Broadband know-how might be built or bought depending on the location of it.

The **CustomerFacingServiceSpec** represents all the Service Provider's **know-how** of non-tangible goods at a functional level. It would be more appropriate to name it KonwHowSpec or ProductFacingServiceSpec, but the name is not changed as it is well known.

The ProductSpecification corresponds to a sub-set of the Service Provider's know-how according to what marketing people want to sell. Therefore, a ProductSpecification is a restriction of CustomerFacingServiceSpecs.

When the ProductSpecification corresponds to a tangible good, then it is realized as a ResourceSpecification.

The **ResourceFacingServiceSpec** represents the **technical solutions** that the Service Provider can implement for CustomerFacingServiceSpec.

Each technical solution (ResourceFacingServiceSpec) requires ResourceSpecifications to be realized and might require buying part of the solution to a supplier (for example the Service Provider might have to buy the Local Loop to provide a broadband access).

Not presented in the diagram: Sometimes the company is not able to realize the ProductSpecification on its own; in these cases, they will have to buy all or part of the technical solution. This is the case for MVNOs that always have to buy what they sell from a Supplier.

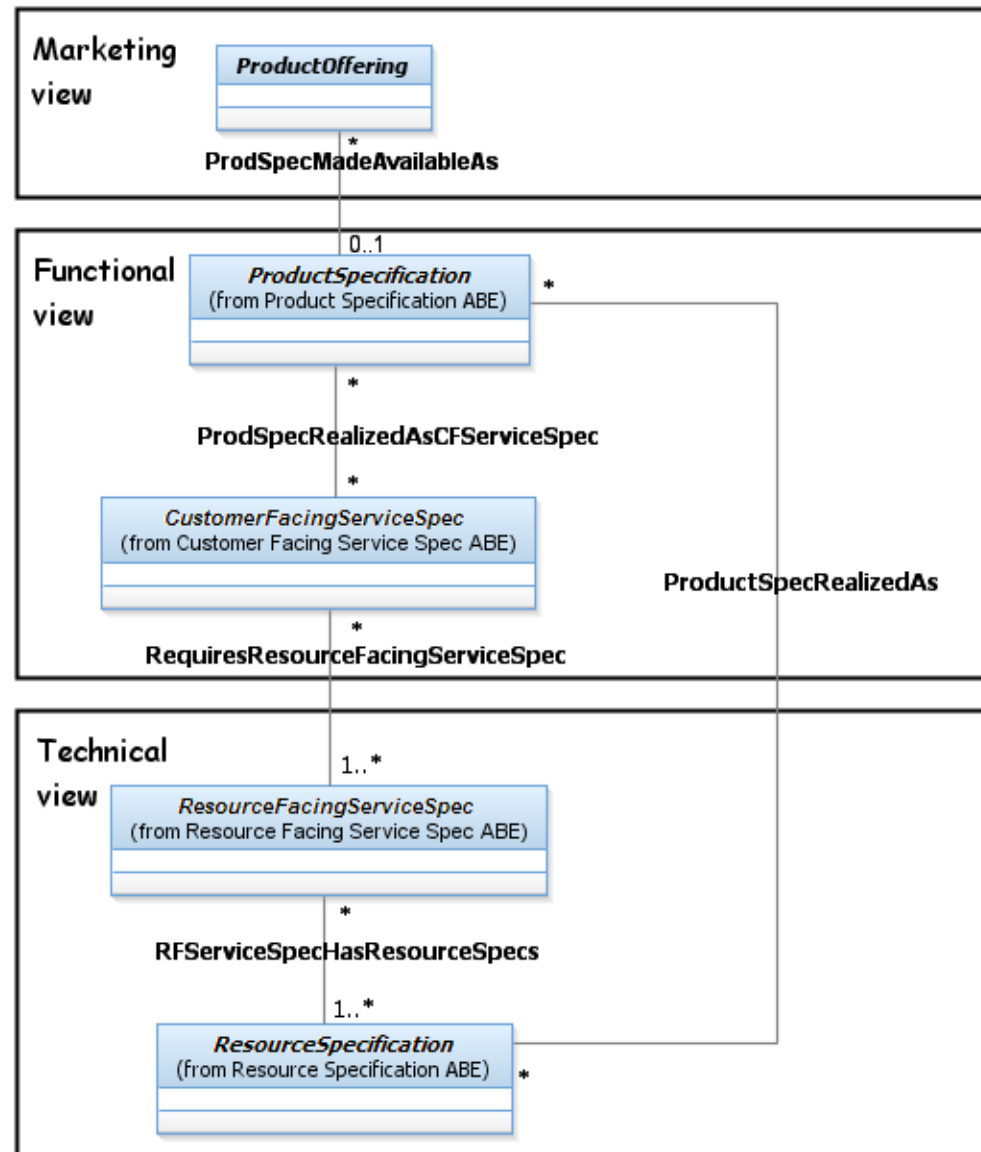


Figure Pr.02 - Product, Service, Resource Specification Relationships

The Catalogue Lifecycle Management process builds the specifications starting with the technical view, then the functional view and finally the marketing view.

On the contrary, the Order Handling process starts looking for what can functionally answer to the customer requirements and then find the ProductOfferings making available the ProductSpecifications.

The Delivery process has to identify the know-how (CFSSpec) corresponding to the ProductSpecification, then identify the different technical solutions (RFSSpec) corresponding to the know-how, chose the technical solution according to rules, and finally implement the ResourceSpecifications corresponding to the technical solution (RFSSpec).

Like this,

- OSS people specify functionally what can be done with OSS (Know-How i.e. CFSSpec)
- IT and Network are isolated from Marketing changes
 - Changing ProductOfferings and/or ProductSpecifications does impact only the order handling and catalogue management
 - Marketing people can define new ProductSpecifications without impacting the know-how (CFSSpec)
- The delivery process is only impacted by CFSSpec / RFSSpec and Resource changes
 - Technical solutions might be changed without changing what is seen by the Customer (ProductSpecification and ProductOffering)
 - The know-how might be improved without impacting existing ProductSpecifications

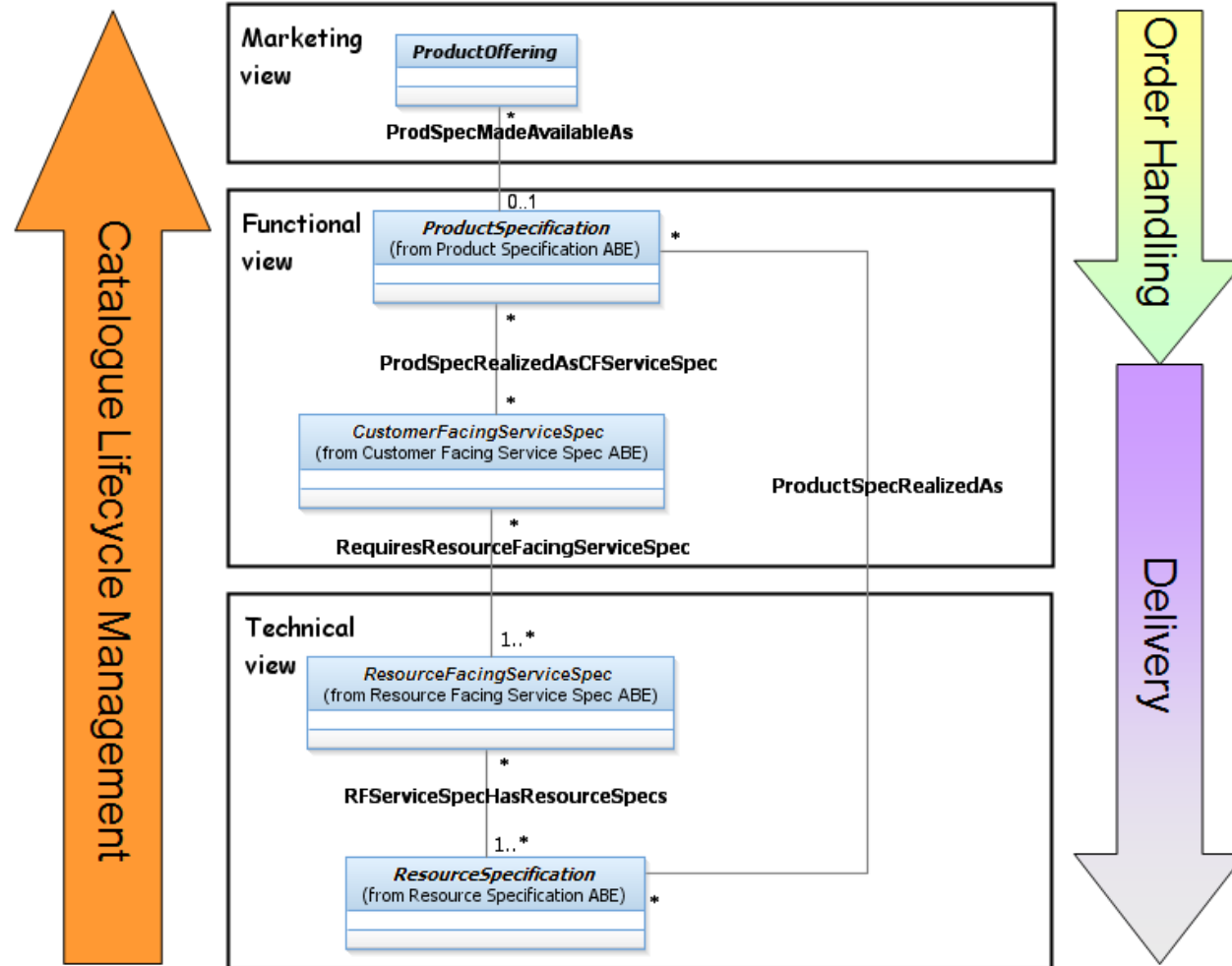


Illustration of ProductSpec, CFSSpec, RFSSpec and ResourceSpec

This illustration shows how the Voice Over IP can be described.

Notes:

- The H323 solution is not described here
- The SIP solution described here is close to the real situation but not complete.

The Voice over IP product is derived from the know-how of the telecommunication operator to be marketed to customers.

The know-how Voice over IP may be either realized as Software SIP solution or as a Software H323 solution.

In case of a Software SIP solution, it always requires a Service platform with type AS Base.

In addition to this, either a Huawei or E/// platform type is needed.

The Huawei platform type requires an IMS Core Huawei resource and the E/// platform type requires an ENUM and HSS resource.

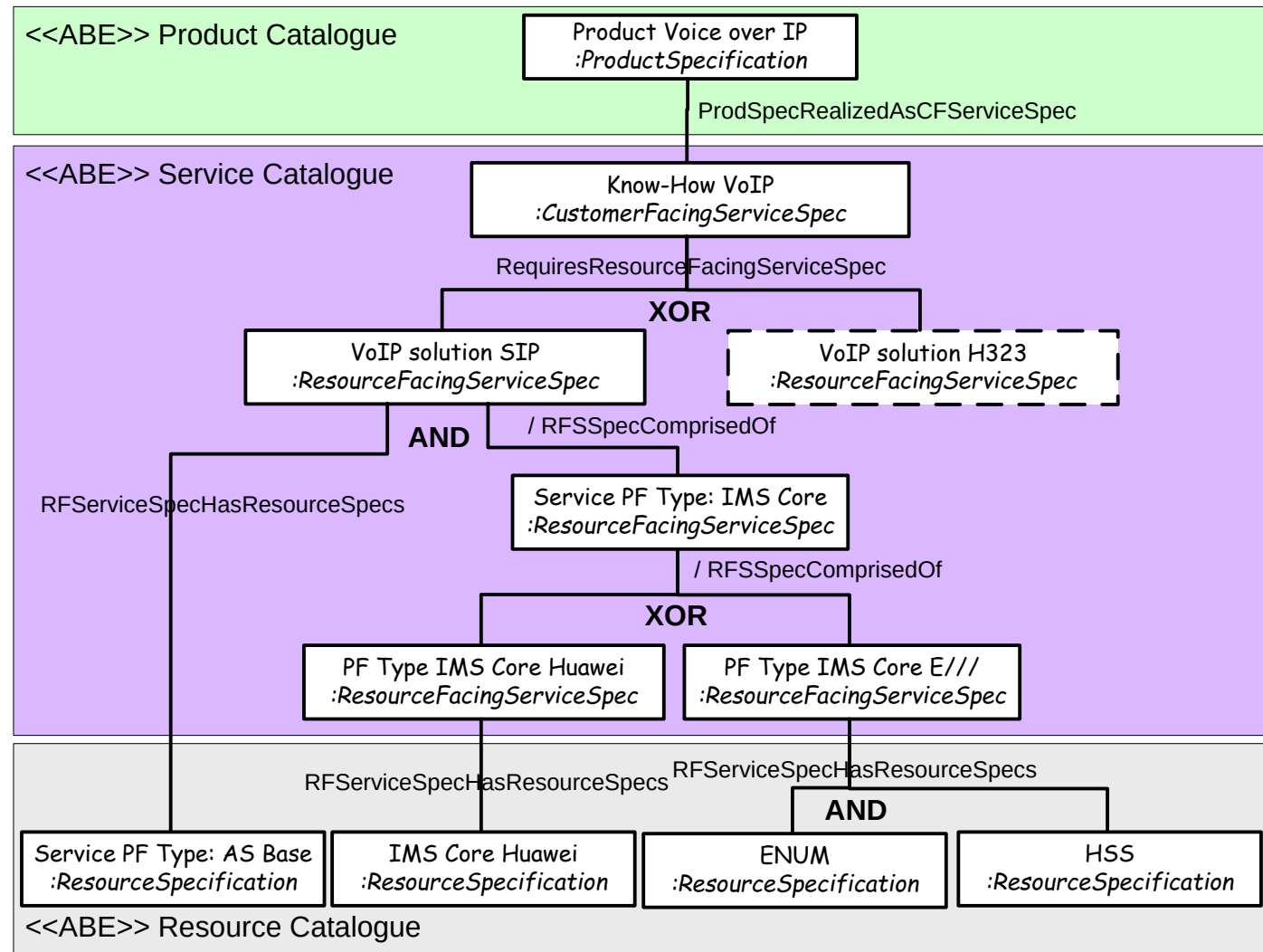


Figure Pr.02 – Illustration 1: Voice Over IP

The figure below depicts the relationship among Product, Service, and Resource domain business entities.

Business entities in the Product domain have a close relationship with business entities in the Service and Resource domains.

While Product business entities represent what the market sees of a provider's offerings, Service and Resource business entities represent the realization of the offerings from a provider's perspective.

For example, a Broadband Internet ProductOffering is realized by two CustomerFacingServices, one is a bandwidth connectivity service to the provider's network; the second is a virtual connectivity service to the Internet Service Provider.

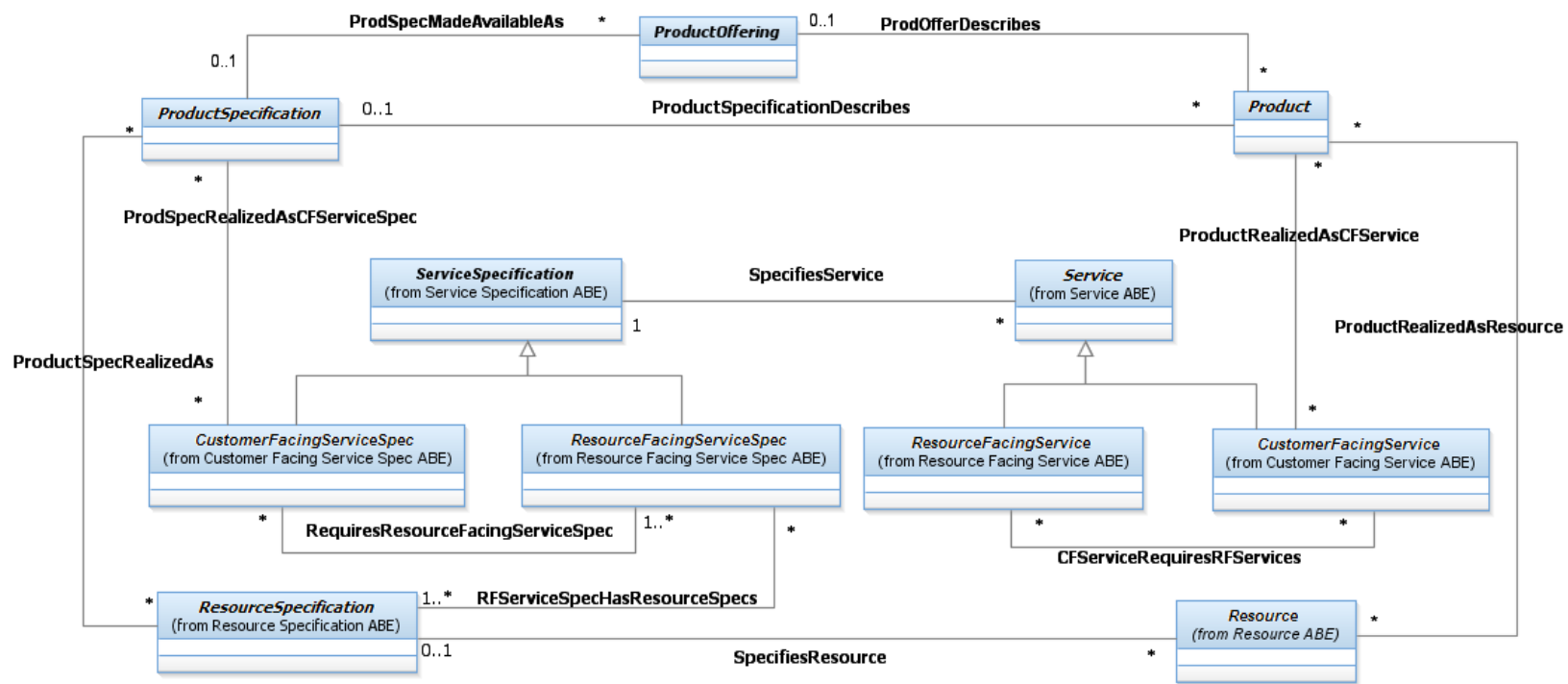


Figure Pr.2a - Product, Service, Resource Domain Relationships

1.2. Product/Product Specification

Products are things (tangible or intangible) which enterprises, such as service providers, market, sell or lease to customers to create profit. In a changing world, monopoly telcos with a custom-built portfolio have been replaced by a value network [eTOM] of Suppliers and Service Providers offering complex communications-based Products. The way in which those Products have to be specified is not as simple as it used to be.

A product serves as a window for customers and users into the world of the services and resources used to realize the product.

Customer/user always deals with products, unless the individual/organization they represent is playing the role of a service technician or some similar role. A user uses a Product to obtain its value promise, e.g. to make a phone call

Neither the Product User nor the Customer can see the CustomerFacingService. They only can see the Product.

Note: previously two views existed and one of them made the CFS visible for the user.

A ProductSpecification may be simple (atomic) or a composition of other ProductSpecifications [Fowler-Specification]. A ProductSpecification may represent a tangible object or something intangible provided to customers, realized as a 'service' but not to be confused with a network service here. The SID modeling of a CompositeProductSpecification covers both the physical build of more than one ProductSpecification and one or more ProductSpecifications supplied together in a collection. In addition, a CompositeProductSpecification may consist of other ProductSpecifications that give rise to separate ProductOfferings, or of AtomicProductSpecifications that are only for sale as part of a single ProductOffering. ProductSpecifications may also exist within groupings, such as product categories or product lines or product types.

Enterprises may assemble products themselves or they may buy them in from other suppliers (through contractual agreements where the enterprise takes the role of customer - see SID Agreement addendum 1A); or they may do a combination of these things.

Product Specification Example - The "3G Mobile super saver pack"

This is sold via intermediaries ("Dog-n-Bones-R-Us" phone shops) as a "BigTelephone" branded product. Nokia supply the phones and provides a 2-year guarantee. The "BigTelephone" network is used for voice and they supply a phone number and a SIM card. The SMS function is provided by "Best Messaging", as "BigTelephone" doesn't have an SMS function. The customer signs the contract with "Dog-n-Bones-R-Us" for 2 years with minimum monthly calls and Dog-n-Bones-R-Us bills the customer directly. Dog-n-Bones-R-Us has a separate contract with "Best Messaging" to cover the use of their SMS function. In looking at the detail of ProductSpecification, the following concepts are modeled by the figure below:

- Relationships between different ProductSpecifications for bundling or composite specifications;
- Different attributes for different types of ProductSpecification; for example, some Products the customer can take home with them, others they experience as a Service; for example, one delivered across a network;

- Atomic ProductSpecifications which cannot be broken down further.
- CompositeProductSpecifications which comprise a number of other ProductSpecifications, any of which may provide more than one instance per CompositeProductSpecification.

Note: A ProductSpecification cannot contain itself, but it can reference other ProductSpecifications.

At least one action is allowed for a ProductSpecification (AllowedProductAction) such as create, update...that might be limited to one or many SalesChannels. An AllowedProductAction has a type represented by a ProductTypeAction.

Note: Another way of specifying the actionType would be to have a relationship with Process.

For ProductOffering a Customer could upgrade their offering on the web, but to downgrade they need to come into a store or contact a service representative.

An AllowedProductAction might imply changes from one ProductConfigSpec to one or many ProductConfigSpec.

Thus, it is possible to define actions such as upgrade or downgrade and specify which SalesChannel enables these actions.

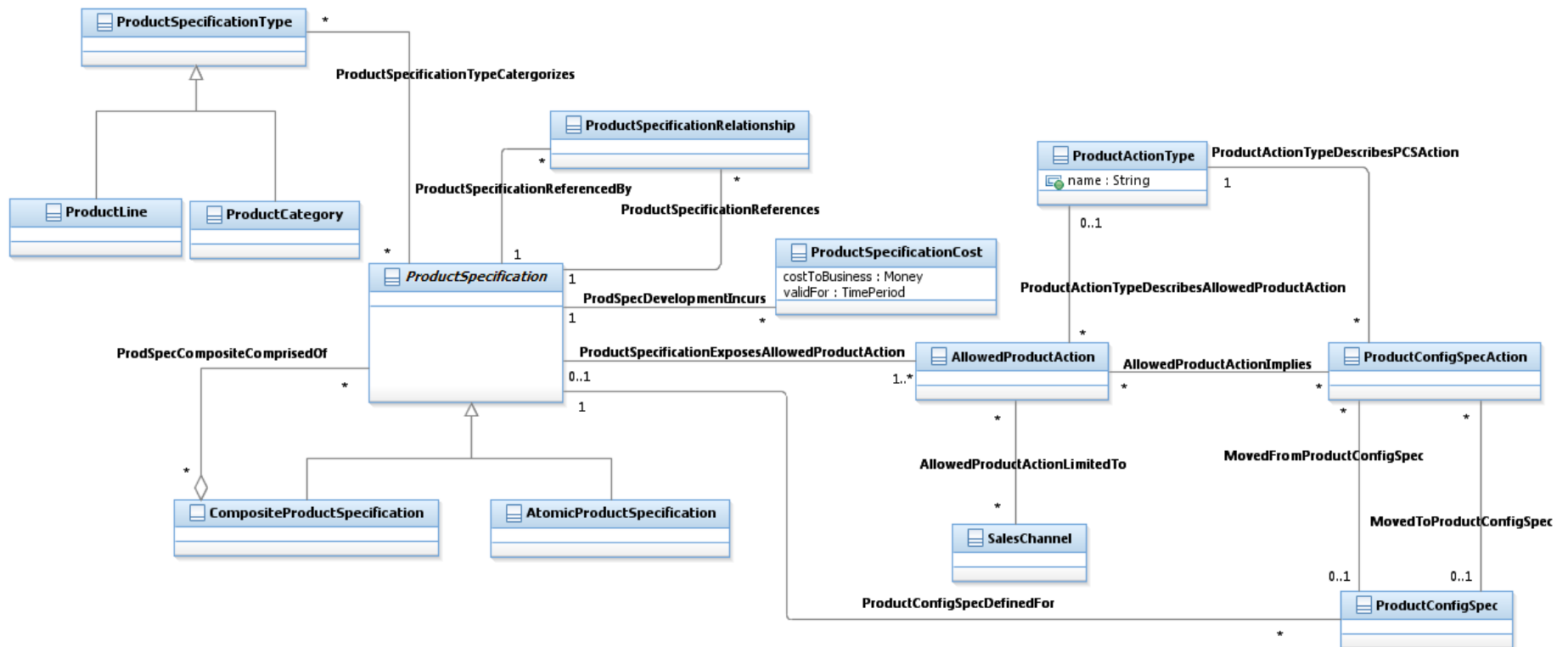


Figure Pr.03 - Product Specification – Basic Entities

In all combinations of products, there must be clearly defined relationships between the products in order to ensure consistency of service to the customer. Some examples of such ProductSpecificationRelationships would be migration, exclusivity, dependency and substitution, as indicated in Figure Pr.3 above.

ProductSpecifications can be grouped based on common characteristics or how the specs are marketed. Groupings are supported by ProductSpecificationType entity and its two subclasses, ProductLine and ProductCategory. For example, a ProductCategory could group all wire-line ProductSpecifications; a ProductLine could group all routers sold to a home-based consumer.

There are also costs associated with creating, developing and maintaining a ProductSpecification; the enterprise records these against the ProductSpecification. There is an implied link between the ProductSpecificationCost entity and business information but this has not been explored in this area of the SID model.

ProductSpecification examples

The following diagram gives examples of subclasses of AtomicProductSpecification that can be defined when it is required to specify particular relationships or attributes for specific ProductSpecification. The specificities of these subclasses have not been yet described in the model.

In SID 16.0, only the particularity of LoyaltyProductSpecification has been detailed.

The **NetworkProductSpec** defines non tangible products. It represents the common behaviour and description of an installed network product that will be provisioned in the network and that enables usages, like a Mobile Line.

A **FixedQuantityPackageProdSpec** defines buckets of usage, such as a 2 hours national calls. A FixedQuantityPackageProdSpec requires one or many NetworkProductSpecs from which Usages will debit the bucket.

A **RatePlanProductSpec** defines criteria to be used to gain special usage tariffs like the period (Ex: evenings and week-ends) or phone number (Ex: friends and family). The ProductOffering packaging a RatePlanProductSpec defines the Usages' ProductOfferingPrice conditioned by the criteria. A RatePlanProductSpec requires a NetworkProductSpecs.

The **GoodsProductSpec** corresponds to the specification of material goods.

The **LoyaltyProgramProdSpec** defines the LoyaltyRules that have to be checked to identify the actions to apply.

The **FieldOperationProductSpec** is a ProductSpecification corresponding to the intervention of an employee (from the marketing operator or one of his Suppliers) such as "Internet installation at home", "Internet training"...It requires an Appointment.

A **DirectoryProductSpec** is a ProductSpecification corresponding to publication of information (like associated subscriber name and address) of a telecommunication line with its phone number in a telephone directory.

A **ShippingProductSpec** defines the possible choices of configuration for the shipment of one or several GoodsProductSpecs like the place to ship, the delay...

A **ServiceLevelProductSpec** is a type of ProductSpecification that defines KPI that have to be evaluated and the threshold that must be respected. These KPIs are described by a ServiceLevelSpec. Note: Service Level Spec ABE has to be updated according to Metric ABE.

There is an alternative approach to support ServiceLevelProductSpec that is used to support adjustments made if a customer's service level is violated. There can be a ProductOfferingPrice that represents an adjustment associated to a ProductOffering. This price would be associated with a customer's Product. There would be a PolicySet associated with the ProductOffering that is evaluated by the triggering of a PriceEvent if a customer's service level is violated. The policy rules associated with the PolicySet determine if the adjustment should be made.

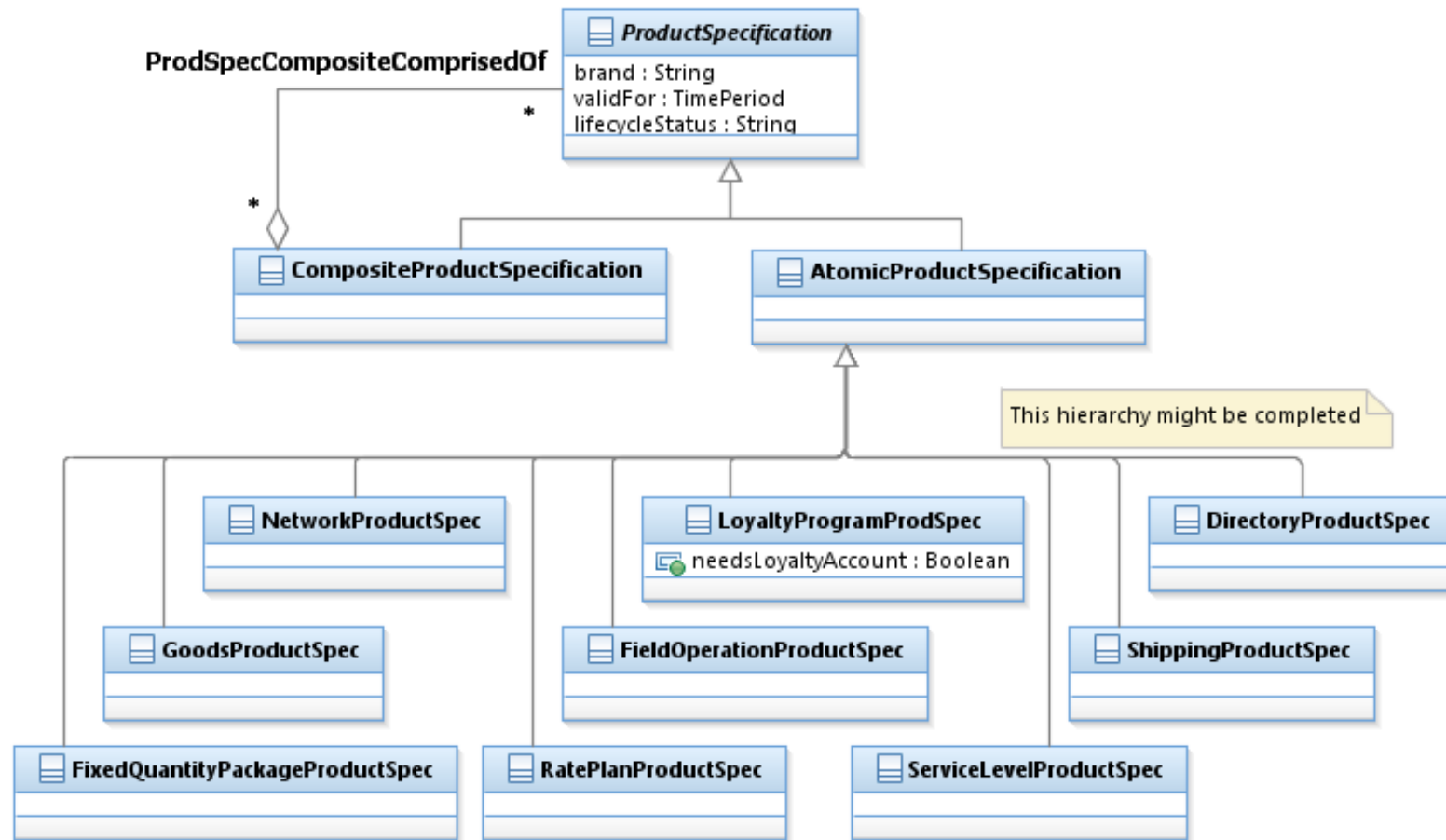


Figure Pr.03a - Product Specification - Examples

Illustration of ProductSpec subclasses: Mobile Line and related ProductSpecifications

This instantiation of the model intend to illustrate the use of ProductSpecification subclasses and ProductSpecificationRelationships.

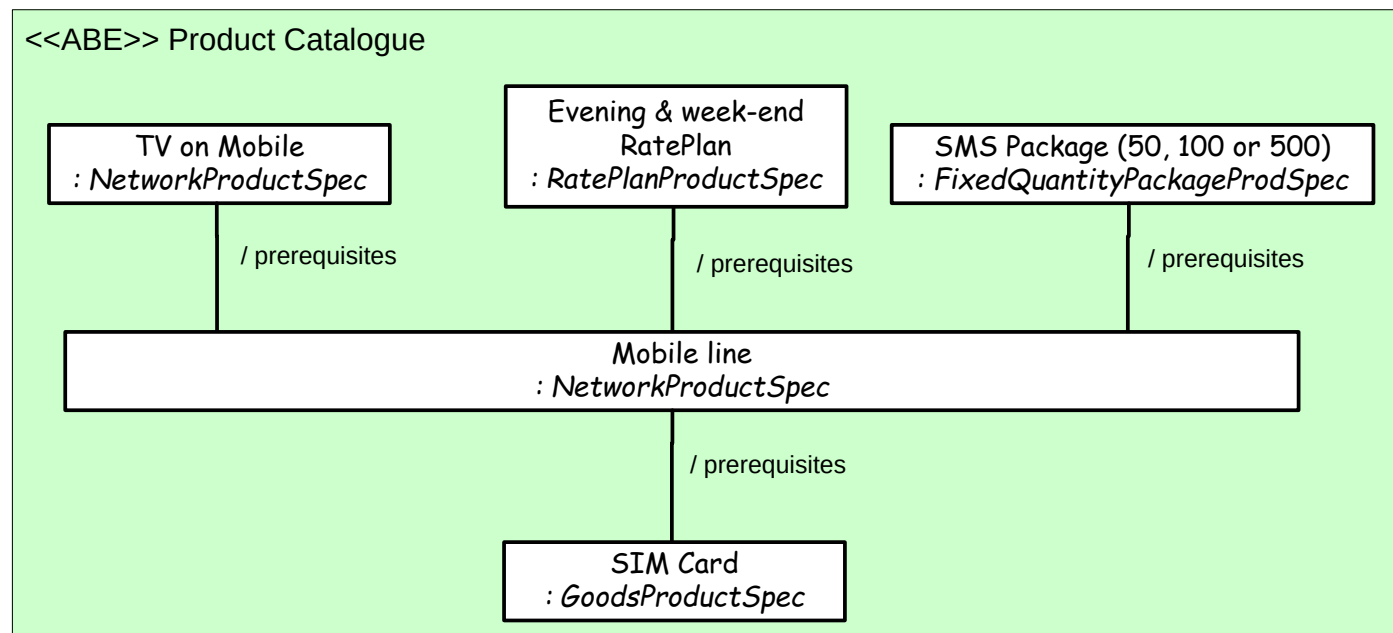
A Mobile line ProductSpecification enables ProductUsageSpecs (like Voice, SMS, MMS...), therefore this is a NetworkProductSpec.

A Mobile line needs a SIM card to function. A SIM Card is a tangible good, therefore this is a GoodsProductSpec.

An evening & week-end product defines the usages' tariff to be applied to the usage of the Mobile line, therefore this is a RatePlanProductSpec.

A SMS package defines a predefined quantity of product usage, therefore this is a FixedPricePackageProdSpec.

A TV on Mobile ProductSpecification enables ProductUsageSpecs (like watch channel...), therefore this is a NetworkProductSpec.



Product Characteristics

ProductSpecifications also have a variety of characteristics, such as a printer's dimensions, paper capacity, print speed, color, and so forth, and possible values associated with the characteristics. The SID uses ProductSpecificationCharacteristic and ProductSpecCharacteristicValue to represent the properties for a ProductSpecification [Fowler-Properties] as shown in Figure Pr.4 on the following page. Additional information regarding this pattern can be found in Gb922 Addendum 1R – Root Business Entities.

Characteristics can be grouped into three different types:

- If a characteristic has a discrete value (e.g. color or business coding), it may be derived from a range of allowable values;
- If a characteristic describes the product within a parameter range (e.g. dimension), it may be defined with a 'high value' and a 'low value' [Fowler-Range] ;
- If a characteristic value is 'derived', it will use other parameters in a formula. N.B. There are business rules implied here that require more work in a future phase of the SID Product definition.

Characteristics can also be bundled together into packages by using the ProductSpecCharRelationship. For example, a number of electrical characteristics can be grouped together using an "electrical properties" characteristic that represents a composite of the detailed properties, such as power requirements, plug requirements, and so forth. This entity can also be used for a number of other relationships between characteristics, such as mutually exclusive, inclusive, and so forth.

CharacteristicValues can reference other CharacteristicValues.

As ProductSpecifications are upgraded, and as the market and technology move on, they are subject to version control. If there is a significant change of capabilities, a new ProductSpecification is created; if the existing characteristics are revised in a way that doesn't affect the ProductOffering, then a new version of the existing ProductSpecification can be created. A ProductSpecification may have a number of different versions that represent minor revisions relating to its characteristics, cost and so forth. A description of these revisions is captured as attributes of the appropriate ProductSpecificationVersion.

When a product is created based on a ProductSpecification (see **Product Offering** below), the party may also have the option to choose from a set of properties values, such as color, or may be able to choose one from a number of product properties. Where a ProductSpecificationCharacteristic is interchangeable with others in this way, it is modeled as a ConfigurableProductSpecificationCharacteristic.

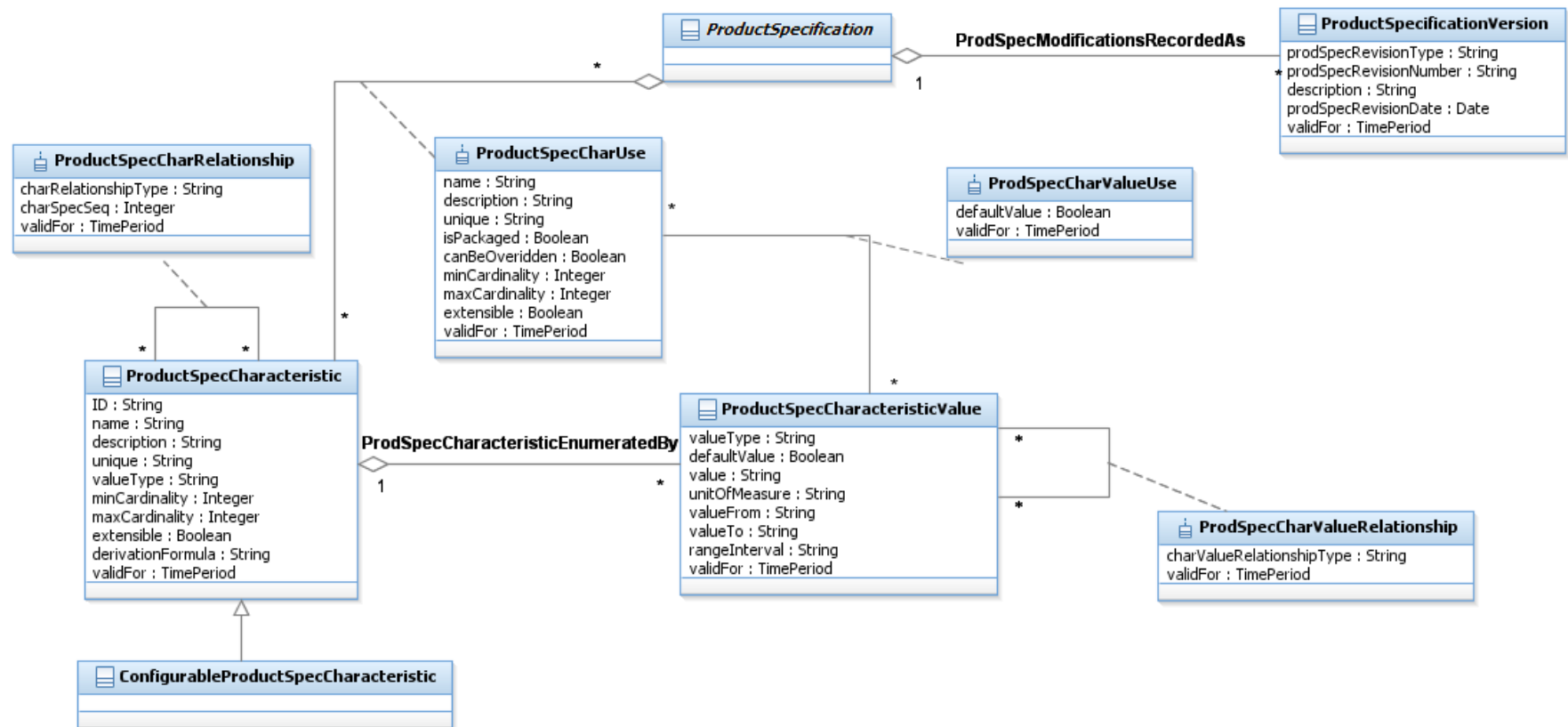


Figure Pr.4 - Product Specification - Characteristics and Versions

Product Characteristic Example – Email ProductSpecification

ProductSpecCharacteristics and their related ProductSpecCharacteristicValues can be used to support features provided by ProductSpecifications. For example, an Email may have a number of features, such as Spam protection, storage, message size, virus scanning and cleaning, mail forwarding, archiving, and so forth.

One way this type of product can be introduced into a model is by sub-classing the Product entity and including each feature as an attribute of an EmailProduct subclass. But this means that each time a new type of product is introduced the information architecture and most likely the application that supports the Product ABE would have to be changed. This is sometimes viewed as a static extension to an information architecture.

A more dynamic approach is to consider these features as instances of a ProductSpecCharacteristic. In this example each feature has a value associated with it. The value associated with Spam protection is "SpamBlock"; the value associated with storage is "1GB"; the value associated with message size is "10BM"; the value associated with virus scanning and cleaning is "included"; the value associated with mail forwarding is "included"; the value associated with archiving is "included. Note that all features or characteristics do not have to have values. For example, user ID and password are features, but the values are assigned when an instance of a Product is created.

The example will be continued in the Product section of this addendum.

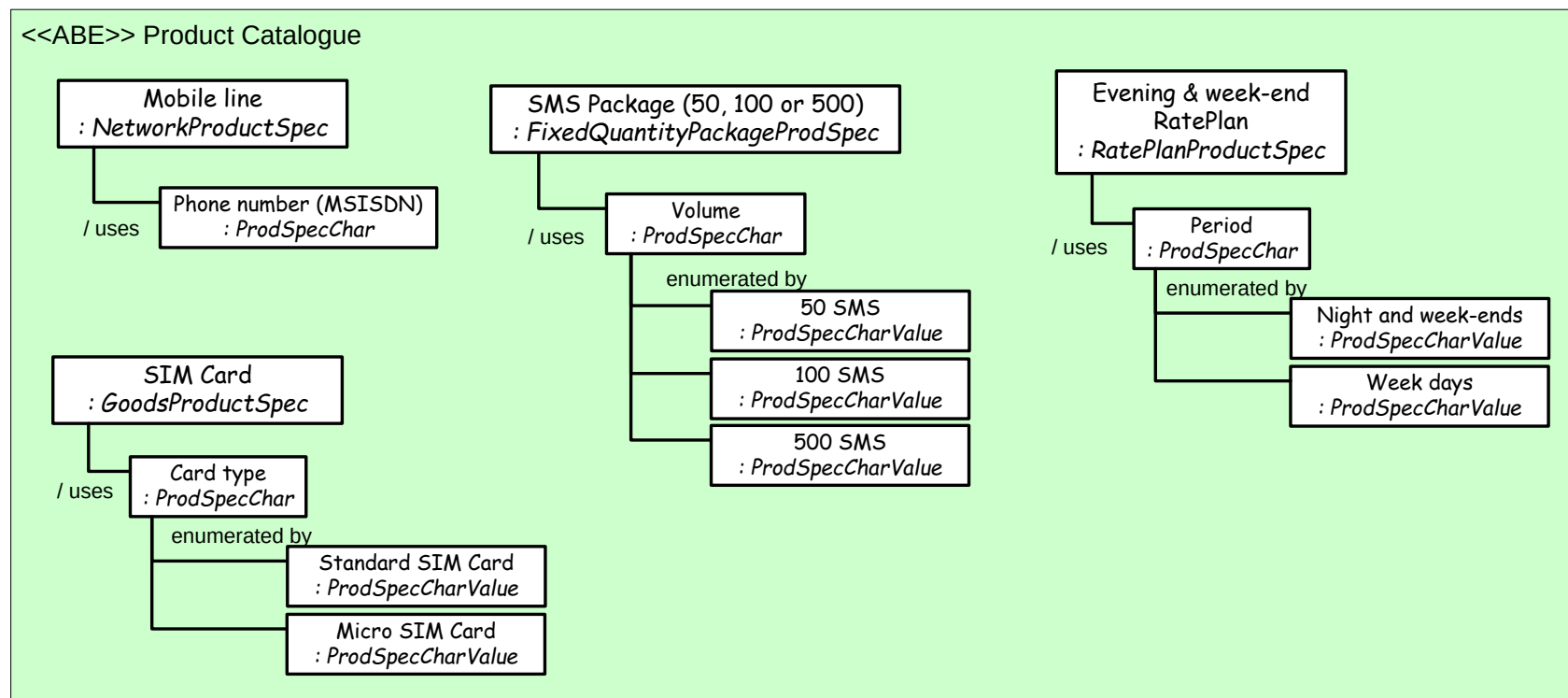
Illustration of ProductSpec characteristics related to Mobile Line and related ProductSpecifications

The phone number (aka MSISDN) of a Mobile Line corresponds to a ProductSpecCharacteristic without any predefined value.

The SMS Package has a Volume with predefined values (50, 100, 500). The value will be chosen by the customer during the order handling. The configuration chosen will impact the delivery and will be evaluated in the rules that determine the price to apply.

The Evening & Week-end Rate Plan has a Characteristic Period with two possible values “Night and week-ends” and “Week days”. The configuration chosen will be evaluated in the rules that determine the usage price to apply.

The SIM Card has a type characteristic that is used in the delivery process to provide a physical resource corresponding to this choice.



There are cases when an instance of a ProductSpecCharacteristic and/or a ProductSpecCharacteristicValue must be translated to a corresponding ServiceSpecCharacteristic and/or a ServiceSpecCharacteristicValue. For example, there may be a ProductSpecCharacteristic that represents the level of service provided as defined by associated Platinum, Gold, and Silver instances of ProductSpecCharacteristicValues. Each of these values may be translated into specific upload and download bit rates as defined by ServiceSpecCharacteristicValues.

Figure Pr.4a shows the associations between characteristics in the Product and Service domains that enable this translation. Similar translations can be made between characteristics in the Product, Service, and Resource domains as shown in Figure Pr.4b.

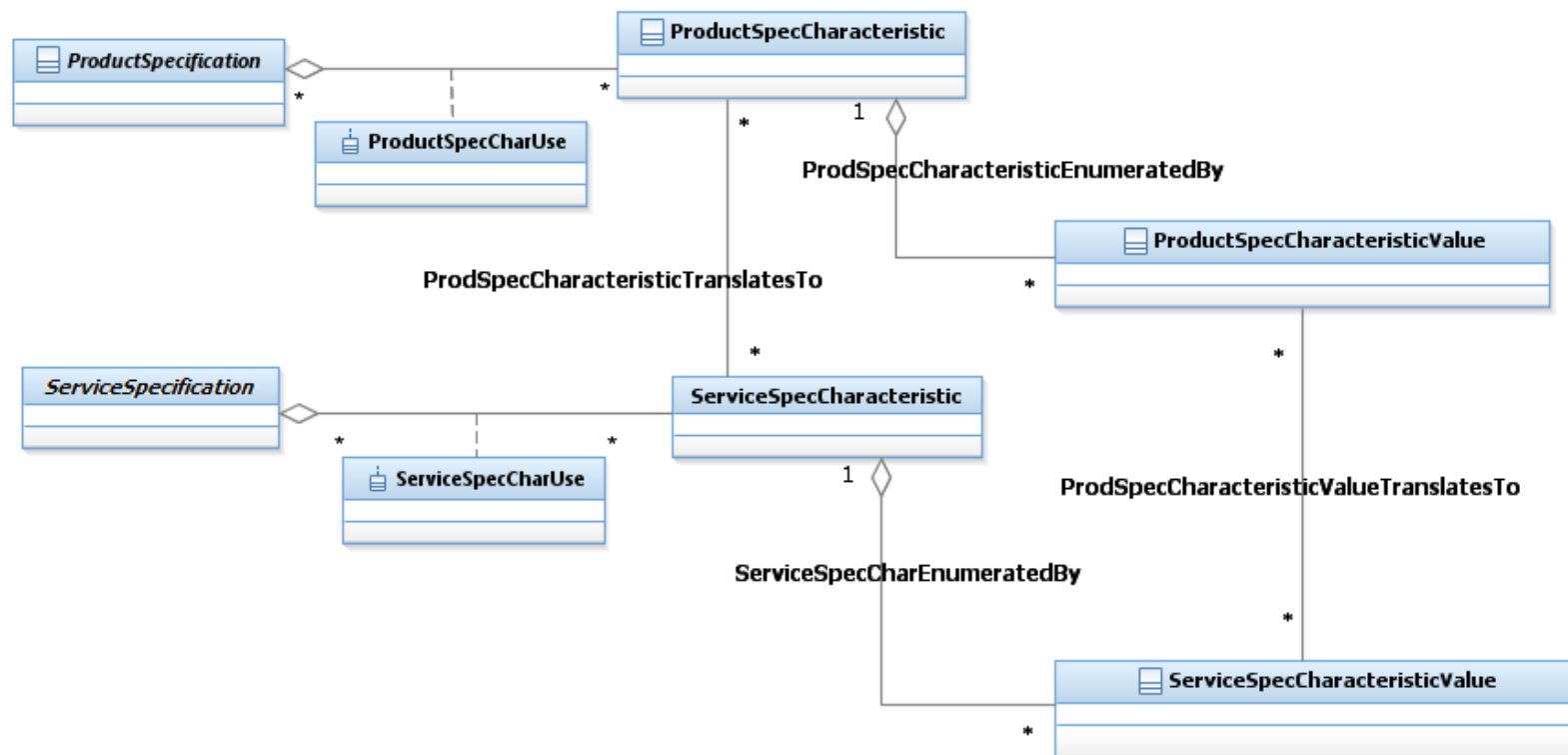
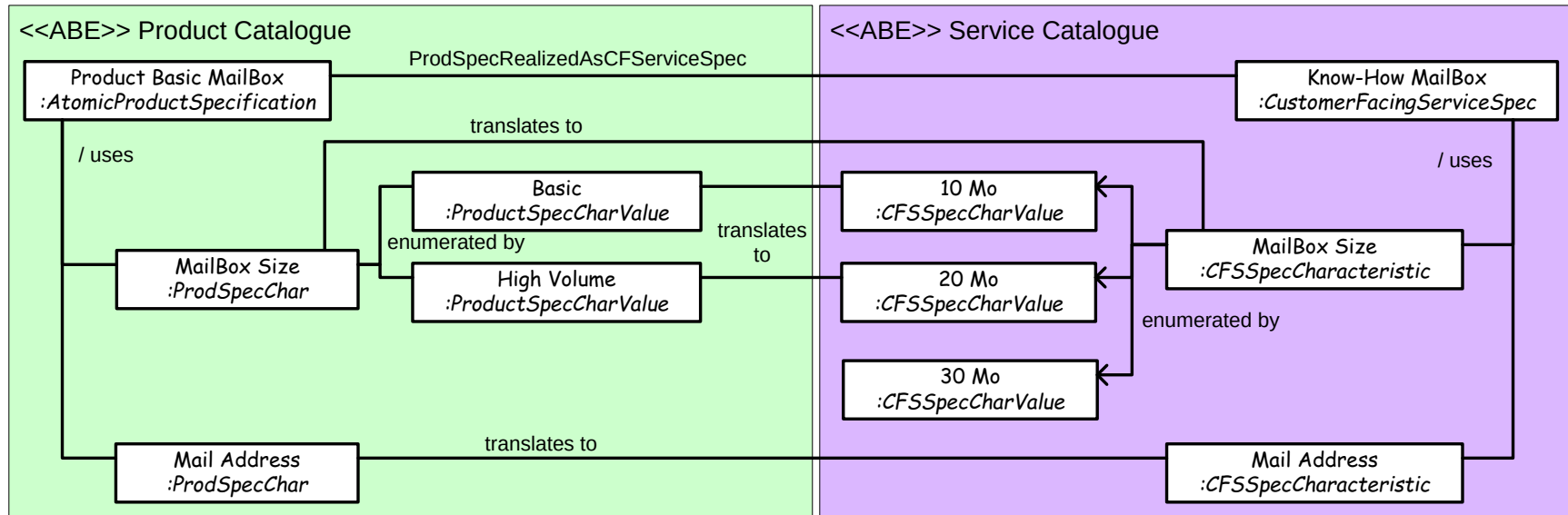


Figure Pr.4a - Product and Service Characteristic Associations

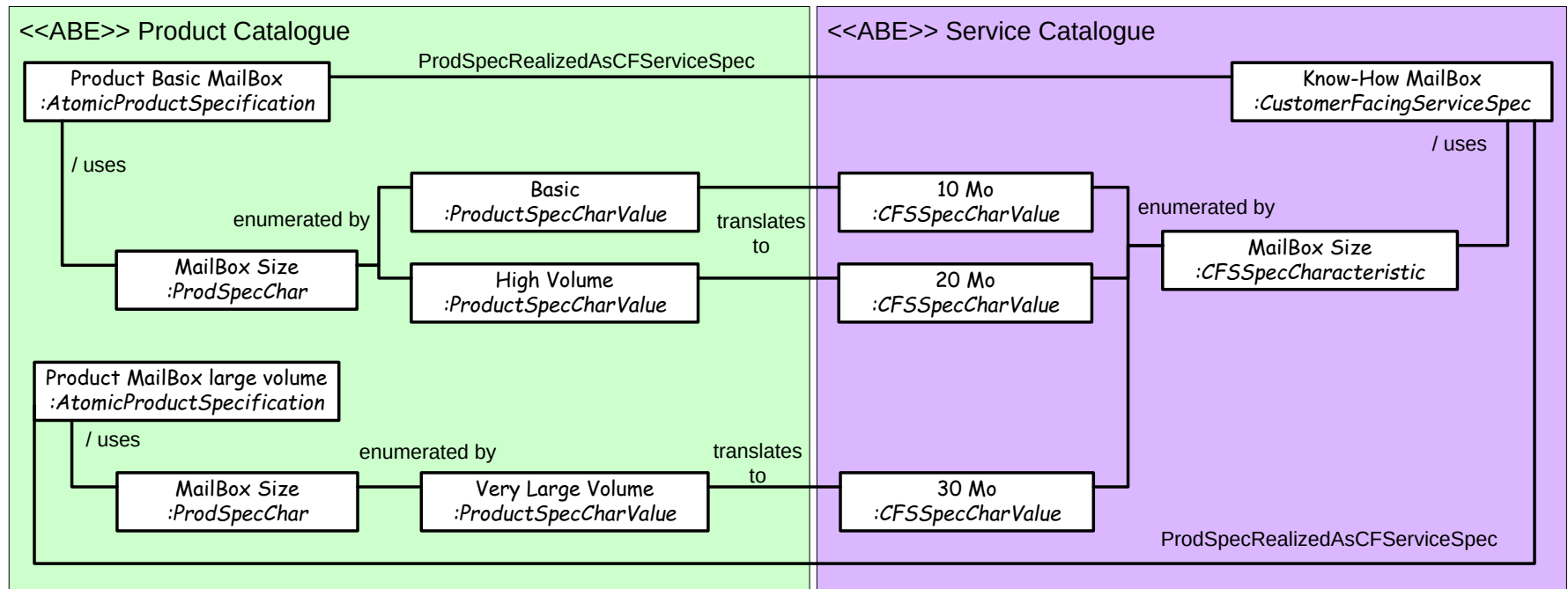
Illustration of ProductSpec characteristics translated to ServiceSpec characteristics

In a telecommunication operator's know-how, a Mailbox may have a size of 10Mo, 20Mo or 30 Mo.

But marketing may decide to market only Mailbox with a size of 10Mo and 20Mo and use more marketing wording (Basic and High Volume).



And based on the same CFSSpec, many ProductSpecification might be built up to being able to sell them in different ProductOffering with different possible configurations.



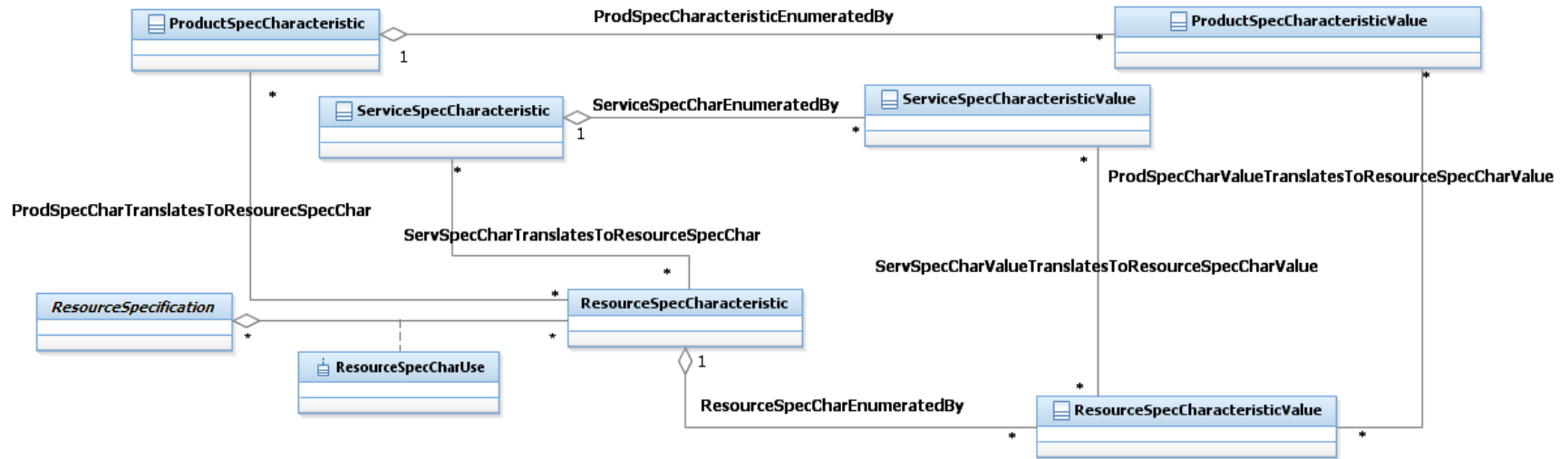


Figure Pr.4b - Product and Resource Characteristic Associations

Figure Pr.4c shows the association between characteristic use and ProductOffering. This association can be applied, if the ProductOffering determines the applicable characteristic values – e.g. the ProductOffering “silver plan” only allows bandwidths 2000 Mbit and 5000 Mbit out of the values 1000Mbit, 2000Mbit, 5000Mbit and 8000Mbit that are possible for the ProductSpecCharValueUse “bandwidth” in general.

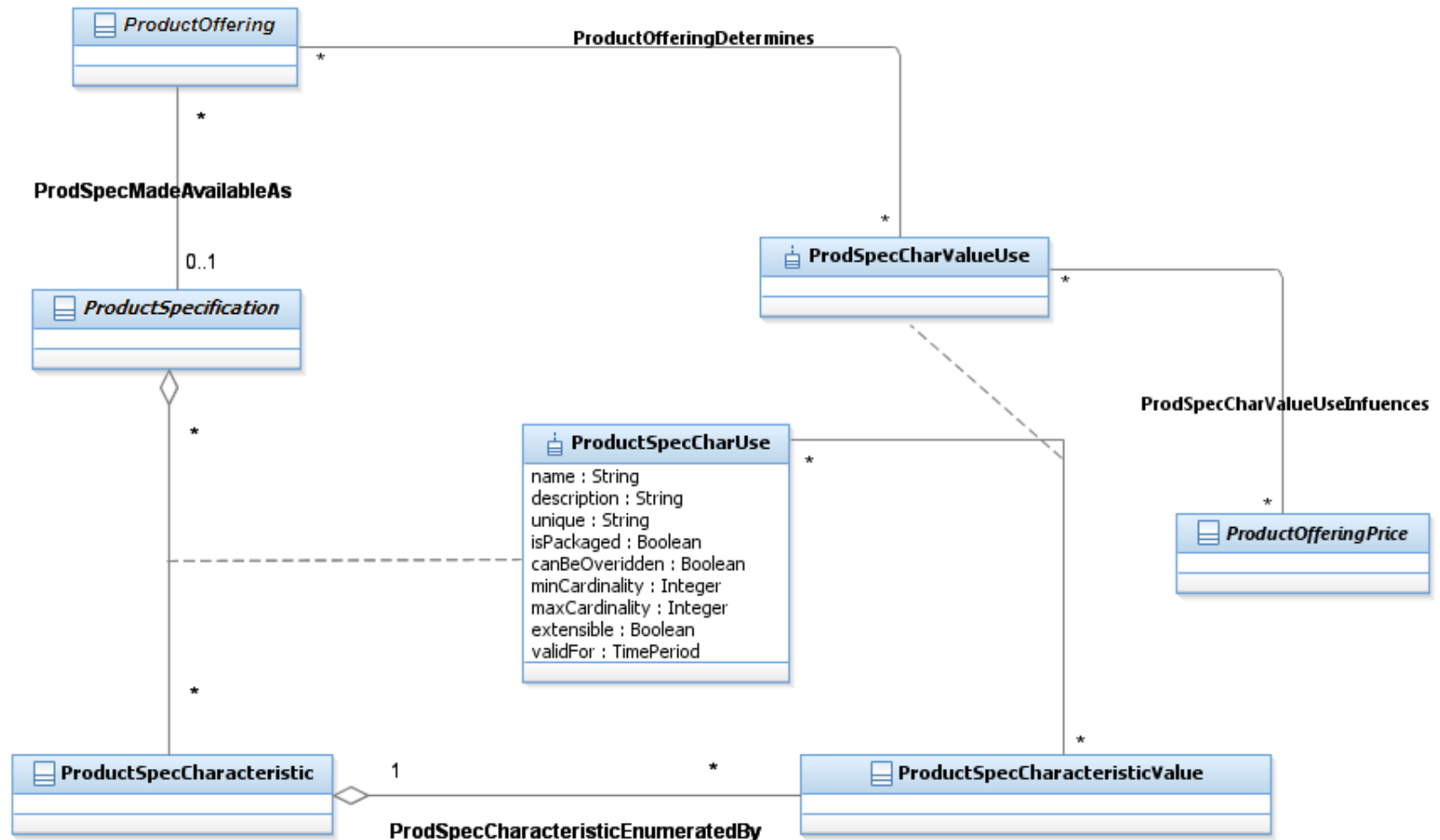


Figure Pr.4c – ProductSpecCharUse ProductOffering Associations

1.3. Product/Product Offering

ProductOfferings are targeted at one or more MarketSegments based on the appropriate MarketStrategy. Enterprises create ProductOfferings from ProductSpecifications that have additional market led details applied over a particular period of time. The ProductOffering may also be defined by the Location(s) at which it is available (i.e. Place). Locations in the SID model are represented by the Location business entity. Customers do not need to know anything about the detailed ProductSpecification, as the enterprise tailors the ProductOffering to their needs.

Where a business market strategy dictates, ProductOfferings may be bundled together and re-priced as necessary. In addition, the time period for a BundledProductOffering will depend upon those for its component offerings.

An enterprise puts ProductOfferings into the marketplace by developing and maintaining a catalog of ProductOfferings across one or more (market) DistributionChannels, for example a catalog on a website and a glossy brochure have different attributes and may also contain different sets of ProductOfferings. The ProductOfferings set out in the ProductCatalog are ProductSpecifications with additional details that enable a contract to be struck for their sale/purchase. This includes not only the ProductCharacteristics but also other product-dependent details, such as SLA parameters, invoicing and shipping details.

There are typically a number of ProductOfferings associated with a single ProductSpecification, for example to reflect offers to the market for different time periods. In cases where a ProductSpecification's value to the business is solely as a component part of a CompositeProductSpecification, it does not have a ProductOffering associated with it as it is not be sold on its own. Cases where these items are supplied to the Customer separately as 'spares', e.g. under warranty, are covered under **Product** below.

When Customers select Products from the ProductCatalog, it is the ProductOffering details that they are looking at and which are reflected in what they agree to contractually.

Each ProductOffering is marketed by a PartyRole. If need be more processes responsibilities such as order handling or billing can be specified using Party Product Specification and Offering (see. Engaged Party guide book for more details).

Each ProductOffering might be valued by ProductOfferingPrices that are asked for or allowed when a ProductOffering is bought, rented, or leased.

At least one action is allowed for a ProductOffering (AllowedProductAction) such as subscription, update...that might be limited to one or many SalesChannels. For example, a ProductOffering might be upgraded directly by the Customer on the web, but to downgrade it they need to come into a store or contact a service representative.

An AllowedProductAction has a type represented by a ProductTypeAction.

Note: Another way of specifying the actionType would be to have a relationship with Process.

The main ProductOffering entities are shown in Figure Pr.5 below.

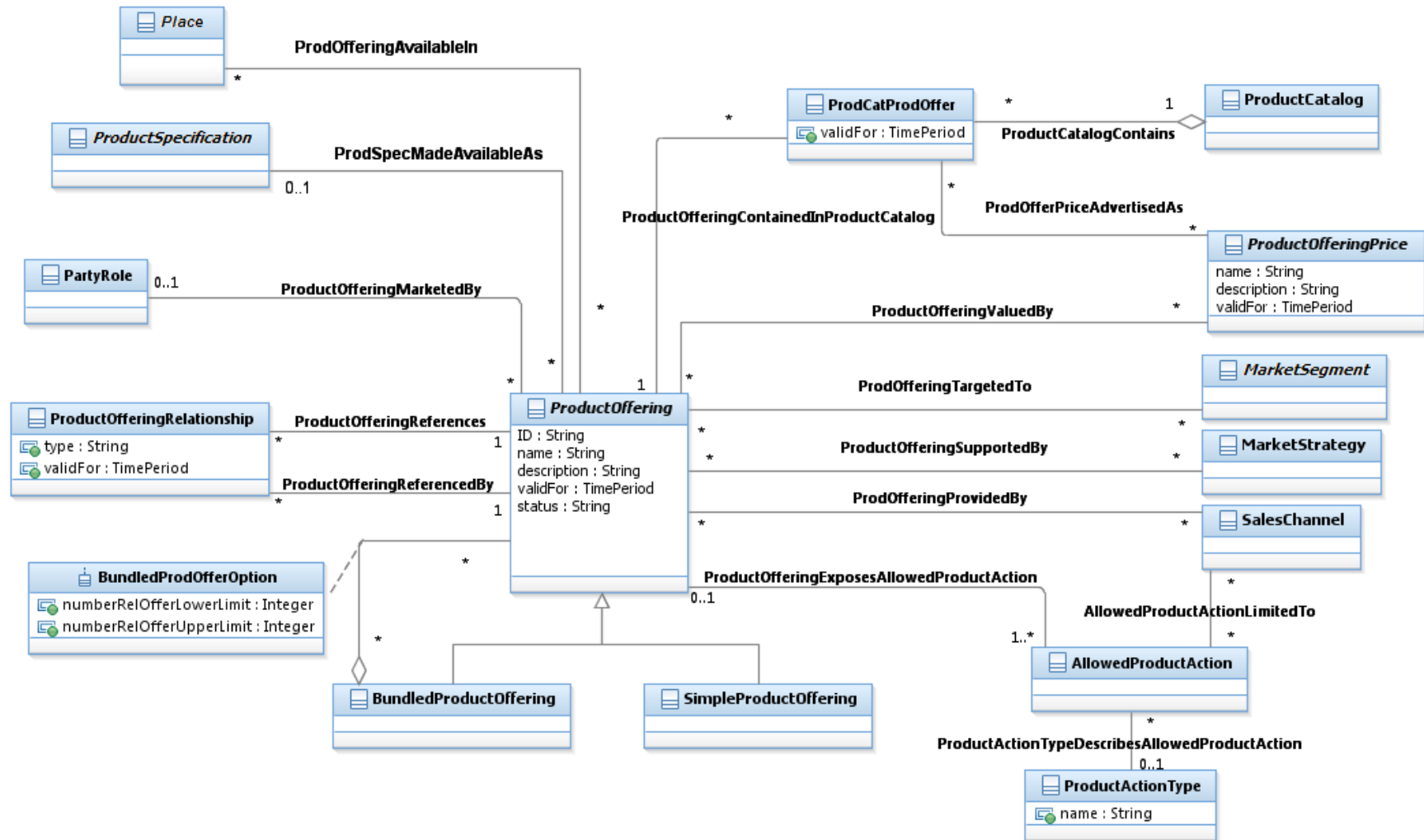


Figure Pr.5 - Product Offering - Basic Entities

Product Offering Example

Smooth Telco normally markets a range of mobile phones as individual items. For a limited period around February 14th, it decides to market, a BundledProductOffering called the Valentine Special, one model with a special red cover, boxed in pairs and accompanied by two champagne glasses. As an added attraction, the deal includes a discount for calls made between the telephone numbers allocated to each pair for a set period. It is a huge success and likely to be repeated the following year when a new ProductOffering with the new valid time period will be created. These ProductOfferings exist side by side with the standard ProductOffering for the mobile phone model when the phones are sold individually.

A question arises with this example. Should the Valentine Special or any BundledProductOffering have an associated ProductSpecification? The current model, shown in Figure Pr.5a, allows for this by representing the association between ProductOffering and ProductSpecification as 0, 1. But this association could lead to a SimpleProductOffering not having a related ProductSpecification.

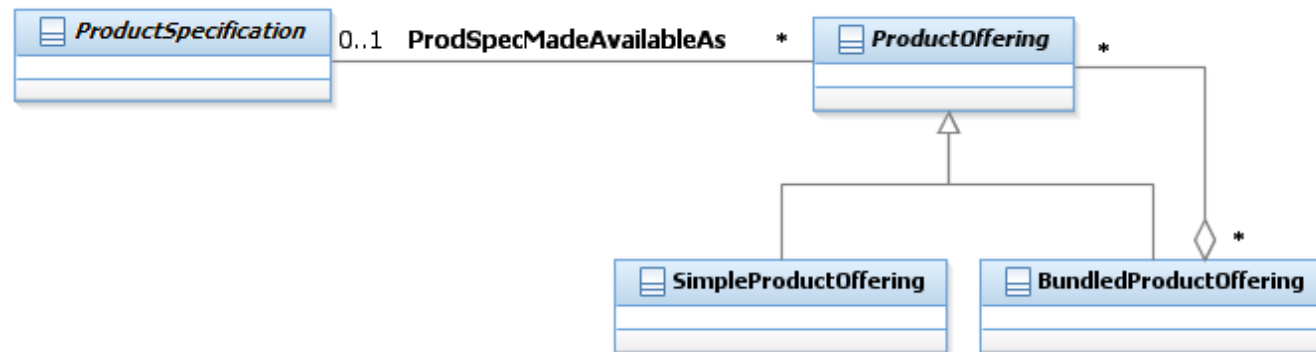


Figure Pr.5a – Product Spec – Product Offering Relationship – Alternative 1

A better representation of this association would be to only relate a SimpleProductOffering to a ProductSpecification as shown in Figure Pr.5b. This association allows a BundledProductOffering to not have a related ProductSpecification while requiring that a SimpleProductOffering has one.

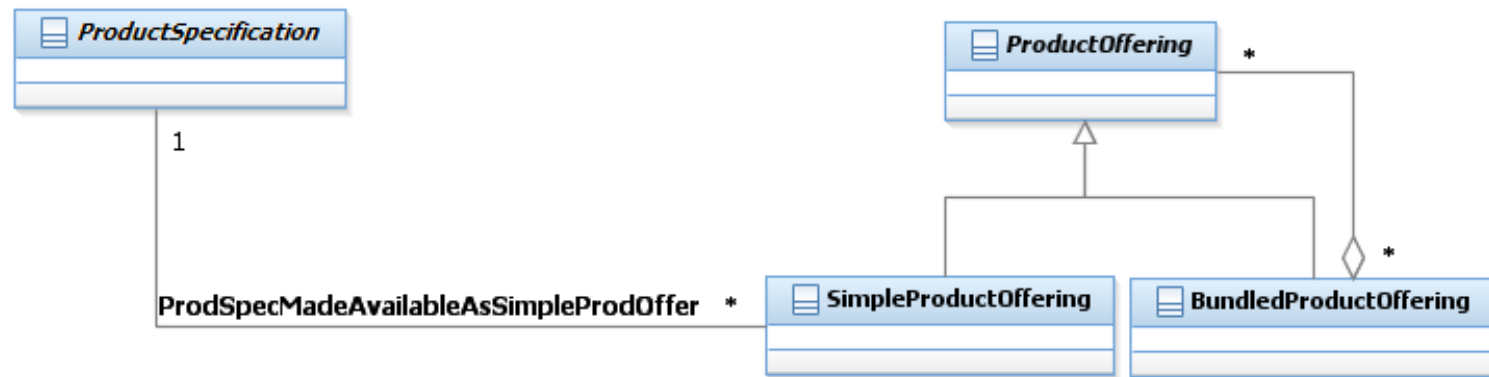


Figure Pr.5b – Product Spec – Product Offering Relationship – Alternative 2

In some cases there are limits to the number of ProductOfferings that can be procured as part of a BundledProductOffering. For example, when subscribing to a mobile telephone service up to five phones can be procured at a special price. The BundledProdOfferOption handle these cases by allowing lower and upper limits to be set for ProductOfferings that comprise a BundledProductOffering as shown in Figure Pr.5c.

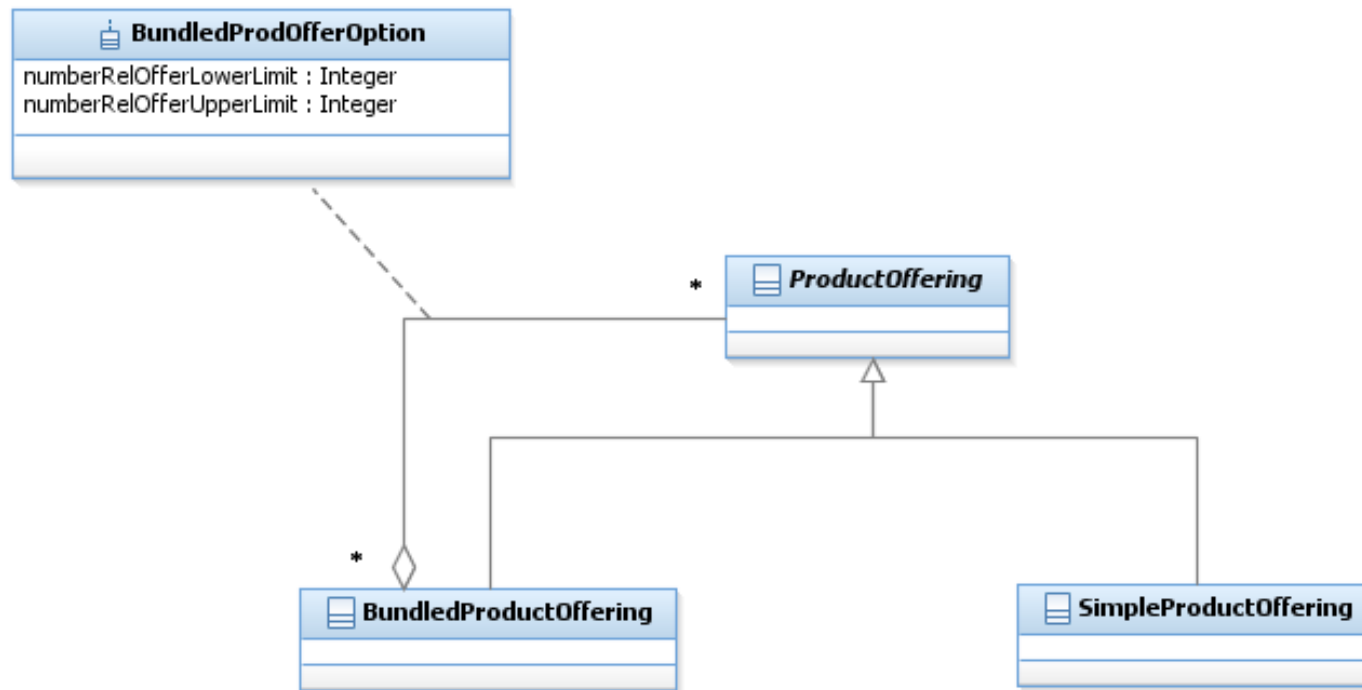


Figure Pr.5c – Bundled Product Offering Option

ProductOfferings that are part of a BundledProductOffering should not be individually procurable. For example, a free SMS offering cannot be procured by itself. A separate SimpleProductOffering would be created to allow for add-on SMS service. From an implementation perspective a rule can be established that does not allow a ProductOffering that is part of a BundledProductOffering to be procured. This rule would check to see if the offering is part of a bundle. Another implementation option would add an attribute to ProductOffering that indicates whether the offering was part of a bundle. The choice of these two options is up to the implementer.

Product Offering Terms

The ProductOffering is primarily the way to position products in the marketplace to create profit; the amount of profit depends upon both the original pricing of the product and the eventual terms of the sales agreements struck for that product offering.

The ProductOfferingPrice is based on both the basic cost to develop and produce products (see **Product Specification**) and the enterprise's policy on revenue targets. The pricing policy mechanism is represented in this model by 'term'. Terms are the conditions under which the ProductOffering is made available to Customers. For example, product finance term is only one of many which may come into play. In addition to a definition of the list price, financial terms may also include acceptable methods of payment for this type of product. As examples of other types of ProductOfferingTerm, the SID model could also include product service term and product shipment term.

When a price has been created, it will appear as the list price for ProductOffering in the ProductCatalog. For a SimpleProductOffering, this may also be the effective list price but, if ProductOfferings are bundled together, the overall price will not necessarily be the combination of component list prices and pricing policy will be required. The Customer will refer to the list price in selecting a Product but, depending upon their relationship with the enterprise making the ProductOffering available, this price may be further revised through discounting, such as for bulk buying, loyalty and so forth. This is modeled through the AgreementTermOrCondition class, shown in Figure Pr.6.

The price, applied for a ProductOffering may also be influenced by the ProductOfferingTerm, the customer selected – e.g. a ProductOffering can be offered with multiple terms, like commitment periods for the contract. A 12 month contract may be cheaper than a contract with a 24 month commitment period.

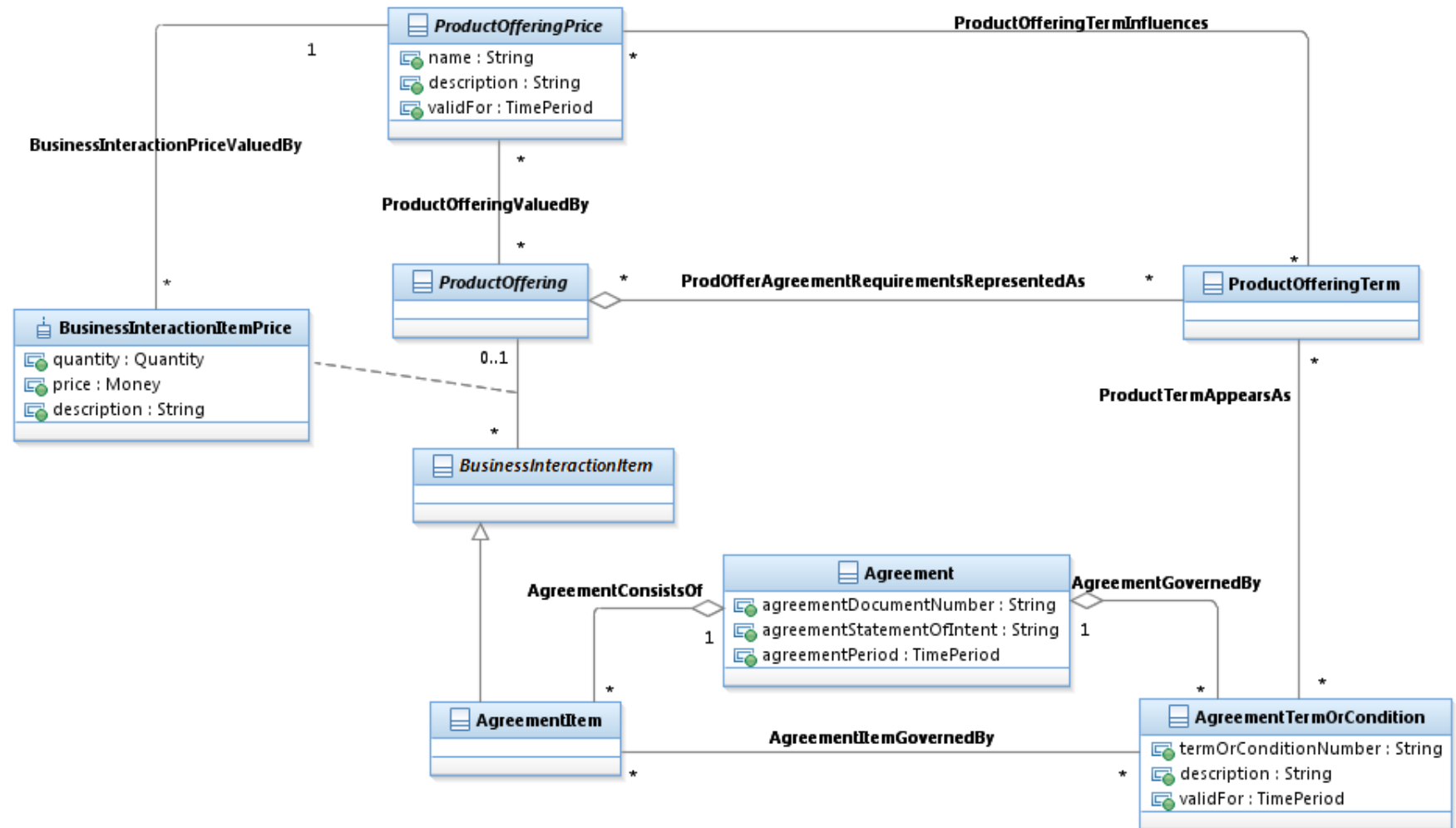


Figure Pr.6 – Product Offering Terms

1.4. Product/Product

When A Customer makes a selection, they may also have a number of options to select from. These are associated with the ProductOffering via its association to a ProductSpecification and modeled in ProductCharacteristicValue (see Figure Pr.7 below), which defines the Product delivered to the Customer. The ProductCharacteristicValue comprises the detail of the ProductSpecCharacteristics, such as color, selected by the Customer.

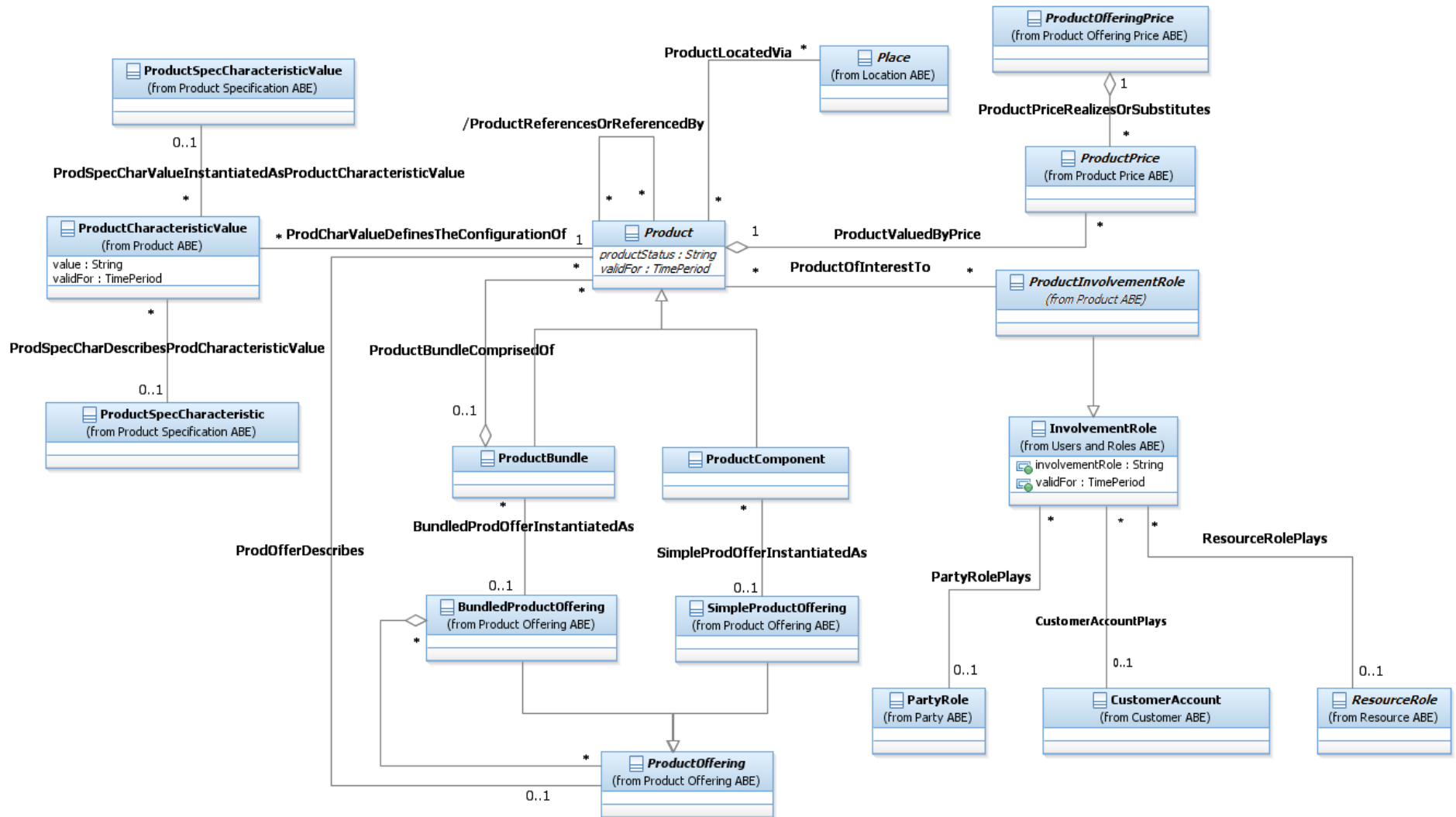
When an enterprise agrees to deliver a Product to a Customer, there is a significant amount of information that needs to be recorded to implement that Product on the provider's infrastructure or Resources and to activate the Service via with the Product is realized. This is modeled in **Product** and associated entities below.

A Product bundle in an instantiation of the BundledProductOffering that was used when the customer purchased a product or products from the provider. The **ProductBundle** holds the information (usually business information) that relates to the specific product offering (such as special prices or commitment terms) in case the customer changes its products and is no more eligible to the special business terms that are attached to the ProductOffering.

If the customer, purchased, as example a triple play bundle (VoIP, High speed internet and Cable TV) and as a result is eligible for a 10\$ monthly discount on the Cable TV charges, then this information should be kept so when the customer cancels the VoIP product the TV discount is removed. In the same way the bundle can also have changes on commitment terms and other business terms. The information which BundledProductOffering was used for the purchase is kept in the ProductBundle entity and when a product is cancelled, possibly the ProductBundle may change (to reflect the fact that now the product business terms are derived from a different ProductOffering

A Product component in an instantiation of the SimpleProductOffering that was used when the customer purchased a product or products from the provider. The **ProductComponent** holds the information (usually business information) that relates to the specific product offering (such as special prices or commitment terms) in case the customer changes its products and is no more eligible to the special business terms that are attached to the ProductOffering.

If the customer, purchased, as example a VoIP line and as a result is eligible for a low recurring charge because of a special promotion in order to penetrate a new city, then this information should be kept so when the customer changes her address the special rate is replaced with a normal monthly rate. In the same way the product component can also have changes on commitment terms and other business terms. The information which SimpleProductOffering was used for the purchase is kept in the ProductComponent entity and when a product is cancelled, possibly the ProductComponent may change (to reflect the fact that now the product business terms are derived from a different ProductOffering



ProductOfferings may be obtained by the Customer in a number of ways. Two different examples are Products are equipment obtained directly by the Customer, compared to Products that are realized by Services provided across the provider's Resources and which the Customer accesses and uses, rather than owns. In some cases, a provider may give the Customer access to a Product resident on a third party's resources. Where a third-party Product is involved, it may be branded/labeled as coming from another provider or it may be re-branded/labeled as the first provider's Product. Whichever of these is applicable, once the customer is using the Product, we are talking about a realization of the Product Offering that is based on a standard Product Specification.

The price someone pays, the rate charged for usage of a Product, discounts and allowances related to a Product are represented by the ProductPrice entity. ProductPrice is a separate SID ABE that is further detailed in a later section of this addendum.

We are also talking about the user/recipient that is entitled to use the Product. This is represented by a PartyRole, which is in turn related to the Product using the Product Involvement Role business entity and its involvementRole derived attribute.

The Product that the Customer has obtained is not truly realized until it is in use. The user may be the signatory to an agreement, particularly if the Customer is an Individual, but they may also be an agent or Employee of the Customer where that Customer is a large Organization.

Finally, a CustomerAccount may also be involved with a Product. For example, the role could be the CustomerAccount that pays for a Product; the CustomerAccount to which a Product was charged; the CustomerAccount associated with ordering a Product. This is represented by the CustomerAccountProductInvolvement business entity.

ProductCharacteristicValue Example – Email Product

The features of a Product to which a Customer subscribes are represented by ProductCharacteristicValues. In the email example, the features Spam protection, storage, message size, virus scanning and cleaning, mail forwarding, and archiving all have fixed values. These values are sometimes referred to as invariant characteristics. User name and password are variant characteristics in that the values are specified for each Product. The values may come from a list of permitted values specified in ProductSpecCharacteristicValues or may be simply be entered Product by Product. User name and password are examples of the latter. In the example, user name may take on a value of "GSmith" and password may take on a value of "mom001".

1.5. Product/Product Offering/Product Offering Price

The price of a ProductOffering or a bundle may be simple or quite complex. A simple price could be the charge associated with the monthly rental of a cable modem. Or, a price could encompass many different components. A typical introductory wireless offering, for example, will have an activation cost, a monthly cost, a number of free minutes, a cost for extra minutes, and some sort of promotional component. It might also include a corporate discount for business users. Each of these elements must be described in order to paint an accurate picture of the overall price charged for any ProductOffering.

This part of the SID model presents a set of components that can be combined to offer a complete and accurate description of the price charged for an offering. A secondary consideration is to define those components in such a way that they can be readily implemented in a typical billing system.

A ProductOffering may have any number of prices, each of which is made up of a single component or a composite of a number of components. The Product Offering Price model employs the composite/component modeling pattern used throughout the SID to represent simple and complex prices, as illustrated in the figure below.

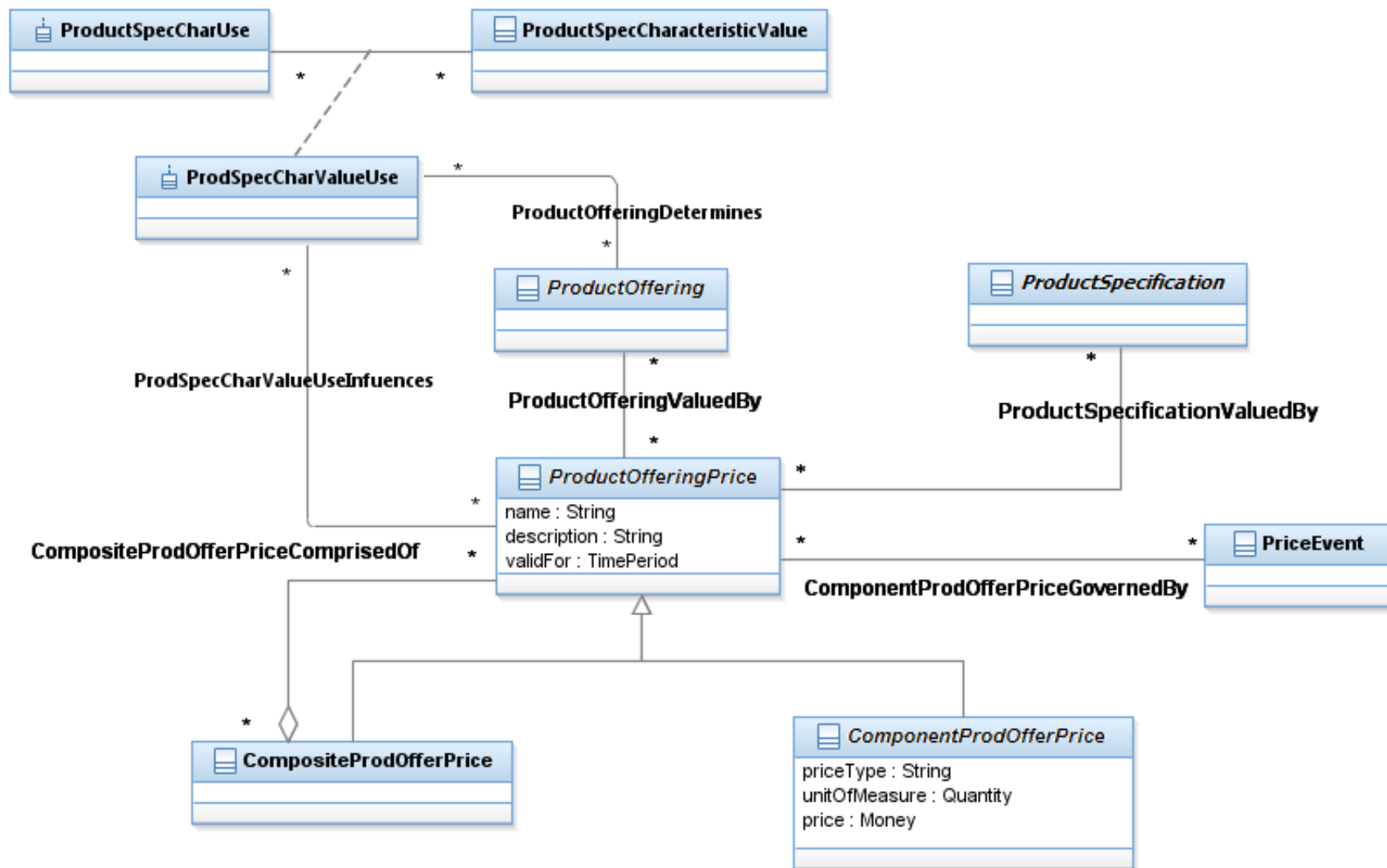


Figure Pr.8 – Product Offering Price

ProductOfferingPrices can be dependent upon a ProductSpecCharValueUse as shown in the figure above. For example, a provider may offer a photo printing service for mobile phone pictures. The price may depend on the size of the photo or the

type of finish of the photo; \$.20 for a 3 by 5, plain finish, \$.25 for a 5 by 7, plain finish, and so forth. The values associated with the size (3 by 5, 5 by 7, and so forth) and finish (plain, glossy, matte, and so forth) can be stored as ProductSpecCharacteristicValues.

ProductOfferingPrice, which may have a complex structure (composition) and involve complex billing / rating capabilities, could be defined in a "pricing catalog" as a reusable component pre-defined for given ProductSpecs and then selected for use by the ProductOffering to be published in the ProductCatalog.

The ProductSpecification to ProductOfferingPrice association allows the use of the ProductOfferingPrice entity as a "pricing specification component" defined for ProductSpecifications. For a given ProductSpecification, the "pricing catalog" may contain several pricing component candidates for use. When defining the ProductOffering, pre-defined pricing components (ProductOfferingPrices) for ProductSpecs, can be selected to value the ProductOffering. Existing pricing components may also be individually altered or new ones defined for a given ProductOffering.

Figure Pr.8 also shows an association between ProductOfferingPrice and PriceEvent. The PriceEvent business entity represents an occurrence (happening) or schedule that triggers the creation of a billing charge for a ProductOffering. For example, a PriceEvent instance may represent the end of a month when monthly recurring billing charges are created. In some cases, the creation of a charge associated with a ProductOfferingPrice may be triggered by multiple events. For example, a penalty (ProductOfferingPrice) may be triggered by over- or under-usage of a service. Under- and over-usage are represented as two separate PriceEvent entity instances.

The price paid for a ProductOffering may appear in ProductCatalogs, may be offered via certain DistributionChannels, or in certain GeographicAreas. The figure below depicts the associations a ProductOfferingPrice has with these business entities.

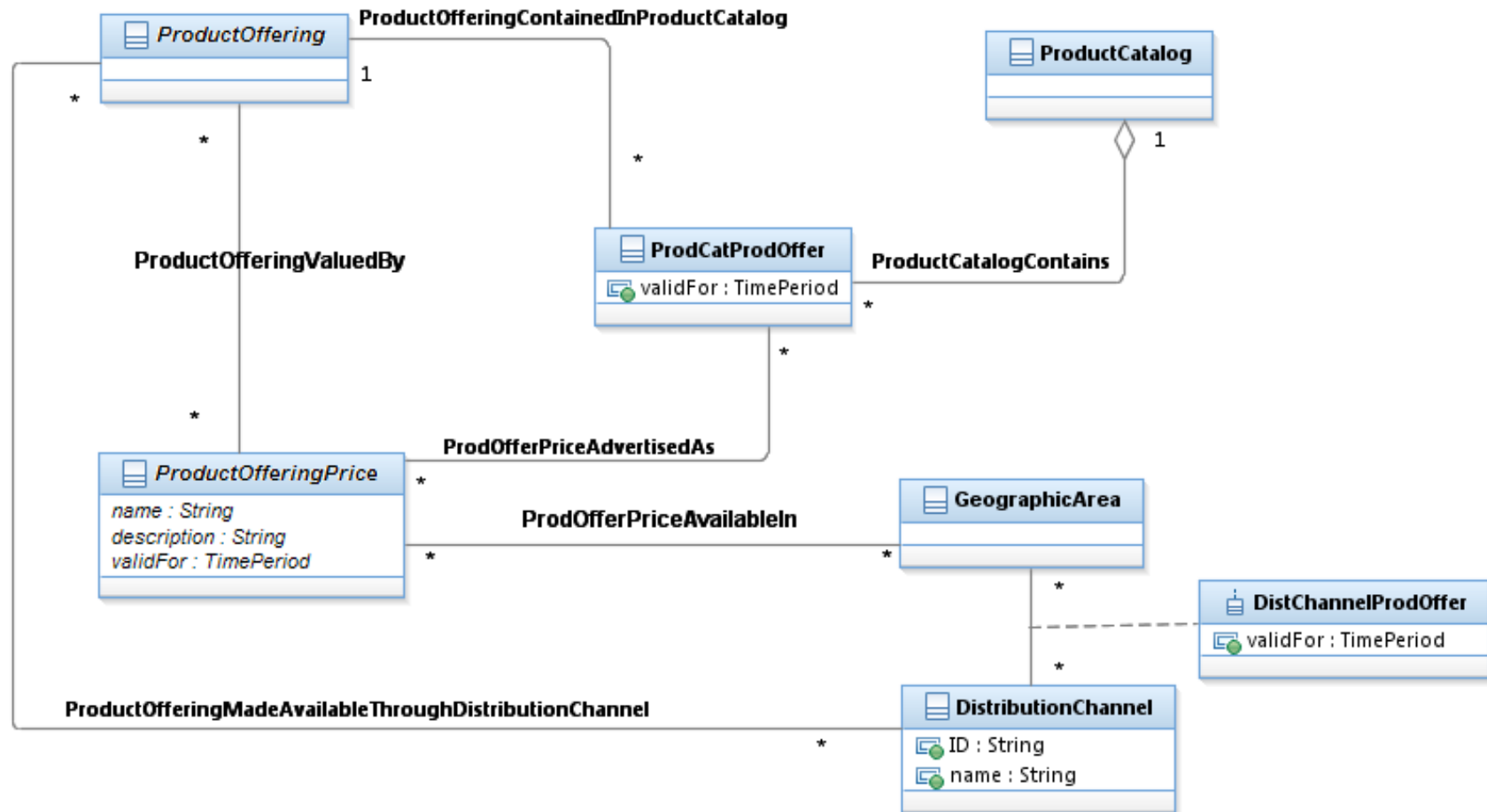


Figure Pr.9 – Product Offering Price’s Associations With Other Business Entities

Each component of a ProductOfferingPrice can take on one of two forms. A component may represent the amount charged to a party playing a role. For example, a monthly internet connection fee. Or, a component may represent an adjustment that will be made to the amount charged. For example, an installation allowance of \$49.95 to the first month's bill, which offsets an installation charge. The figure below depicts the two types of ComponentProdOfferPrices.

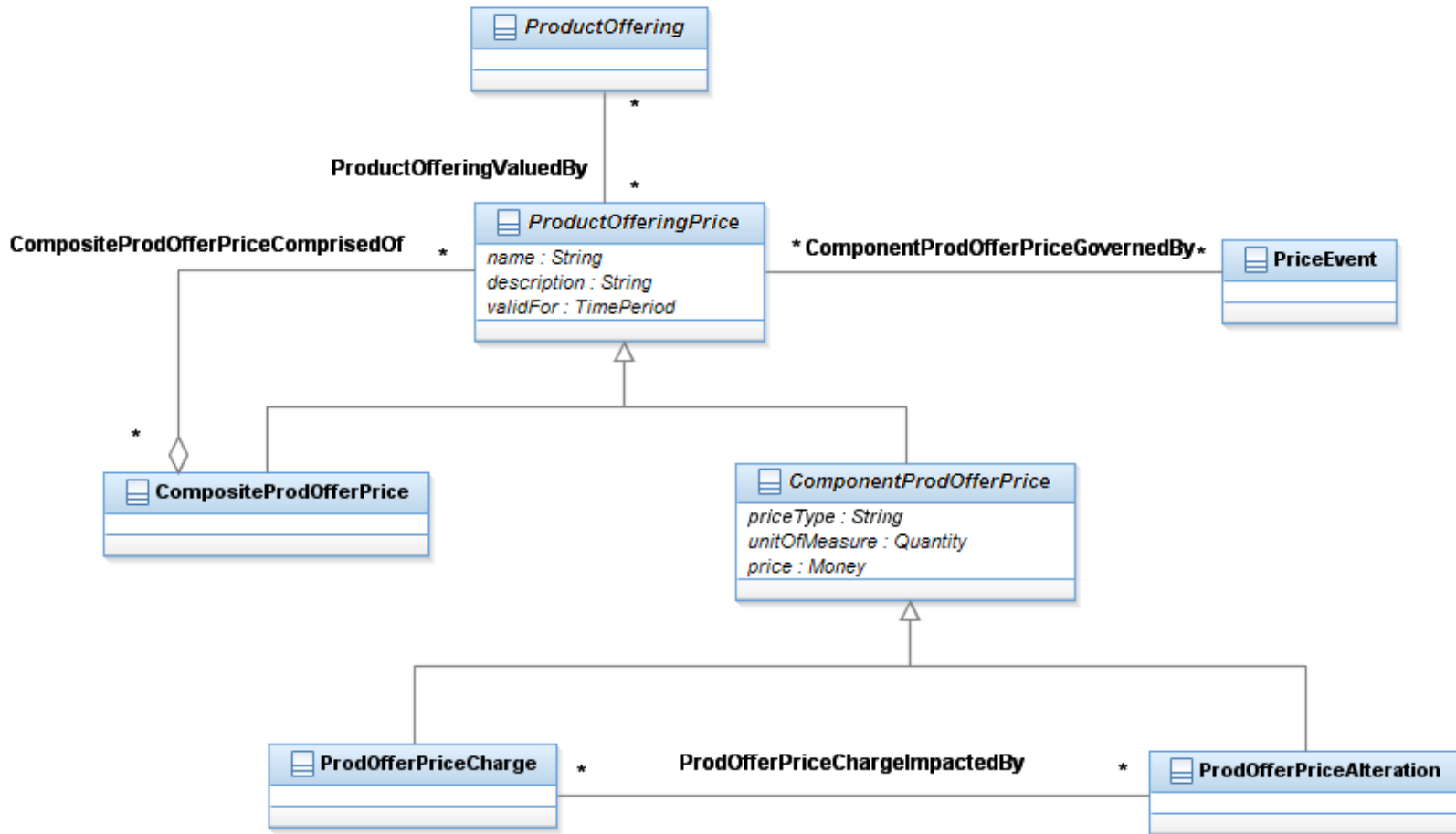


Figure Pr.10 – Product Offering Price Components

Charges incurred for a ProductOffering can take many different forms, such as

- **Recurring Charge** – A recurring charge is levied repeatedly. It has a period, such as monthly, quarterly, or yearly. An example would be a monthly charge for long distance.
- **One Time Charge** – A one-time charge is levied once. An example is the cost of a cell phone handset.
- **Fee** – A particular type of one time charge, such as an activation fee or installation fee.
- **Usage** – A charge levied for usage, based on measuring events. This model distinguishes between simple usage and usage where the charge is described by a tariff.
- **Simple Usage** – Simple usage describes a rate where the charge is the same for all events. A 10 cents a minute for long distance plan is one example.
- **Tariff Usage** – Tariff usage describes a rate where there are different charges for different types of events. This could include plans, which have different prices for evening minutes and anytime minutes. This usage aggregates a number of simple usages.
- **Alternate Price Charge** – A price charged in lieu of another price. For example, purchasing two of a single product offering may result in a discounted price for both.

A representative set of these types of charges is shown in the figure below. The set by no means represents the complete set of charges that can be incurred for a product offering. The Product Price Model provides a framework into which other types of charges can easily fit.

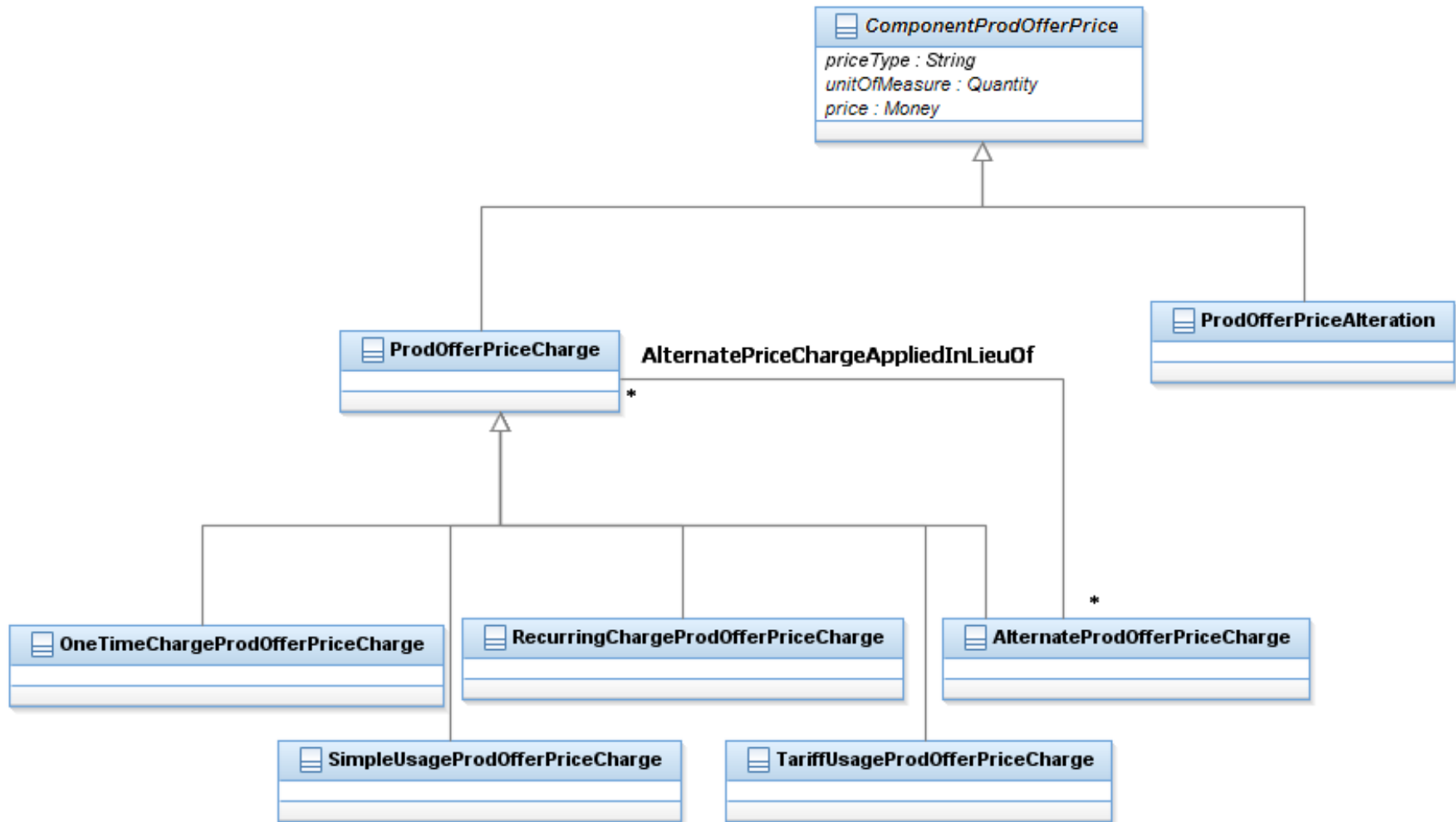


Figure Pr.11 – Types of Product Offering Price Charges

Similarly, alterations made to charge(s) incurred for a ProductOffering can take many different forms, such as

- Discount – A discount is a reduction in price. It could be a flat fee such as 10 dollars or a percentage, such as 50% off. A discount always applies to, and thus is associated with, one or more other ProdOfferPriceComponents. So a discount would be 10 dollars off the installation fee or 50% off the per-minute rate.
- Allowance – An allowance is a number of something allowed before charging begins. 300 free minutes per month is an example of an allowance. Another example would be a minimum monthly charge, where a ten dollar per month recurring charge would be combined with a ten dollar allowance. The ProductOfferingPrice contains attributes that describe what the allowed unit is (i.e. minutes, dollars), the number of units as well as the period (per month, per call). Note: the unit and number are represented by the composite attribute quantity.

A representative set of these types of allowances is shown in the figure below. The set by no means represents the complete set of allowances for a product offering. The Product Price Model provides a framework into which other types of allowances can, like charges, easily fit.

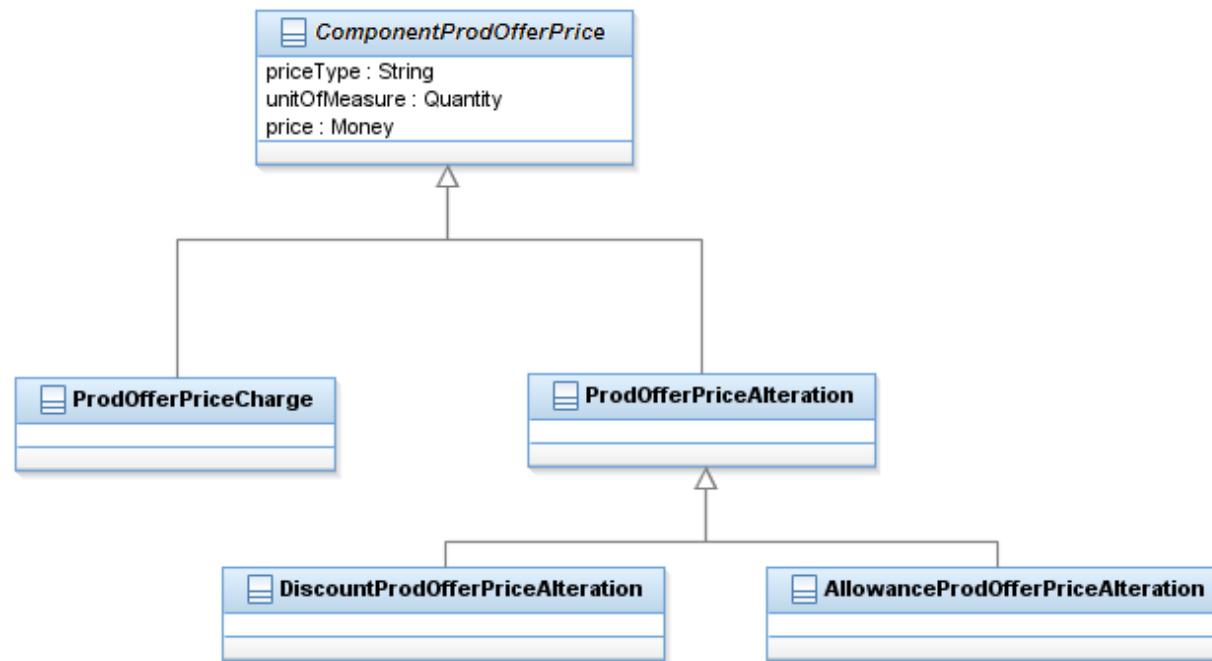


Figure Pr.12 – Types of Product Offering Price Adjustments

Putting charges and allowances in a single example provides a complete view of a ProductOfferingPrice. A simple wireless plan includes a 75 dollar phone, a 50 dollar activation fee, a 20 dollar monthly charge, 250 free local minutes per month, 10 cents a minute for additional minutes, and 25 cents a minute long distance. The components are

- A 75 dollar One Time Charge (charge)
- A 50 dollar Fee (charge)
- A 20 dollar Recurring Charge per month (charge)
- A Tariff Usage comprising a 10 cent per minute Simple Usage and a 25 cent per minute Simple Usage (charge)
- An Allowance of 250 minutes (adjustment)

Adding on to the example, suppose Sue is an employee of a company, that has a volume discount agreement with the service provider. The agreement is to waive the installation fee and provide a 20 percent discount on local minutes. This would manifest itself as:

- A composite price that associated the 50 dollar Discount with the 50 dollar Fee
- A composite price that associated the 20 percent Discount with the 10 cent per minute Simple Usage

The complete ProductOfferingPrice model is shown in the figure below.

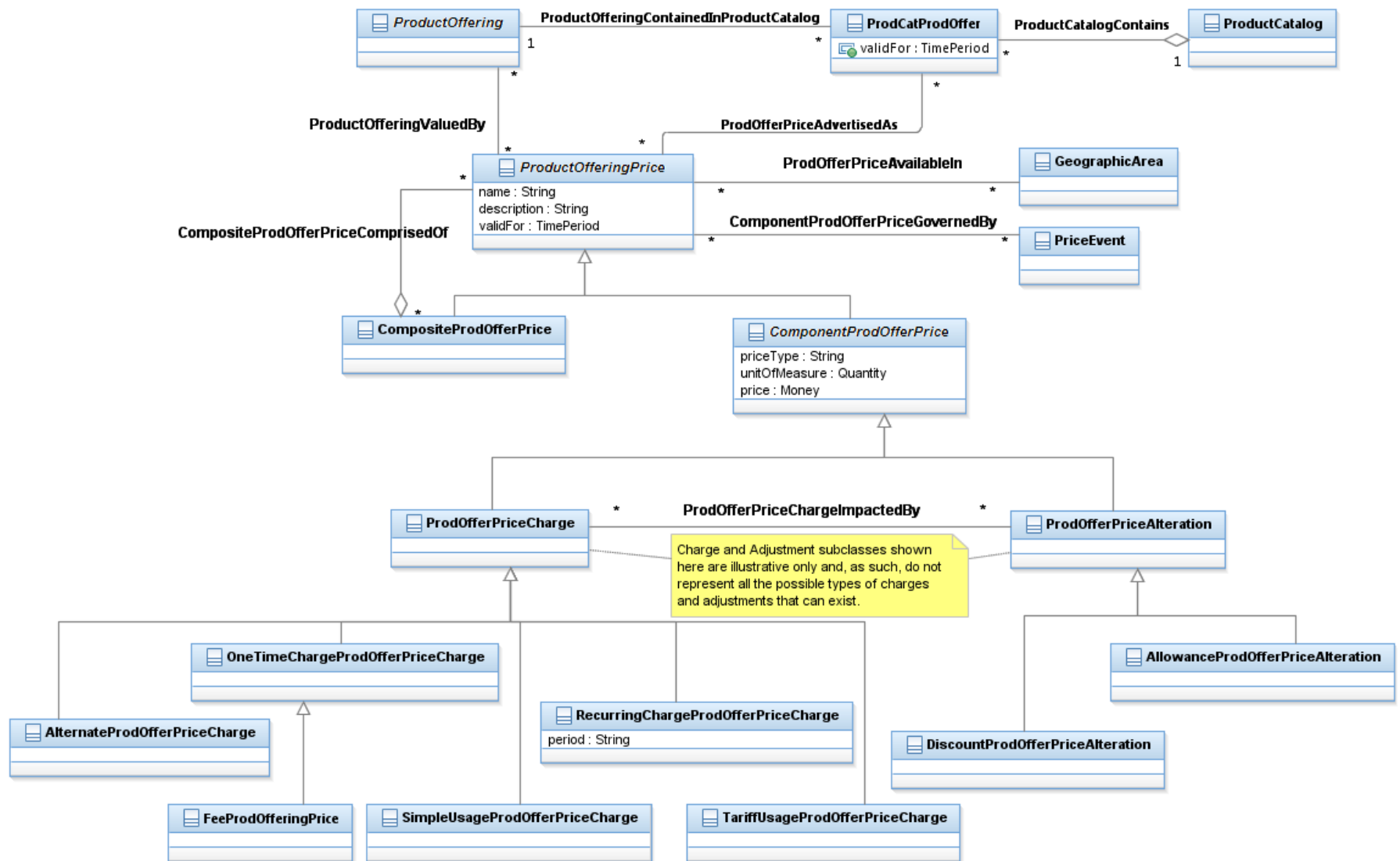


Figure Pr.13 – Complete Product Offer Price Model

1.6. Product/Product Offering/Product Offering Price Rule

The price of a Product is often dependent upon the procurement of another Product for a particular price, another ProductOffering, or another ProductOffering of the same type. For example, a Family Plan mobile phone service includes two free phones. Within this offer, the cell phones are associated with a zero value price. In another offering, there may be a choice of some number of phone accessories that are offered at no or greatly reduced cost.

Both of these offers require the employment of a policy rule that governs the price of the accessories. In the first case, the PolicyRule is used to ensure that two of the phones are provided at zero cost, as long as the terms of a particular ProductOffering are met. In the second case, the PolicyRule defines which accessories can be purchased at what price.

Price rules such as these require another set of business entities in addition to the more basic rules set forth in Product Offering Price and its components.

The intent of this part of the SID model is to present a model that can be used to define simple cross-product price policy rules. This will be done by extending the SID Policy model, as documented in [1-POL]. As such, the model presented here is not intended to provide for elaborate pricing schemes. In these cases, it may be necessary to employ business entities as yet to be defined within the policy model or that may require the development of code specific to the rule. Still, it is significant that the existing SID Policy model, whose primary use case was to support the definition of networking policies, can be easily extended to meet the needs of a very different domain.

The figure below shows the relationship that a price rule has with ProductOfferings and with Policy.

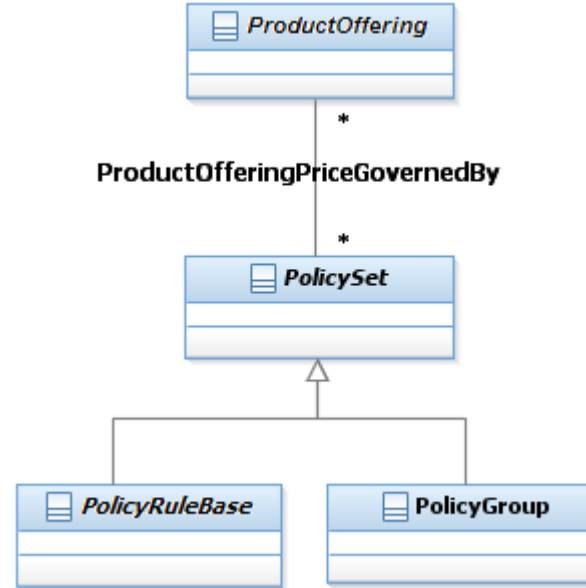


Figure Pr.14 – Product Offering Price Governed By Policy

The price of a ProductOffering is governed by a PolicySet. The PolicySet may be a single PolicyRule or a PolicyGroup which ties together a number of PolicyRules. The use of both single rules and grouped rules will be explained via examples.

For example, suppose there is a ProductOffering called “Subscribe for a Year, Get Two Accessories Free”. The free accessories can be selected from list of available accessories. In this example, the accessories are a cell phone re-charger, hands-free holder, cell phone wallet, or earphone and microphone. Included in the offering are the cell phone service and the accessories. There is an instance of PolicySet rule related to this offering.

All PolicyRules consist of three types of components, which define the semantics of the rule. (Please see [1-POL] for more information.) This is shown in the figure below.

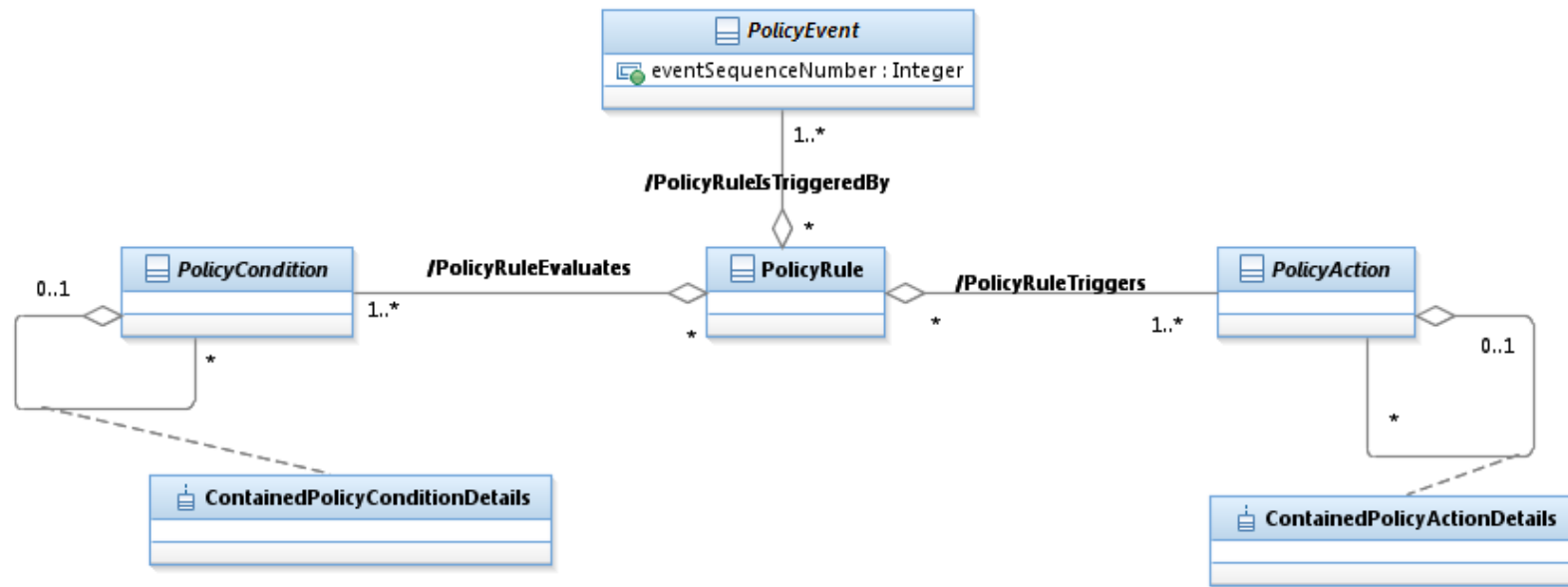


Figure Pr.15 - Policy Rule Components

The semantics of a rule that governs the price of a ProductOffering builds upon this generic structure. The first component is a PolicyEvent, which triggers the evaluation of the PolicyRule. The second defines the PolicyCondition(s) of the PolicyRule being evaluated. The third defines the PolicyAction(s) that are taken if the PolicyCondition(s) of the rule are met. In the “Subscribe for a Year” offering:

The PolicyEvent is the time of purchase

The PolicyCondition is the particular type of cell phone service offering being purchased (since different pricing rules could be attached to different types of cell phone offerings)

The PolicyAction taken as a result of this particular condition being true is that two of the accessories are free.

Note that a PolicyRule aggregates at least one or more PolicyConditions and at least one or more PolicyActions. (Addendum 1-POL defines a general set of semantics that specify how a PolicyRule is evaluated, including the concepts of rule priority and a decision strategy for combining different PolicyConditions.) In addition, there are other constructs, such as a Policy Group, that can aggregate multiple PolicyRules. This enables more complex pricing rules to be developed by defining appropriate subclasses of PolicyRule, PolicyCondition and PolicyAction. Note that PolicyCondition and PolicyAction both use the composite pattern; please see Addendum 1-POL for more information.

Next, a subclass of PolicyConditionAtomic, called ProdOfferPriceCondition will be defined. Its purpose is to add semantics that determine if one or more PolicyAction(s) should be taken.

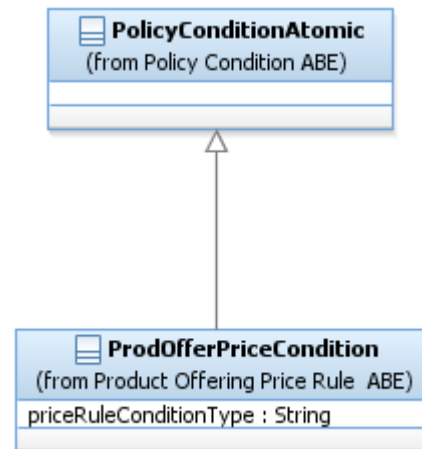


Figure Pr.16 – Product Offering Price Condition

The attributes of the above policy subclasses do not completely define the semantics of the condition governing the price of a ProductOffering. To complete the definition of the condition, associations to other business entities specify what must be procured or what must have been procured. The related business entity may be a ProductOffering, a similar type of product, or a product at a particular price. In the “Subscriber for a Year” offering example, the ProdOfferPriceCondition instance is associated to the cell phone service instance of ProductOffering.

In the example, the condition type (which is represented by the priceRuleConditionType attribute) is “buy”. Other values for the type will be described in later examples.

The condition is fully defined using the PolicyStatement entities in the Policy model, as shown in the figure below.

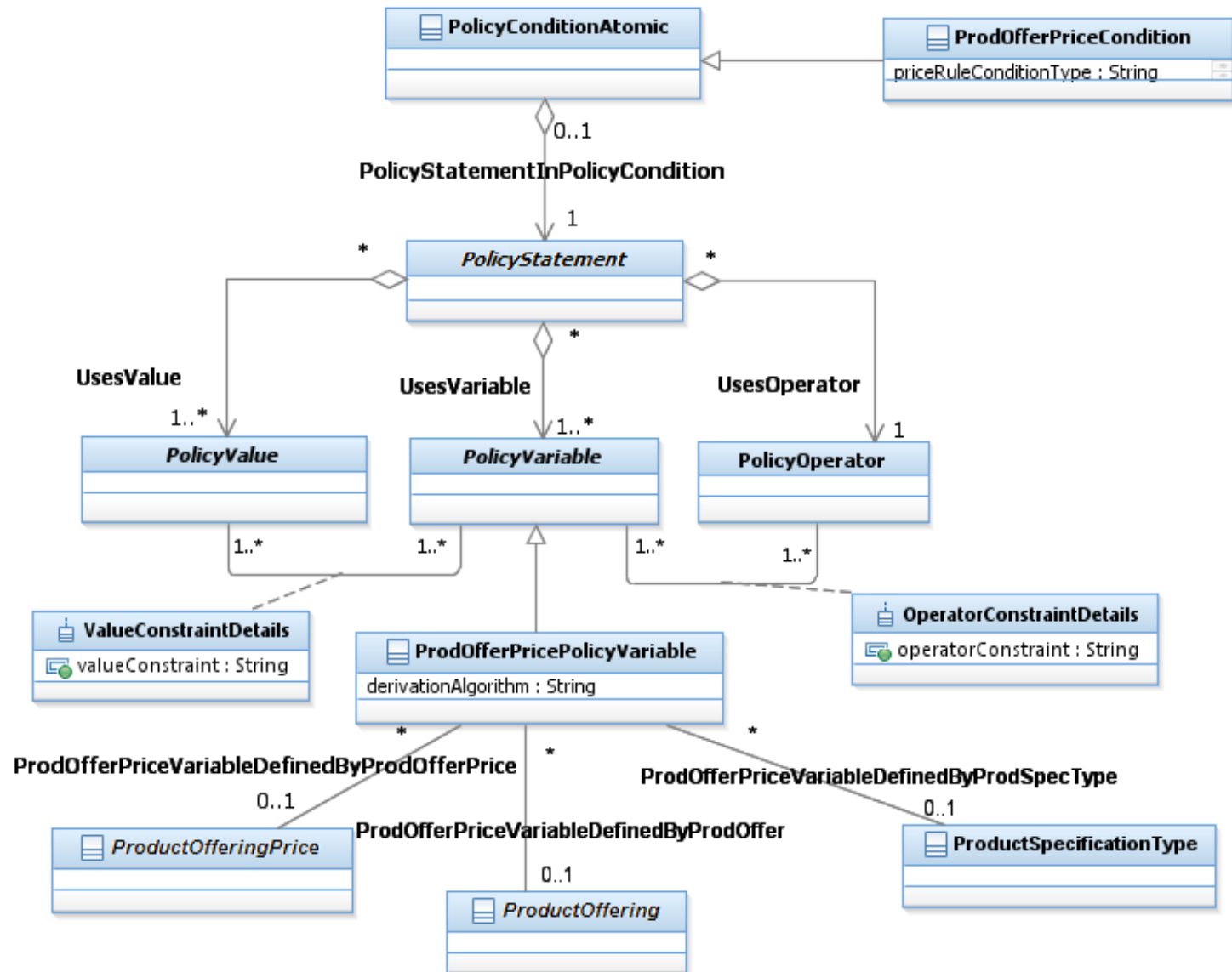


Figure Pr.17 –Product Price Condition and Policy Statement

The complete PolicyStatement for this condition is: The ProdOfferPolicyVariable is the quantity ordered for the cell phone instance of ProductOffering. The PolicyOperator is “equals”. The PolicyValue is 1. The condition is interpreted as “if one instance of the cell phone is ordered” then some action will be taken. The power of representing a PolicyStatement in this way is that each part of the PolicyCondition is object-oriented and reusable. This sets the stage for representing more complex pricing rule conditions.

Satisfying a condition may govern the entire price of an offering or one or more of the components of the offering. In this example, it governs the entire price of the offering for the accessories. That is, two can be chosen for free.

The action to take is defined by ProdOfferPriceAction, which is a subclass of PolicyAction. These entities are shown in the figure below, along with the association between PolicyAction and PolicyStatement.

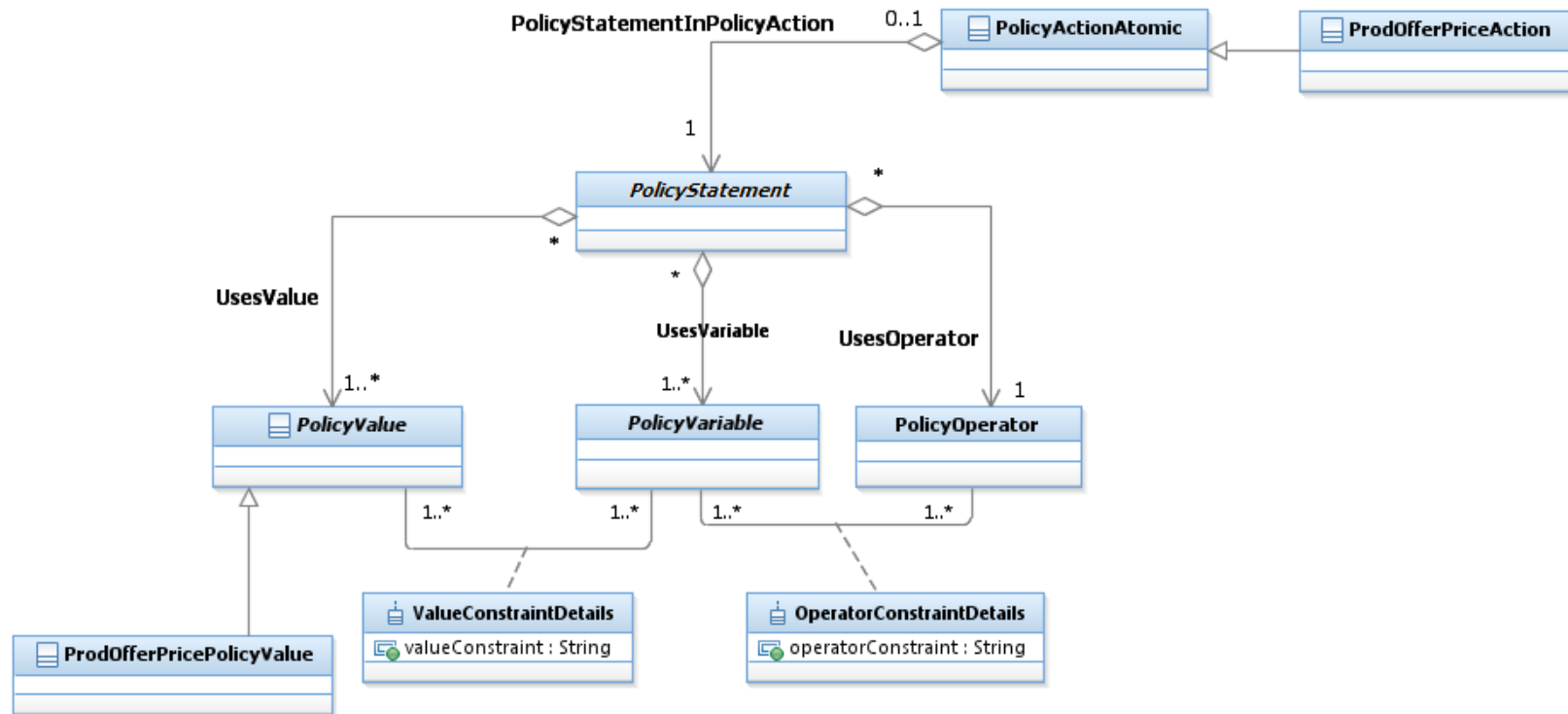


Figure Pr.18 – Product Offer Price Action and Policy Statement

The complete PolicyStatement for this action is: The Prod Offer Policy Variable is the quantity for the accessories. The Policy Operator is “equal to or less than”. The PolicyValue is “2”. The action is interpreted as “set the quantity allowed for free accessories to no more than 2”.

There is one missing variable in this action. The missing variable is the price (0) to be paid for the accessories and what accessories are free. This variable is expressed using an association between ProdOfferPriceAction and ProductOfferingPrice. Instances of this association specify the ProductOffering and the Price associated with the action. In this example, the instance of the action would be associated with instances of zero value prices for cell phone re-charger, hands-free holder, cell phone wallet, and earphone/microphone. The figure below shows this association.

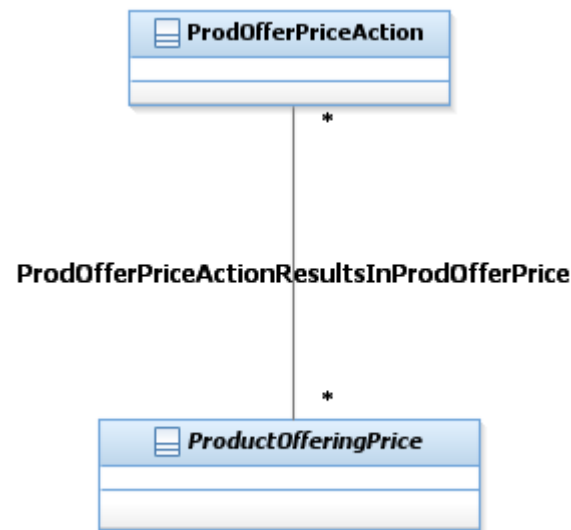


Figure Pr.19 – Product Offer Price Action and Product Offering Price

The figure below shows the instances of all business entities that define product price rules within the context of the model for the Subscribe for a year with Free Accessories example.

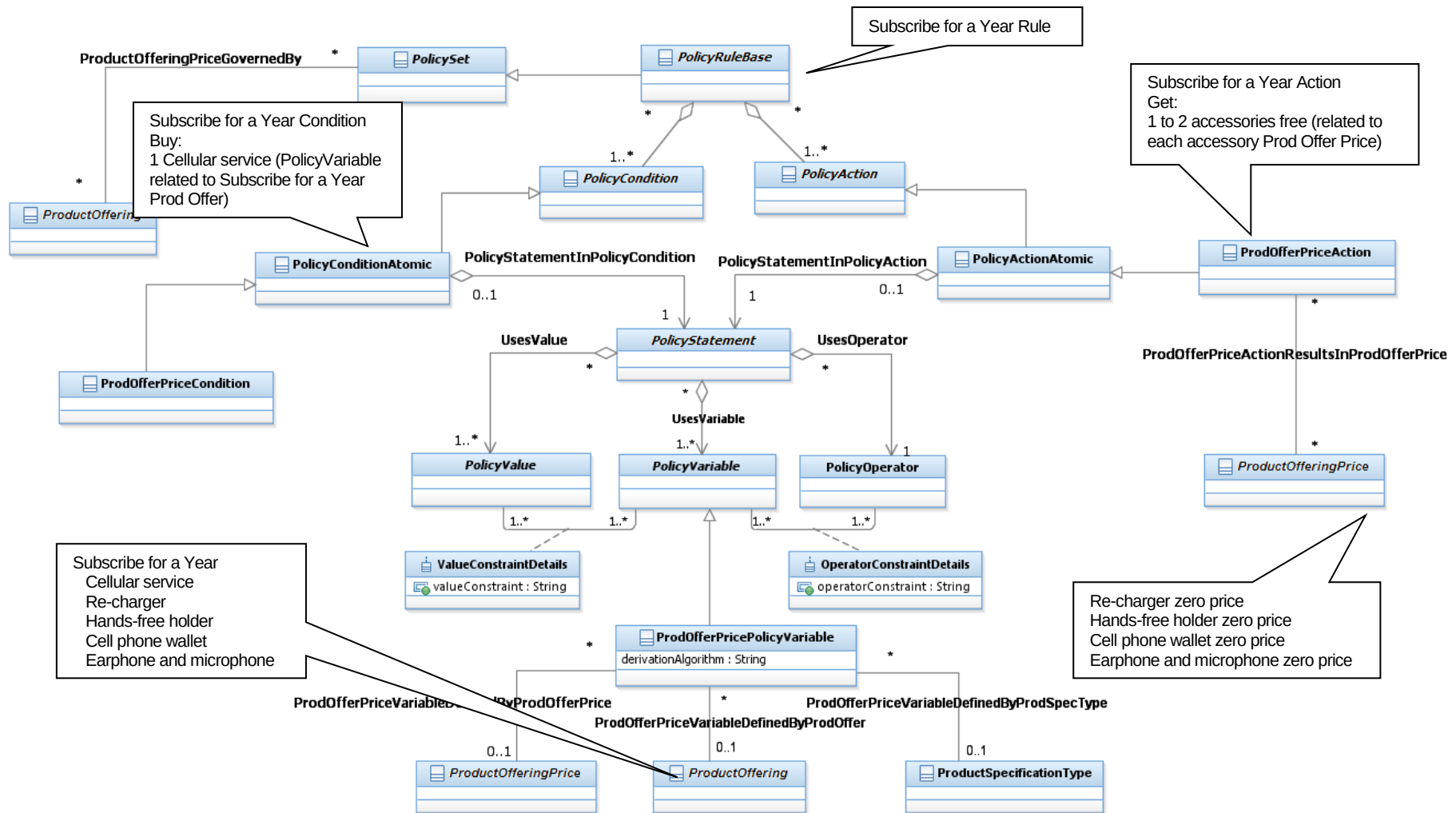


Figure Pr.20 – Subscribe for a Year with Free Accessories

It is not always necessary to relate an instance of ProductOffering to a price rule. In some cases, obtaining a product offering at a specific price may enable the procurement of other product(s) for a particular price. This price may be represented by an instance of an entire ProductOfferingPrice or may be represented by an instance of a ProductOfferingPriceComponent.

For example, suppose that if a customer subscribes to wireless Local Area Network (WLAN) access at the regular price, the customer is entitled to a discount on their DSL internet access or a discount on their cellular phone service.

Handling this rule requires the use of a PolicyGroup. The PolicyGroup subclass of PolicySet brings together multiple PolicyRules and applies them as an atomic set of rules. This enables different PolicyRules to be defined to serve stand-alone needs, yet be reused in other applications. The first rule checks to see if the price of the ordered WLAN access is equal to or greater than the regular price. If this condition is true, then an event is generated that triggers an option that allows the user to select one of the two items offered at a discount. The second rule, also triggered by this event, determines which choice of options has been made and sets the price to the discounted price.

Expressed as a PolicySet, the rules' conditions and actions are:

User chooses WLAN access in GUI screen #1. This selection triggers an event, which fires the following rule:

PolicyRule 1: WLAN_Access_Sale

```
ON EVENT customer_choice DO
  IF (order >= min_price) THEN
    SET Event_WLAN_Access_Options to TRUE
  ENDIF
ENDEVENT
```

Setting the Event to TRUE initiates the display of GUI screen #2. The user chooses an option, which is recorded in a choice variable, which initiates the following nested PolicyRule (remember, a PolicyRule can contain a PolicyRule):

PolicyRule 2: Determine_Option_Choice

```
ON EVENT WLAN_Access_Options DO
  IF (choice == DSL) THEN
    SET Business_Interaction_Item_Price = Discounted_DSL_Price
  ENDIF
  IF (choice == Cell) THEN
    SET Business_Interaction_Item_Price = Discounted_Cell_Price
  ENDIF
ENDEVENT
```

To support the action associated to this complex rule, another extension is added to the Policy model. The value of the discounted price is not a fixed value, but one associated with a specific Product Offering Price. This can be modeled by first creating a subclass of PolicyValues, called ProdOfferPricePolicyValue, and then creating an association between ProdOfferPricePolicyValue and ProductOfferingPrice, as shown in the figure below.

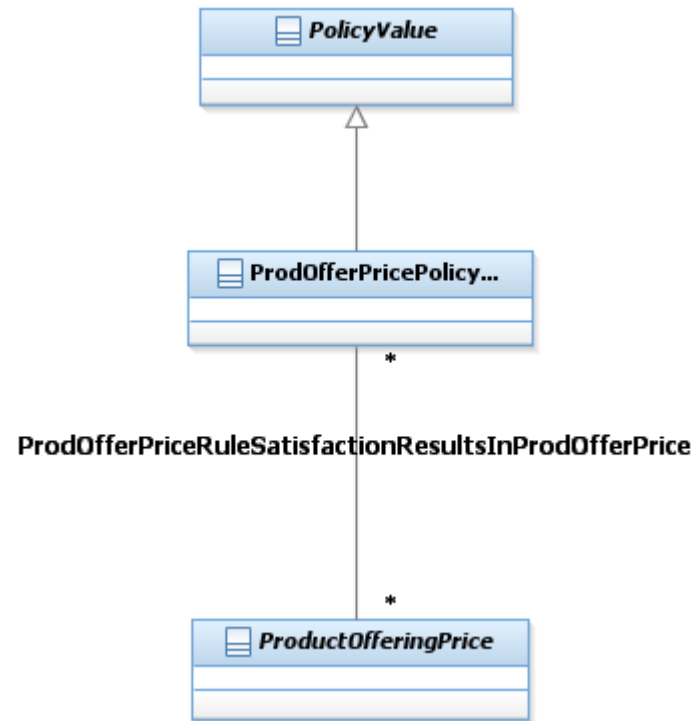


Figure Pr.21 – Product Offer Price Policy Value

Note: In all the pricing rule examples, if a condition is not TRUE, then nothing happens.

The figure below shows the instances of these business entities for the WLAN example within the context of the model.

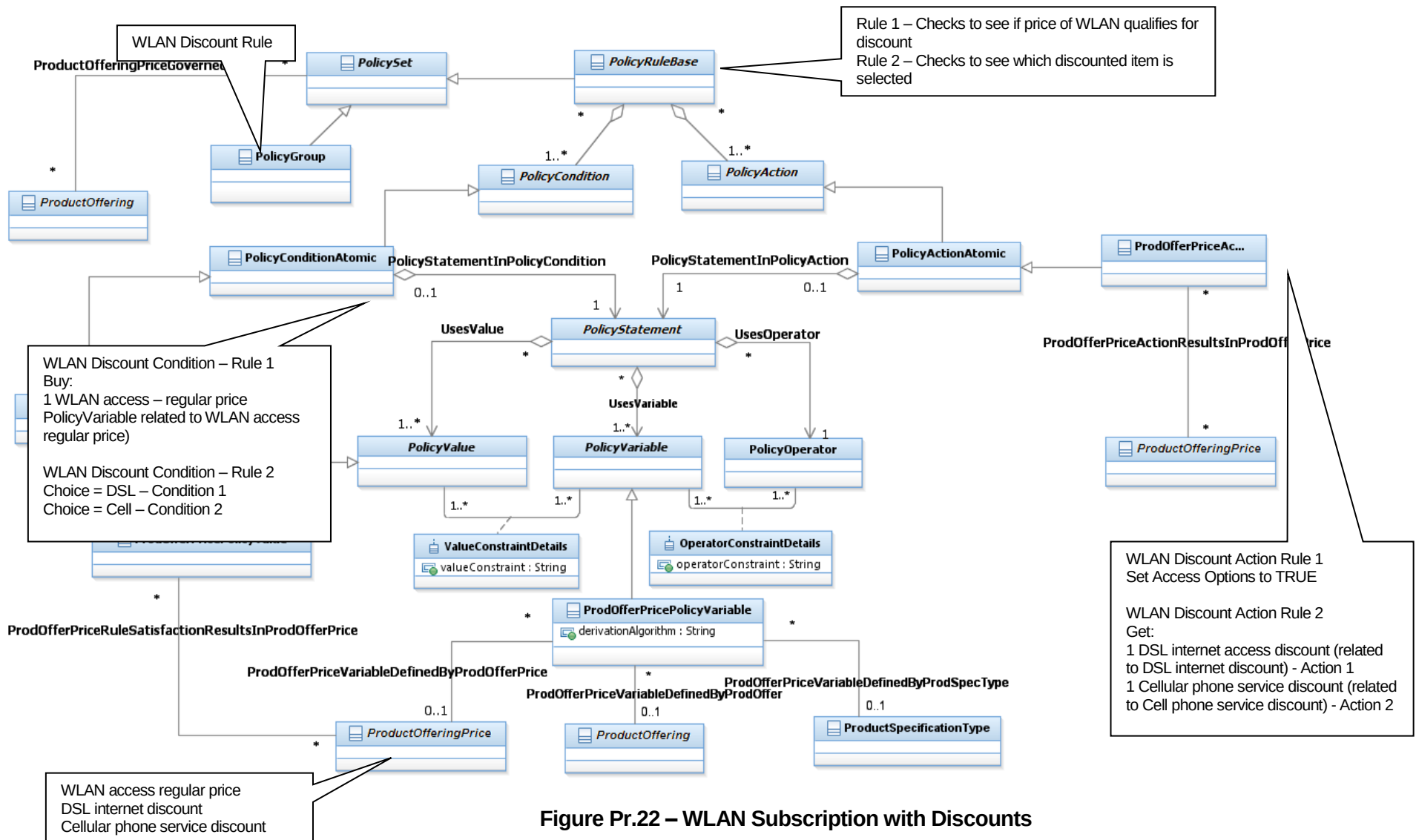


Figure Pr.22 – WLAN Subscription with Discounts

The next example is a ProductOffering that combines ordering DSL service with one of several types of routers that offer varying functionality. If both the DSL service and either of the routers are purchased, a subscriber (customer) gets a fifty-dollar rebate on the router of choice (we'll keep it simple and assume that there are only two types of routers to choose from in this example). In this case the ProductOffering instance, "Buy DSL Service and Router, Get \$50 Back!", has as its product offering components, the DSL service and one of two different routers.

Since the price rule involves purchasing two products, a composite PolicyCondition is employed. The model for representing a composite PolicyCondition follows the Composite/Component pattern used throughout the SID model and in the Product domain, so the PolicyConditionComposite class is not shown here.

The price rule has six condition instances, three atomic condition instances and three composite condition instances (the composite instances aggregate two or more atomic instances). The first atomic condition's ProdOfferPolicyVariable is related to the quantity for the DSL service. The PolicyOperator is "equals". The PolicyValue is "1". The second atomic condition's ProdOfferPolicyVariable is related to the quantity for one type of router. The PolicyOperator is "equals". The PolicyValue is "1". The third atomic condition's ProdOfferPolicyVariable is related to the quantity for the other type of router. The Policy Operator is "equals". The Policy Value is "1".

In order to represent the completed composite PolicyCondition, two of the composite conditions join the DSL service with the routers using an "and" Boolean operator. The third composite joins these two composites using an "or" Boolean operator. This entire condition is: (DSL service "and" router #1) or (DSL service "and" router #2).

The ProdOfferPriceAction is related to the \$50 rebate for the DSL service. The action is: "Set allowed quantity for the DSL service rebate equal to 1". The description of the price indicates it is a rebate on the purchased router. The figure below shows the instances of these business entities within the context of the model.

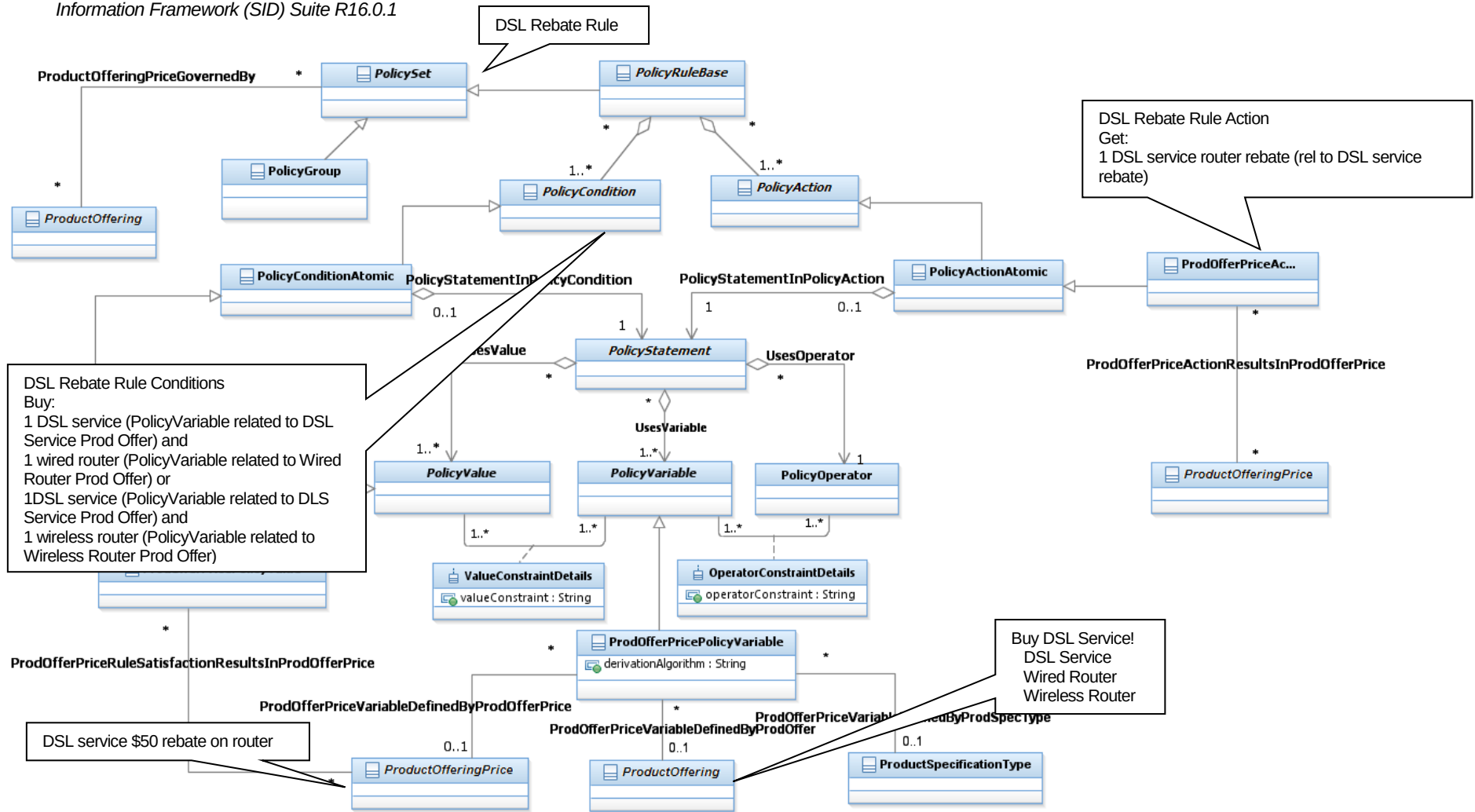


Figure Pr.22 – Product Price Rule Model - DSL Service and Router Rebate

The next example shows a price rule that allows a discount on like products. Suppose that for any mobile phone accessory purchased at the regular price, one whose price is equal to or less than the accessory purchased at the regular price may be purchased at a 50% discount. In this case the ProductOfferingPrices represent regular and discounted prices. In this example, rule condition types of “buy” and “price” are used. The “buy” condition specifies that two accessories must be obtained. The “price” condition specifies the relationship between the prices of the two accessories; the price of the discounted accessory must be less than or equal to the price of the accessory obtained at the regular price.

Of course, the policy model can’t represent all of the behavior implied in this example; application code would have to be written to interpret a rule such as this. For example, when a customer purchases a cell phone wallet (an accessory), the application would have to check to see if there are price rules associated with compatible accessories. It would find the “buy one get one at 50%” rule. The customer could be queried to see if a purchase of a second accessory is of interest. The figure below shows the instances of these business entities within the context of the model.

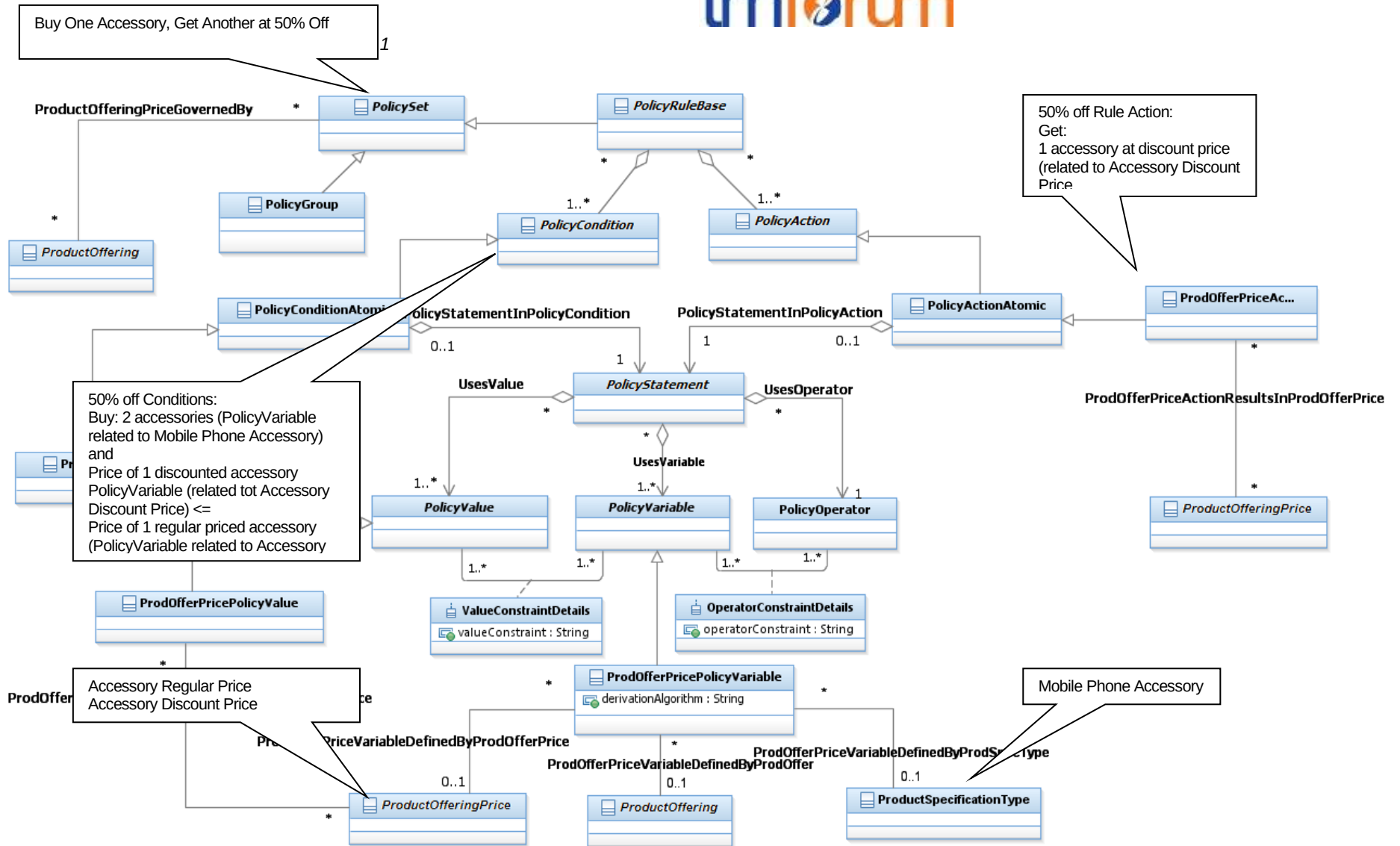


Figure Pr.22 - Product Price Rule Model — Price Rules for Like Products

The ProductOffering price rules described here can also be used to represent many other types of pricing rules. Some additional examples include:

- Defining conditions based on the prior procurement of a product offering possibly within a certain time period
- Defining quantity based discounts
- Defining quantity-tier based discounts

As can be seen by the examples provided here, this simple model provides a variety of pricing options that can be used by a provider.

Prices are often associated with ProductOfferings involved in BusinessInteractions, such as CustomerOrders, Agreements, RequestsForQuotes, and so forth. Additionally, these prices are sometimes influenced by rules that govern ProductOfferingPrices. Figure Pr.25, below, shows these associations.

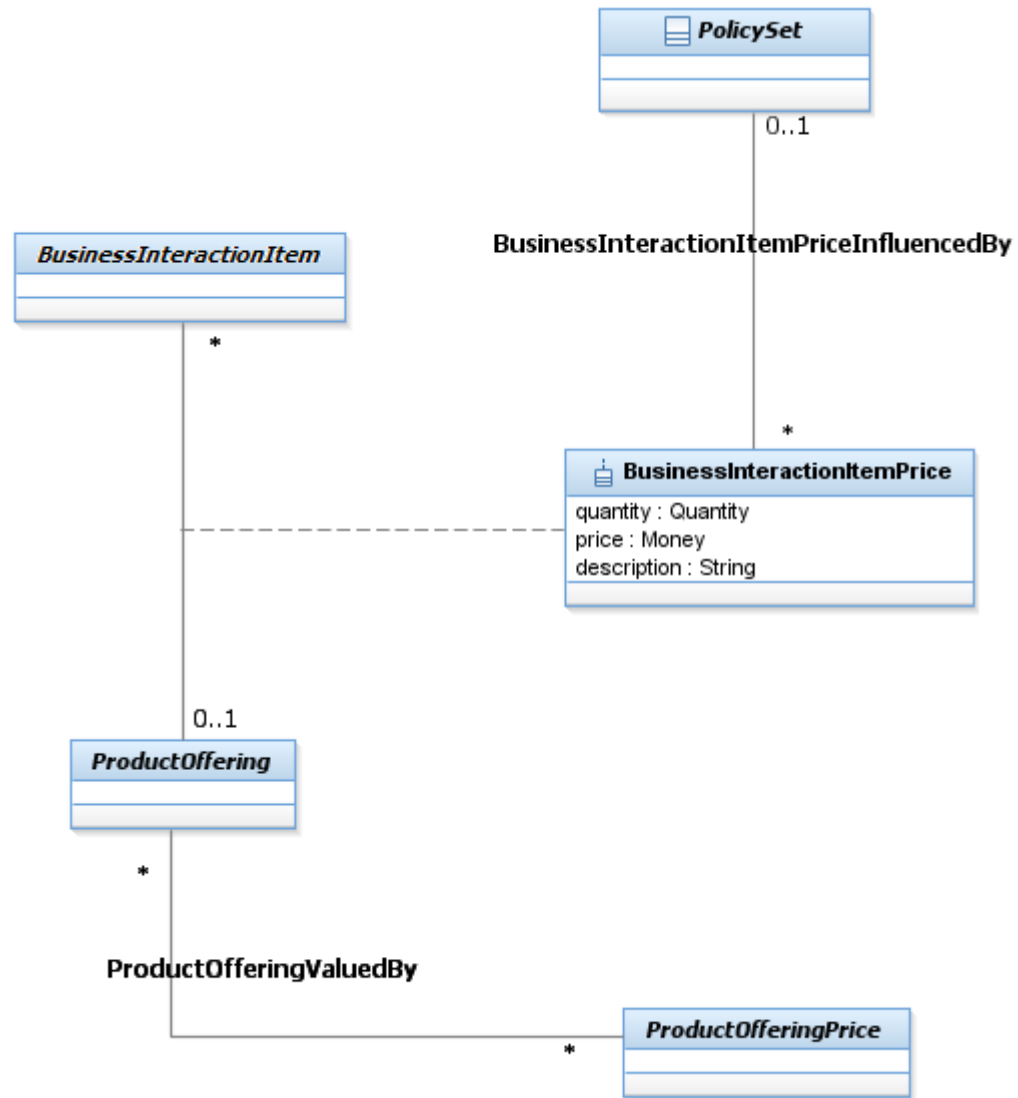


Figure Pr.25 – Product Offering Prices Associations With Business Interactions

1.7.Product/Product Offering/Pricing Logic Algorithm

The PricingLogicAlgorithm (PLA) entity is designed to support the link to a “black box” that is used for rating events. This is the common case for usage rating but this scenario exists also for rating recurring and one-time charges. In such cases the price is not kept in the catalog nor the rules used to determine the price, only the interface to the black box algorithm is modeled and prices are determined by sending the relevant information using the interface, and getting the rate for the event from this black box algorithm. The PricingLogicAlgorithm and its related entities (especially PricingLogicAlgorithmSpec) are designed to model flexible interface definition in order to provide maximal reuse for existing rating algorithms.

Use Case 1: Linear Call Rating

This use case describes one of the simplest possible rating algorithms for call usage events – a linear rating according to call duration. Such an algorithm may require two parameters in order to be instantiated (just an example):

Rate for a minute

Interval for calculating the rate

This algorithm is charging the call based on call duration in intervals according to the second parameter, so if the interval is 5 seconds, a call between 1 and 5 seconds would be charged for 5 seconds, between 6 and 10 seconds would be charged for 10 seconds and so on. In any case the charge will be for a whole number multiplication of 5 seconds.

The following table illustrates such an algorithm – in all cases the rate is \$0.30/minute

Call Duration (secs)	Interval (secs)	Call charged for (secs)	Call cost (\$)
20	5	20	0.10
20	6	24	0.12
32	6	36	0.18
32	10	40	0.20
32	16	32	0.16
32	20	40	0.20

Using such an algorithm, instantiation of \$.12/minute and interval of 5 seconds will always yield a whole number (in cents) and so \$.10/minute and 6 seconds interval. On the other hand a rate of \$.07/minute and 1 second interval will almost always yield charges with fractions of cents.

Note that in order to calculate the rate you need to set the rate and interval (which is something that is communicated to the customer and typically done during the customer order), and also pass the Usage event details (Usage class in SID) to the rating algorithm (in order to get event relates data, in our use case only the call duration is important).

Use case2: Time and Customer Information Sensitive Algorithm

This use case describes a more complex algorithm in which there are more parameters which are also more complex. Let's assume we have an algorithm which charges differently for peak and off-peak calls (and can also split the charge of a call that starts peak and ends off-peak or vice versa). Let's also assume that the rate per minute is different according to customer type (student, normal, VIP)

In order to instantiate such an algorithm we need to set the following parameters:

- Define which time is peak and which is off-peak (we can have one instantiation in which peak is 10-15 Mon-Thu and another one which is 8-18 Mon-Thu)
- Define the rate per combination of customer type and peak/off-peak. Such a matrix could look like

	Peak	Off-Peak
Student	\$.15/min	\$.10/min
Normal	\$.20/min	\$.12/min
VIP	\$.10/min	\$0/min (free)

Note – this matrix is not necessarily two dimensional in some cases it has more dimensions (the rate depends on time (peak/off-peal), customer type and local/long distance rates)

More complex usage rating algorithm may include tiers (\$.12/minute for first minute, \$.10/minute for minutes 2-5 and \$.08/minutes for more than 5 minutes) in which case the parameters become more complex.

What we try to model in the SID is the *interface* to these algorithms. The algorithms can be implemented in many different ways by different systems but we need to model the interface to them with all the parameters in order to be able to use these algorithms in our systems

PricingLogicAlgorithm is a classic case of the EntitySpec/Entity pattern as described in the above use cases. This is described in the following diagram

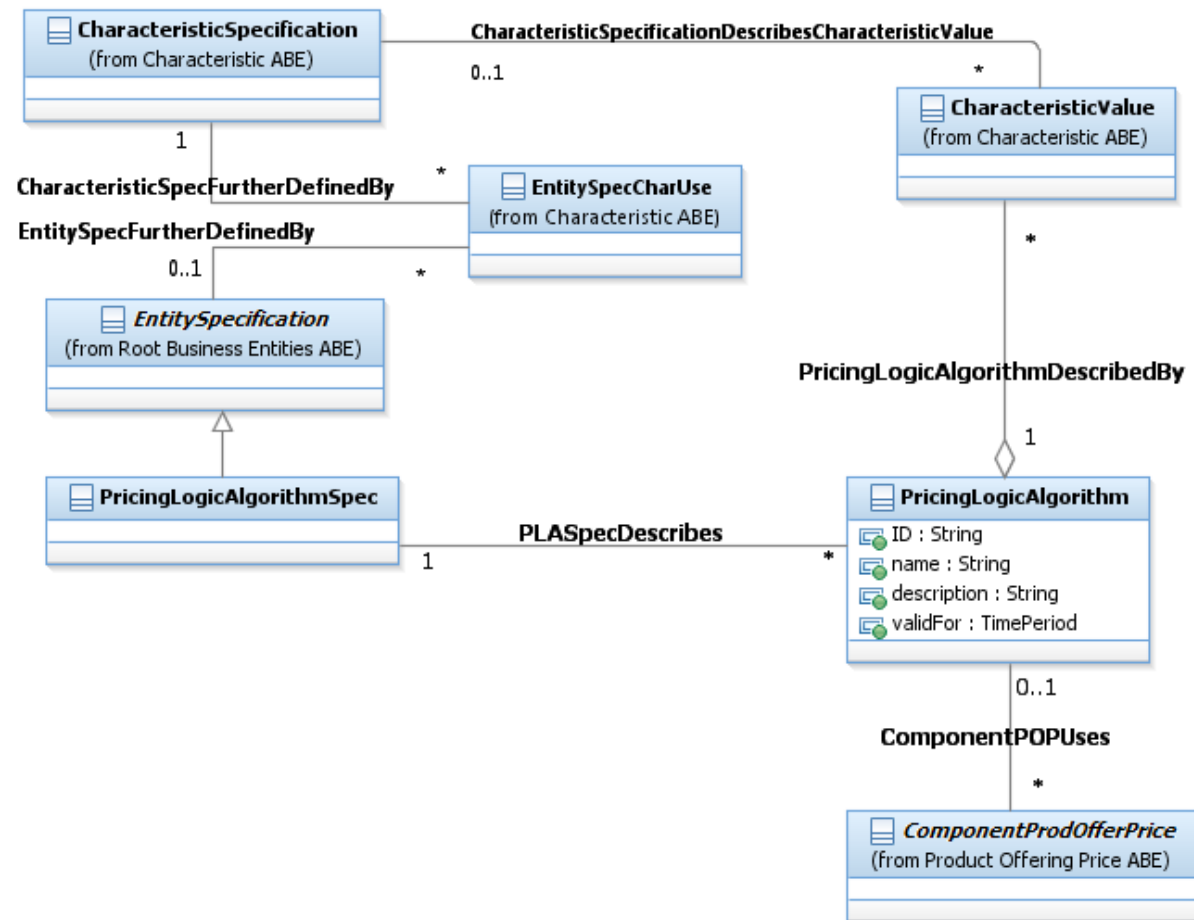


Figure Pr.26 – Pricing Logic Algorithm basic entities

This diagram uses the generic EntitySpec/Entity pattern so that PricingLogicAlgorithmSpec automatically gets the characteristics that are associated with EntitySpecification. The PricingLogicAlgorithm is an instantiation of PricingLogicAlgorithmSpec after the parameters were set, so in the case of the simple use case above it represents linear usage rating, \$0.10/minute with 6 seconds interval. The parameters (\$0.10 and 6) are characteristic values

The next diagram shows the association of the PricingLogicAlgorithmSpec to the other sources of parameters to the algorithm (not only PLA characteristics, but also ProductSpecCharacteritics and UsageSpecCharacteristics)

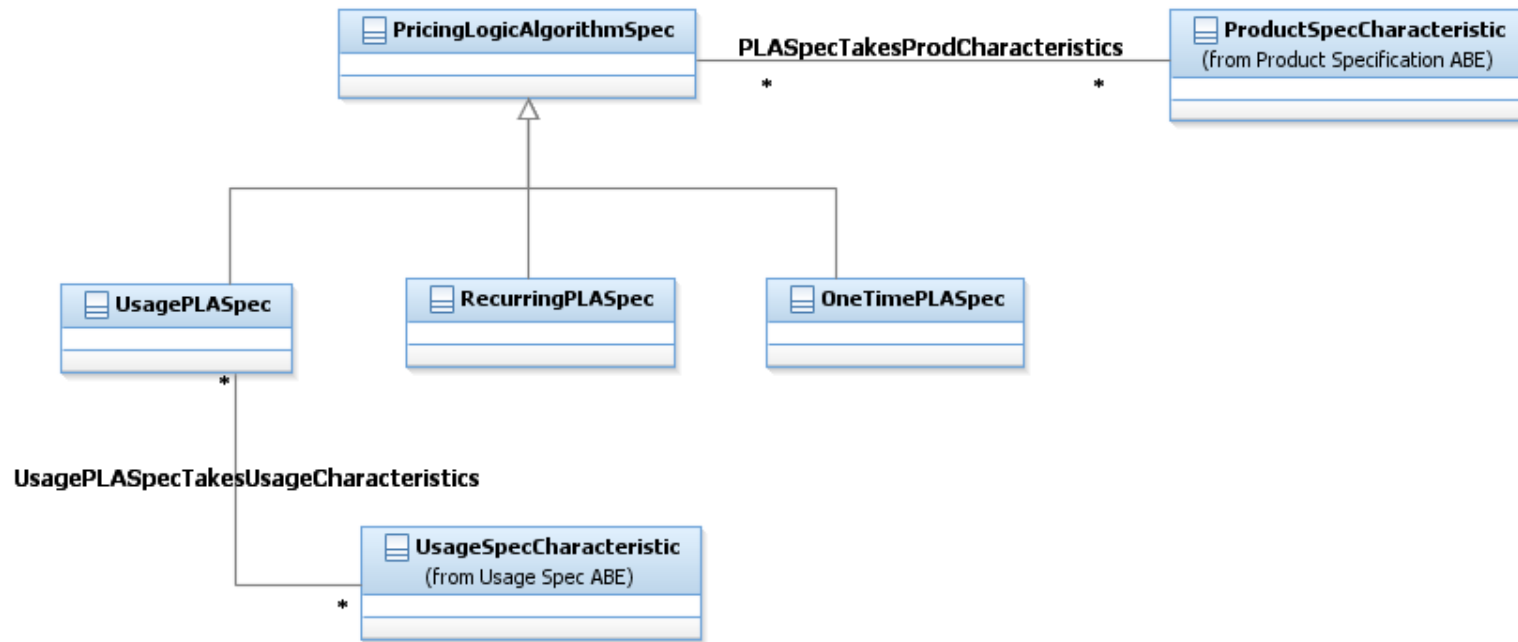


Figure Pr.27 – Types of PLA Spec and associations to different Spec Characteristics

This diagram completes the definition of PricingLogicAlgorithmSpec and together with Figure Pr.26 it shows all the sources of parameters which are needed to be defined and associated with the PLA spec in order to create an instance (PricingLogicAlgorithm) which has enough information (with Product and Usage information) in order to rate an event.

The next diagram shows how the PricingLogicAlgorithm uses the ProductCharacteristicValue and the Usage record information and associated UsageCharacteristicValue in order to generate a charge out of an event:

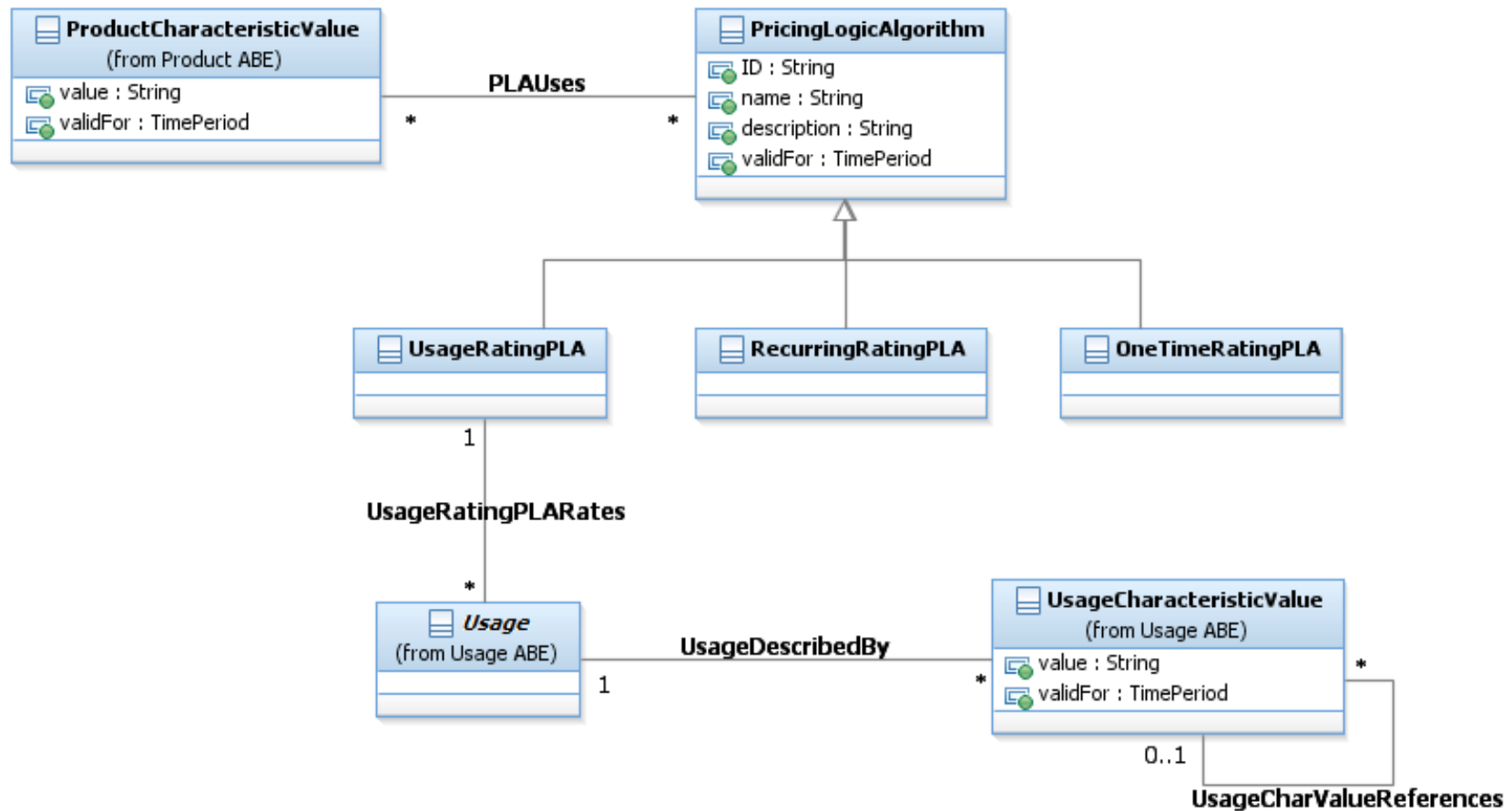


Figure Pr.28 – Types of Pricing Logic Algorithms and associations to different characteristic values

The following diagram shows the entire picture – how PLA Spec is defined, how PLA is instantiated, and the association to the possible event related characteristic values.

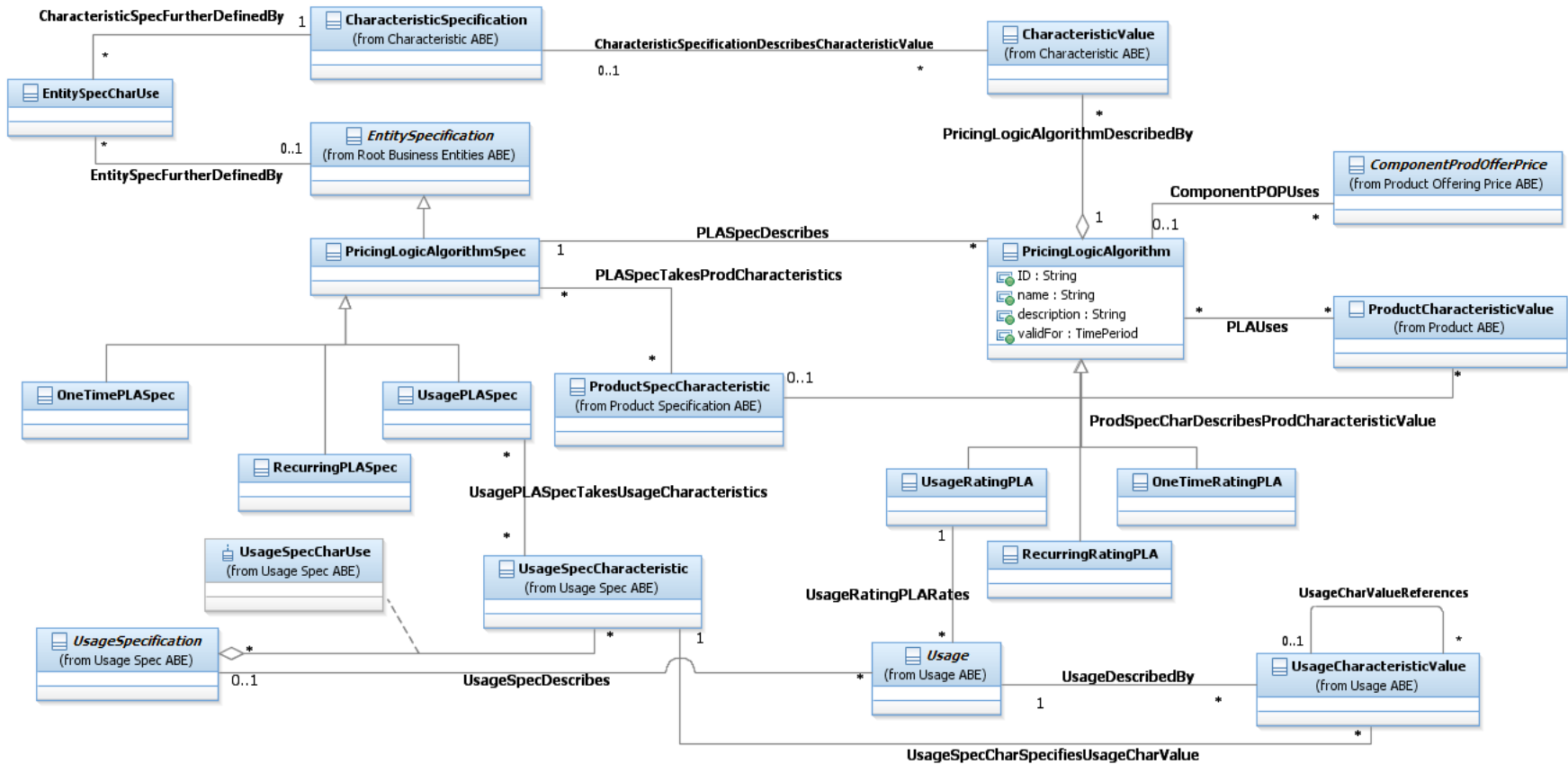


Figure Pr.29 – Pricing Logic Algorithm full picture

1.7.1. Modeling Matrices

As described in the use cases, a typical implementation of PricingLogicAlgorithm requires parameters which are multi-dimensional matrices, in which each dimension is defined by the valid values for a specific characteristic. For example, if we have PLA which rates events based on customer type, we can have one instantiation in which customer type is either residential or commercial (and in which case the size of the dimension of the matrix is 2) and in other instantiation the customer type can be student, normal and senior citizen (in this case the size of the dimension is 3).

Modeling the matrix

As described above, the matrices are defined by the possible valid values of characteristics (from ProductSpec or PLASpec), so we can define the matrix itself using the self-association CharacteristicSpecificationReferences. This was correct if CharacteristicValue and ProductCharacteristicValue were the same entity. Since currently these entities do not share a common base class there is a need to create some more complex modeling in order to create a matrix.

Note: MatrixSpecDimention is associated to a single entity either ProductSpecCharacteristic or CharacteristicSpecification – but it is always associated to one (association to none or both entities is not allowed, so this class (MatrixSpecDimention) is a specification for a single dimension of the matrix.

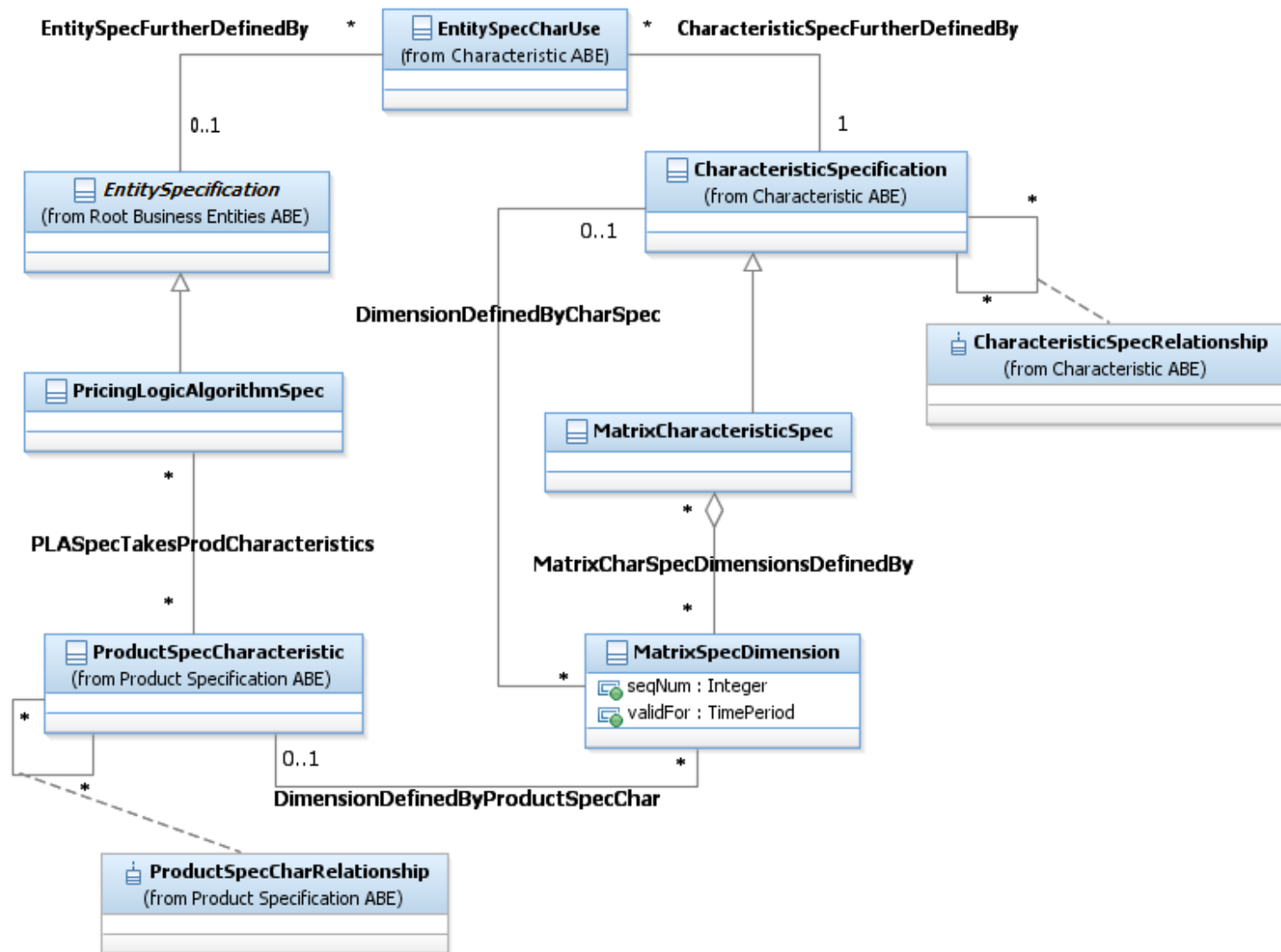


Figure Pr.30 – Modeling a matrix with dimensions based on PLASpec and ProductSpec Characteristics

Modeling the Matrix Values

We can look at each value (cell) of n dimensional matrix as $n+1$ -tuple of Characteristic Values, one describes the cell value and the other n are the indices of the dimensions (not numeric indices, but the values are themselves indices). Since each index can be either CharacteristicValue or ProductCharacteristicValue we cannot use the self-association for the CharacteristicValue class. Instead we define the class MatrixCharValueIndex which associate either CharacteristicValue or ProductCharacteristicValue. In addition we define a class which is derived from CharacteristicValue which represent a matrix value and includes associations to the matrix dimensions (defined by either CharacteristicValue or ProductCharacteristicValue) and aggregates the matrix cell values which are derived from CharValue (to include the value) and associated with the MatrixCharValueIndex for the indices of this cell value.

This is the matrix value diagram

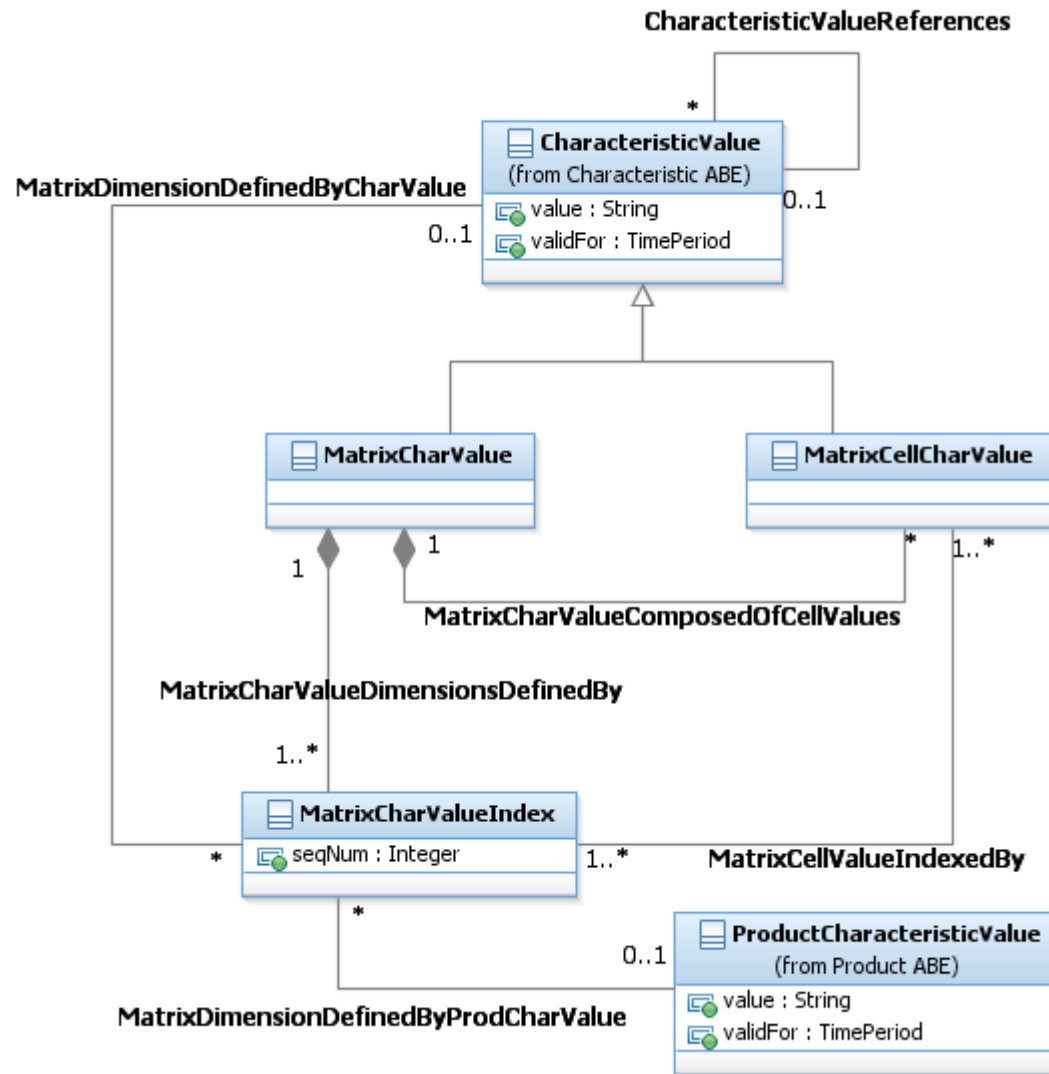


Figure Pr.31 – Modeling matrix values

1.8. Product/Product/Product Price

When an enterprise agrees to deliver a product to a customer, a price is associated with the Product that has to be recorded. The figure below presents the set of entities that are used to hold this information. The ProductPrice represents an amount, usually of money, that represents the actual price paid by a customer for a purchase, a rent or a lease of a Product. The ProductPrice is associated with the Product entity and the ProductOfferingPrice entity. The structure of ProductPrice subclasses resembles the structure of ProductOfferingPrice subclasses.

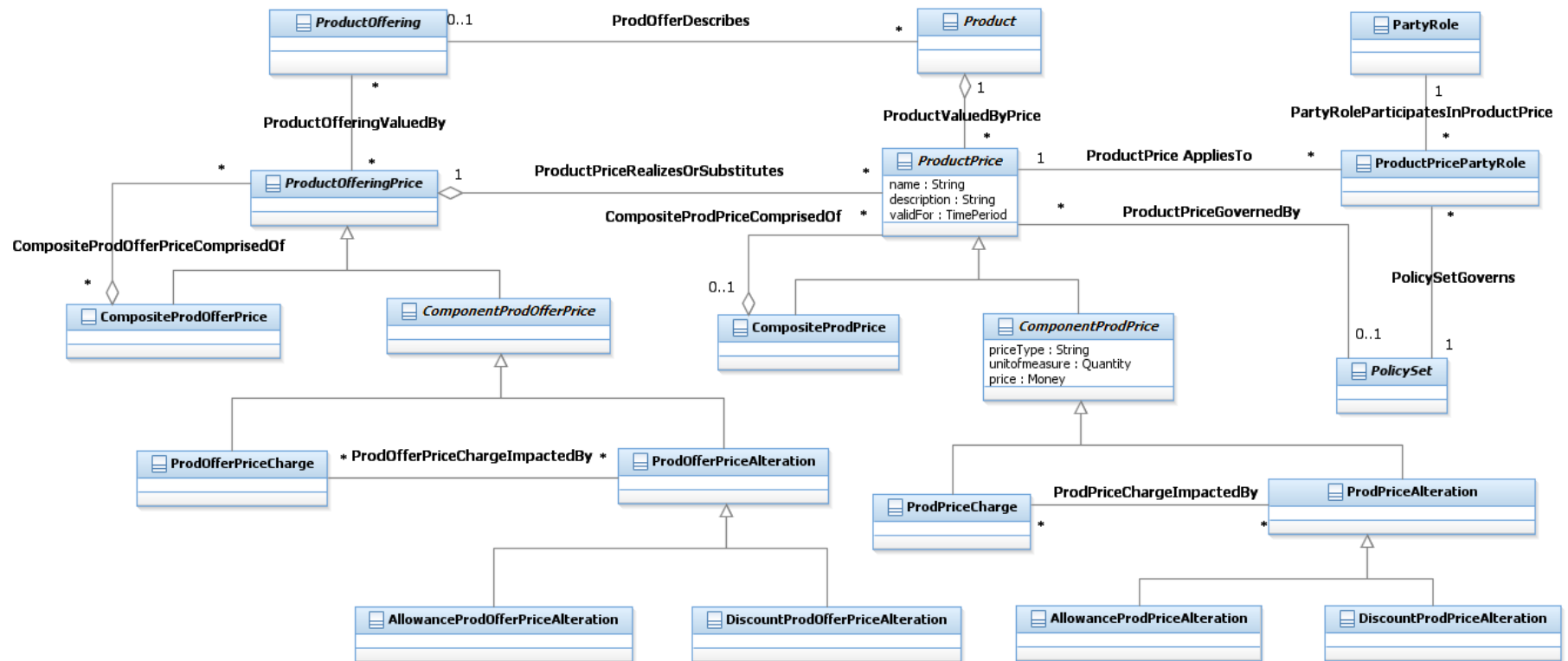


Figure Pr.32 – Product Price

1.9. Product/Product Usage

Put simply, usage is how much service is used, by whom is it used, where and when is it used and circumstances under which it is used. Normally, when a usage event occurs, it is stored in a network element, for instance in a switch, router, gateway or in an application system. Resources (applications, network and computing capabilities) usually store usage data in proprietary formats that are not understood by the billing system. Depending on the polling strategy, a mediation engine connects to resources, collects usage data and formats them into a format that is understood by the billing system. Output of a mediation engine are Usage Detail Records (UDRs). Examples of UDRs are Call Detail Records (CDRs – used to describe usage details of voice call services) and Internet Protocol Detail Records (IPDRs – used to describe usage details of Internet Protocol based services). In this document we will use the Usage abstract business entity to describe any resource-, service- or product-based usage that the billing system can read, update and process.

1.9.1. Usage

Figures Pr.27 and Pr.27a show business entities that are used to model usage. The model shown below is generic; particular usage models should define their own usage business entities by extending the Usage entity.

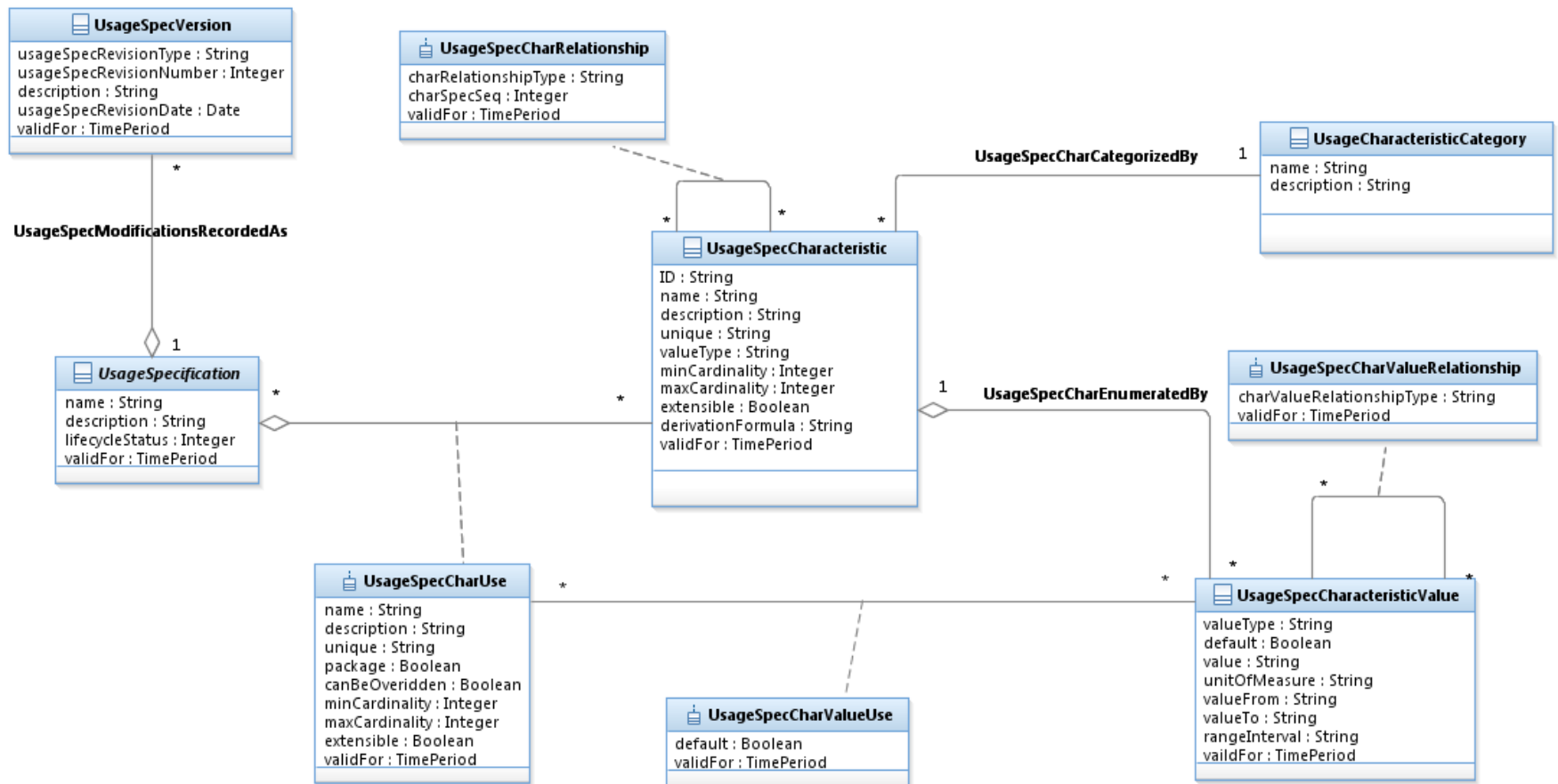


Figure Pr.33 – Usage Specification

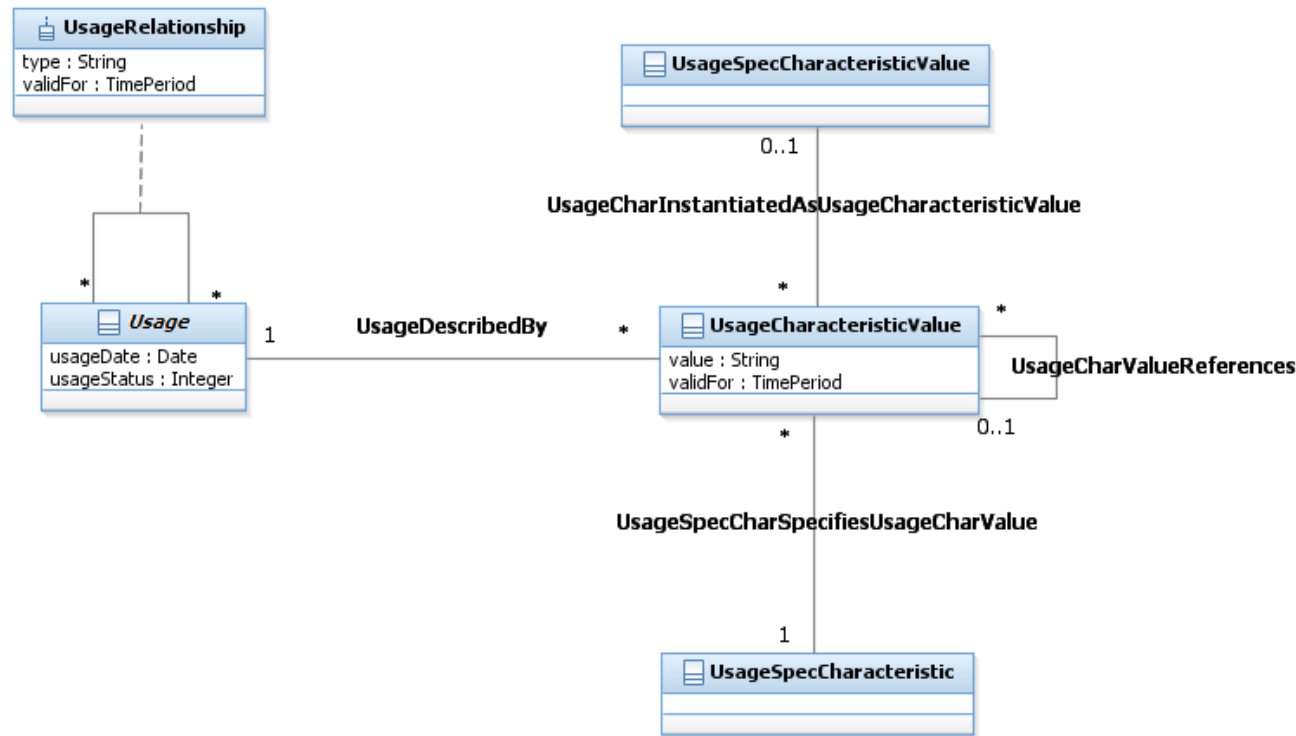


Figure Pr.34 - Usage

The UsageSpecification class describes a type of usage. In order to achieve a flexible structure of the UsageSpecification all its attributes are stored as characteristics. Alternatively, different types of usage could be modeled as sub-classes of the Usage entities (Usage, ProductUsage, ServiceUsage, ResourceUsage).

The UsageSpecification is comprised of UsageSpecCharacteristics that define all characteristics (attributes) known by a particular type of usage. Each characteristic is described by its name, category, presence and a set of allowed values.

Name defines a unique name that identifies the characteristic.

Category defines one of the high-level aspects of the usage information described by the characteristic. These categories are commonly referred to as: Who, When, What, Where, and Why.

Presence defines whether the characteristic is required or optional.

The set of allowed values is modeled with UsageSpecCharacteristicValue. This class describes all allowed values or value ranges.

Additional information regarding the CharacteristicSpecification/CharacteristicValue pattern can be found in GB922 Addendum 1R – Root Business Entities.

The Usage class represents a usage event. It is described by the UsageSpecification and comprised of UsageCharacteristicValues that hold data attributes of a usage specified by UsageSpecCharacteristics. The UsageCharacteristic can either store its value in the usageCharValue property or reference predefined UsageSpecCharacteristicValue.

The UsageSpecification entity has the following subclasses:

- ServiceUsageSpec (that represents a service usage);
- ResourceUsageSpec (that represents a resource usage);
- ProductUsageSpec (that represents a product usage).

1.9.2. Product Usage

A Product is realized as one or more Service(s) and/or Resource(s). A Product is created by instantiating a ProductOffering and adding it to a specific customer account. In order to charge a service (or a resource) usage, the service (or resource) has to be associated to a product.

Figure Pr.28 shows relationships among the ProductSpecification, the Product, the ProductUsageSpec and the ProductUsage ABEs. The ProductUsageSpec is defined for the certain ProductSpecification. As the ProductSpecification has 2 subclasses (the AtomicProductSpecification and the CompositeProductSpecification), the ProductUsageSpec needs 2 subclasses as well: the AtomicProductUsageSpec and the CompositeProductUsageSpec (associated with the AtomicProductSpecification and the CompositeProductSpecification entities respectively).

The ProductUsage describes the usage of an instantiated Product. The Product ABE is extended with the ProductComponent and the ProductBundle business entities. Consequently, the ProductUsage is extended with ProductComponentUsage and the ProductBundleUsage business entities (that are associated with the ProductComponent and the ProductBundle respectively).

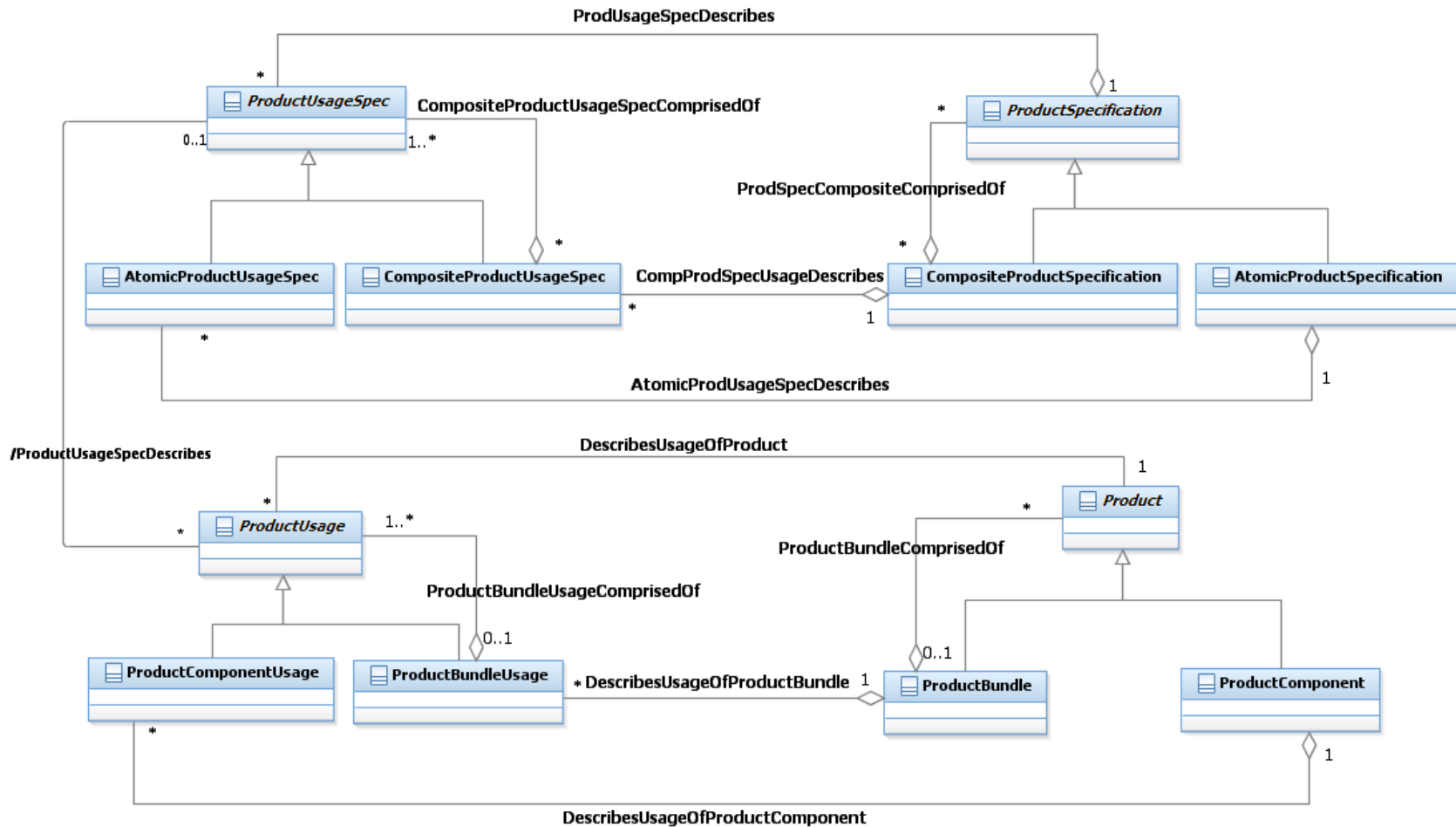


Figure Pr.35 – Product Usage, Part 1

As mentioned above, a product is realized as one or more services and/or resources. Consequently, the *ProductUsageSpec* business entity has to be associated with the *ResourceUsageSpec* and the *ServiceUsageSpec* business entities. The *ProductUsage*, the *ServiceUsage* and the *ResourceUsage* business entities have to reflect these associations as well. The *ProductComponentUsage* can be associated with either *ServiceUsage* or *ResourceUsage* and included in a *ProductBundleUsage*. These entities and associations are shown on the figure Pr.29.

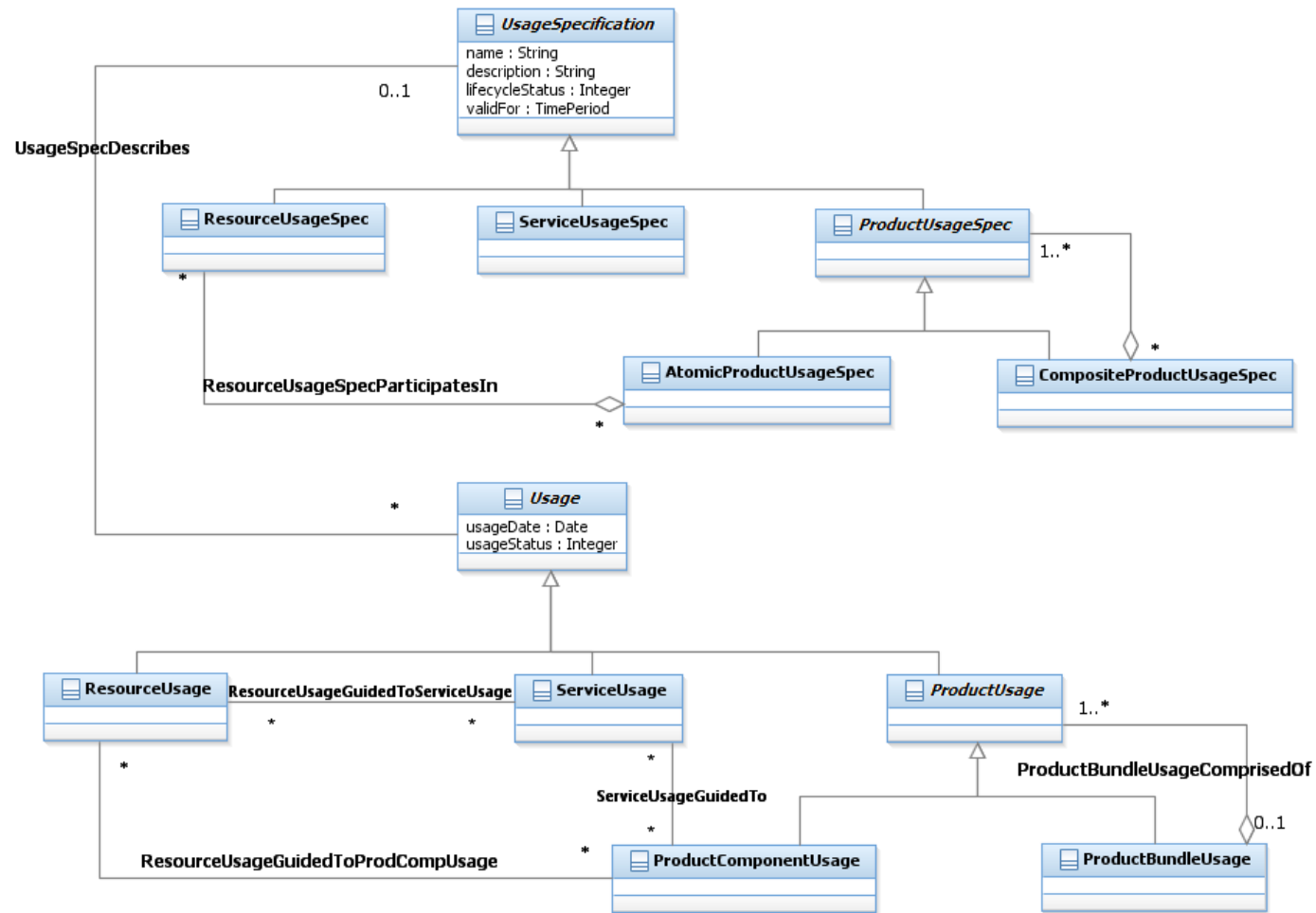


Figure Pr.36 – Product Usage, Part 2

1.10. Product/Product Catalog

From SID v12.5 release onwards, the ProductCatalog documentation and respective entity definition (data dictionary) is available in the SID Catalog guide book.

1.11. References

Normative References	
UML 1.5	OMG UML Resource Page http://www.uml.org/
Non Normative References	
Fowler-AP	Analysis Patterns – Reusable Object Models by Martin Fowler ISBN 0-201-89542-0 http://martinfowler.com/articles.html
Fowler-Time	Patterns for things that change with time http://martinfowler.com/ap2/timeNarrative.html
Fowler-Properties	Dealing with Properties by Martin Fowler http://www.martinfowler.com/apsupp/properties.pdf
Fowler-Range	Range http://martinfowler.com/ap2/range.html
Fowler-Specification	
ACIA	Alliance Common Information Architecture (AT&T, BT & Concert) version 3.1, 15 Dec 2000
eTOM	GB921 version 7.0
Metasolv	Sales & Marketing Business Objects Document for TMF
Larman	Applying UML and Patterns, Second Edition ISBN 0-13-095004-1 http://www.craiglarman.com/book_applying_2nd/Applying_2nd.htm
Arsanjani	Service Provider: A Domain Pattern and its Business Framework Implementation http://st-www.cs.uiuc.edu/~plop/plop99/proceedings/Arsanjani/provider3.pdf
Hay	Advanced Data Model Patterns by David Hay
OMG	ftp://ftp.omg.org/pub/docs/formal/00-11-11.pdf
SID	GB922 Addendum 1-POL Policy Business Entity Definitions
SID	GB922 Addendum 1A – Agreement Business Entity Definitions

2. Administrative Appendix

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired; however a Document History must always be included.

2.1 About this document

This is a TM Forum Guidebook. The guidebook format is used when:

- The document lays out a ‘core’ part of TM Forum’s approach to automating business processes. Such guidebooks would include the Telecom Operations Map and the Technology Integration Map, but not the detailed specifications that are developed in support of the approach.
- Information about TM Forum policy, or goals or programs is provided, such as the Strategic Plan or Operating Plan.
- Information about the marketplace is provided, as in the report on the size of the OSS market.

2.2 Document History

Version History

Document Version	Date	Modified By	Purpose
0.1	Jan 2002	John Reilly	GB915 Addendum Template created from TMF407 v1.1 Guidebook template
	Feb 2002	John Reilly	TAW - updated diagrams & minor changes
0.5	May 2002	John Reilly	Change to Product Offering and Product (Instance) agreed on team calls. General changes resulting from Telstra comments including Version.
1.0	June 2002	John Reilly	Formatted for Member Evaluation release
1.1	Sept 2002	John Reilly	Phase II updates.

Document Version	Date	Modified By	Purpose
1.2	Sept 2002	John Reilly	Updates based on team review.
3.0	May 2003	John Reilly	Added Product Pricing, updated association and attribute names.
5.0	May 2004	John Reilly	Added Product Pricing rules, added data types to figures, added subclasses to ProductSpecificationType.
5.1	August 2004	John Reilly	Synchronized with other Addenda.
6.0	July 2005	John Reilly	Update based on review comments
6.5	November 2006	John Reilly	Included more product offering and characteristic examples; added Product Catalog ABE within Product Offering ABE; minor edits.
6.6	January, 2007	John Reilly	Minor & cosmetic Updates.
7.0	March 2007	John Reilly	Updates to Product Offering, Product Specification, Product, Product Spec Characteristics
7.5	January 2008	John Reilly	Updated for ITU-T submission
7.6	May 2008	Tina O'Sullivan	Converted to correct template and other corrections prior to posting.
7.7	July 2008	John Reilly	Updates to Product Offering, Product Specification, Product, Product Spec Characteristics
7.9	May 2009	Alicja Kawecki	Minor updates to reflect TM Forum Approved status
9.0	Jan 2010	Josh Salomon	Updates to reflect change requests from V9.0
9.0	Feb 2010	Josh Salomon	Added PricingLogicAlgorithm contribution.
9.1	Apr 2010	Alicja Kawecki	Minor cosmetic corrections for web posting and ME
9.2	Jun 2010	Alicja Kawecki	Updated Notice
9.3	Oct 2010	Alicja Kawecki	Updated to reflect TM Forum Approved status
9.4	Jan 2011	Dirk Rejahl	Incorporation of CRs: artf1364, artf2012, artf2079, artf2087
9.5	Mar 2011	Alicja Kawecki	Minor formatting corrections prior to web posting and ME
9.6	Sep 2011	John Reilly	Updated multiplicity on Figure Pr.04c.
9.7	Jan 2012	Josh Salomon	Updates Figure Pr.07 and InvolvementRole modeling
9.8	Mar 2012	Alicja Kawecki	Minor cosmetic corrections prior to web posting and ME
9.9	Oct 2012	Alicja Kawecki	Updated to reflect TM Forum Approved status of R12.0
9.10	Oct 2012	Raj Ravi	Removed the ProductCatalog definitions and respective

Document Version	Date	Modified By	Purpose
			documentation as it is moved to the Catalog guidebook which is introduced as a part of SID12.5 release.
9.11	Oct 2012	Alicja Kawecki	Minor style/cosmetic corrections prior to web posting and Member Evaluation
9.12	June 2013	Cécile Ludwichowski	Incorporation of CR: Artf2357 (Fig Pr30) Artf2381 (Fig Pr2,)
9.13	July 2013	Alicja Kawecki	Updated cover & header, corrected notice & footer prior to posting
9.14	Sep 2013	Alicja Kawecki	Added IPR Mode to cover, updated notice to reflect TM Forum Approved status
9.14.1	Nov 2013	Alicja Kawecki	Updated cover, header, footer & Notice
14.5.0	Oct 2014	John Reilly	Updated relationships shown in Figures Pr.4c and 8.
15.0.0	Apr 2015	Cécile Ludwichowski	Update Figure Pr.05
15.0.1	May 2015	Alicja Kawecki	Updated cover, Notice
15.0.2	Nov 2015	Alicja Kawecki	Updated cover and Notice to reflect TM Forum Approved status
16.0.0	May 2016	Cécile Ludwichowski	Adds - AtomicProductSpec sub-classes - illustrations Update descriptions and diagrams Remove Entities and Attributes descriptions as they are embedded in RSA.
16.0.1	25 May 2016	Alicja Kawecki	Minor cosmetic edits prior to publication for Fx16
16.0.2	8 Sep 2016	Alicja Kawecki	Updated cover, header and Notice to reflect TM Forum Approved status

Release History

SID Release Number	Date Modified	Modified by:	Description of changes
7.0	January 2007	SID Team	
8.0	March 2008	SID Team	Updates to Product Offering, Product Specification, Product, Product Spec Characteristics
9.0	January 2010	SID team	Various updates to diagrams and fix discrepancies in existing model. Added Pricing Logic Algorithm ABE description into this addendum
9.5	January 2011	SID team	Incorporation of CRs: artf1364, artf2012, artf2079, artf2087

9.5	September 2011	SID team	Corrected multiplicity on Figure Pr.04c
12	January 2012	SID Team	Corrected Figure PR.07 based on changes in InvolvementRole modeling as described in addendum 1 Users and Roles
12.5	October 2012	Catalog Team	Removed the ProductCatalog definitions and respective text as it is moved to the Catalog guidebook which is as a part of SID12.5 release.
13.0	June 2013	SID Team	Incorporation of CR: Artf2357 (Fig Pr30) Artf2381 (Fig Pr2,)
14.5.0	October 2014	John Reilly	Updated relationships shown in Figures Pr.4c and 8.
15.0.0	April 2015	Cécile Ludwiczowski	Update Figure Pr.05
16.0.0	May 2016	Cécile Ludwiczowski	Adds - AtomicProductSpec sub-classes - illustrations Update descriptions and diagrams Remove Entities and Attributes descriptions as they are embedded in RSA.

2.3 Company Contact Details

Company	Team Member Representative
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